

# Ancestral Beliefs and Fertility in Sub-Saharan Africa

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November 20, 2024

## Abstract

This paper contributes to the explanation of the puzzle of persistently high fertility in sub-Saharan Africa. I focus on the impact of a belief system that emphasizes the role of ancestors, who influence people's lives and have a strong interest in the continuation of their lineage into which they may be reincarnated. I combine first-hand data with original ethnographic information and both historical and contemporary surveys to show: 1) a strong, positive relationship between ancestral beliefs and fertility in different contexts and time periods; and 2) that this relationship is specifically driven by the motive to continue one's lineage. I test the specific predictions of a simple model of fertility in which children are a public good for a family with ancestral beliefs because they continue the family line. However, whether one's children continue one's lineage depends on the kinship system: while this is the case in a patrilineal system, children only continue the mother's lineage in matrilineal societies. The model predicts that 1) ancestral beliefs have a stronger positive influence on fertility in patrilineal societies; and 2) in groups with ancestral beliefs, very specific free-riding behaviors emerge: in patrilineal societies, male fertility decreases with the number of brothers, whereas in matrilineal societies, female fertility decreases with the number of sisters (but not brothers). These predictions are supported by the data.

Keywords: Fertility, Sub-Saharan Africa, Culture, Supernatural Beliefs, Kinship

JEL Classification: O12, J13, Z12, Z13

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## 1. Introduction

The fertility transition in sub-Saharan Africa (hereafter SSA) has been described as different from that in other low- and middle-income countries. It started later, from higher fertility levels, and has occurred at a slower pace (Bongaarts and Casterline, 2013; Bongaarts, 2017; Shapiro and Hinde, 2017; Schoumaker and Sánchez-Páez, 2024). However, there is no consensus as to why (Casterline, 2017). In this regard, conventional demographic transition theory, which focuses almost exclusively on the importance of material conditions and individual choices, fails to predict Africa's fertility trends, and supply-side interventions aimed at curbing fertility have often been disappointing (Dupas et al., 2024).<sup>1</sup>

The limitations of conventional demographic theory in understanding the factors contributing to the stalling of fertility decline suggest room for alternative explanations that consider the importance and evolution of cultural norms, values, and practices, often rooted in traditional beliefs, to better understand the observed patterns of reproductive variation (Caldwell and Caldwell, 1987, 1988, 1990; Casterline, 2017). In particular, the influence of religious and cultural beliefs has been overlooked in economics, despite their pervasiveness and importance for human behavior (Butinda et al., 2023).

This paper contributes to the explanation of the "puzzle" of persistently high fertility in SSA. I examine the influence of a belief system that, embedded in a social structure organized around descent and kinship, emphasizes the importance of continuing the family line and equates fertility with virtue and spiritual approval. The key feature of this traditional belief system is the belief that ancestors, the spirits of the dead, continue to influence society and have a strong interest in the continuation of their lineage in which they may reincarnate (Radcliffe-Brown, 1922; Fortes, 1965; Caldwell and Caldwell, 1987). The close relationship between fertility and belief in ancestral spirits lies in the nature of African kinship, where the living are only a fraction of a lineage composed of both living members and ancestors whose primary goal is the reproduction of the lineage. To this end, ancestors bless the fertility of their descendants, bring wealth, health, and

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<sup>1</sup>Much of the literature has focused on the role of factors related to the region's (low) economic development: low income, high infant mortality, low education, high participation in agriculture, low urbanization, and low access to health and family planning services (World Bank, 1986; National Research Council, 1993). Yet, conditional on these factors, fertility rates remain about one birth higher in SSA (Zipfel, 2022).

children, and reincarnate. The resulting belief system works to maintain high fertility by placing great emphasis on the continuation of the family line, and has shaped societies to reward high fertility (both socially and spiritually). Most of the powers attributed to the ancestors are therefore related to family survival, and high fertility is morally associated with right living, joy, and ancestral approval, while low fertility, miscarriage, or infertility is associated with sin and misfortune, showing ancestral disapproval and punishment.

I proceed in two steps. First, I build on a rich demographic and anthropological literature and combine first-hand data with novel ethnographic information and historical and contemporary surveys to examine the relationship between beliefs in the influence of ancestors and fertility in SSA. I find that, across different samples, econometric specifications, and time periods, there exists a positive relationship between ancestral beliefs and fertility rates. Second, to show that the positive association found in the first part of the paper is specifically due beliefs in ancestors, I develop a simple model of fertility where the motive to continue the family line enters as a public good. This model provides specific predictions that I test in the data. I show that the patterns observed in the data are consistent with the motive to continue the family line driving fertility upwards, and are inconsistent with alternative conventional explanations of high fertility rates.<sup>2</sup>

I begin by analyzing first-hand data collected among farmers in contemporary southern Benin. This is mostly a rural context, where the clan-based structure of the society is still pervasive and traditional beliefs are widespread despite the success of modern religions. In this context, holding strong beliefs in ancestors is associated with about one additional child, which is comparable to the effect size of having attended primary school, or to the difference in births per women between SSA and other low-income regions.

Second, I focus on the context of post-independence Democratic Republic of the Congo (DRC) and match original information on beliefs in ancestors at the ethnic group

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<sup>2</sup>I will use the term "*ancestor worship*" as a synonym of beliefs in ancestors throughout the paper. However, there is controversy about the adequacy of the term ancestor worship in the African context. For example, Kenyatta (1965) prefers the term "*communion with the ancestors*", as he mentions that communing with the ancestors is an aspect of the everyday life of the African (ancestors are still members of the community), and their attitude towards ancestors is very different than their attitude towards the deity (who is truly worshiped). Similarly, Grande (2024) notes that "Ancestorism is not always a matter of ancestor worship. It is most often a matter of ancestral spirits helping sustain the living family while being sustained by it in return (through rituals)."

level, digitized from Vansina (1966), with individual-level birth histories for about 70,000 women from two unexplored demographic surveys conducted in the west part of Congo between 1975 and 1977. Similarly, I find that ancestor worship is associated with about one additional child, and this relationship holds even when including origin-destination fixed effects.

Finally, I show that these results generalize to the wider context of sub-Saharan Africa by using individual-level information on contemporary supernatural beliefs from 25,000 respondents distributed across 19 SSA countries coming from Pew Research Center (2009), and that the link between beliefs in ancestors and fertility exists in the oral traditions of pre-industrial societies, where ancestral beliefs were widespread.

In the second part of the paper, I show that the positive influence of ancestral beliefs on fertility is specifically due to the motive to continue one's lineage. To do this, I develop a simple model of fertility decisions that incorporates the total number of children in the lineage as a public good, because they continue the family line. Yet, whether one's children belong to one's lineage depends on the kinship system due to the asymmetry in marital allegiances (Fox, 1983; Berggreen and Gokmen, 2023).<sup>3</sup> In patrilineal societies, lineage is traced through male members, while in matrilineal societies lineage is traced through female members. From the men's perspective, the lineage is continued by one's children and by the children of one's brothers in patrilineal societies, and by the children of one's sisters in matrilineal societies. This is because the husband does not belong to the lineage of his wife in matrilineal societies. From the women's perspective, the lineage is continued by one's children and by the children of the husband's brothers in patrilineal societies, and by one's children and the children of one's sisters in matrilineal societies. The asymmetry comes from the fact that women in patrilineal societies do belong to their husbands' lineage. Therefore, one's children always continue one's lineage in patrilineal societies, while children only continue the mother's line in matrilineal societies. The model provides two main predictions. First, the influence of ancestral beliefs on fertility is quantitatively stronger in patrilineal societies, where both spouses and the children belong to the same lineage. Second, very specific free-rider behavior among siblings emerges. In patrilineal societies with strong beliefs in ancestors, the number of husband's brothers contribute negatively to one's fertility, since their children also con-

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<sup>3</sup>See Appendix L for a simple introduction to kinship systems.



tinue the husband's lineage (however, the husband's sisters and their children belong to a different lineage, that of their husbands). Similarly, female fertility is negatively associated to the number of sisters (but not of brothers) in matrilineal societies.

Consistent with these predictions, I find that: 1) the positive influence of ancestor worship on fertility is driven by ethnic groups where patrilineality is the main type of descent, both at the global level in the folkloric data and in contemporary DRC;<sup>4</sup> and 2) there is negative relationship between one's fertility and the number of family members capable of continue one's lineage with their own children. Using survey data from the DRC DHS (where there is information on household composition) and ethnographic information from Vansina (1966) (where there is information on ancestor beliefs) and Murdock (1967) (where there is information on descent types), I show that there is a negative relationship between male fertility and the number of men aged 15-49 living in the household, but only in patrilineal societies with strong ancestral beliefs.<sup>5</sup> Similarly, using the same data (now with accurate information on respondents' siblings), I show that there is a negative relationship between women's fertility and the number of sisters (but not brothers) they have, but only in matrilineal societies with strong beliefs in ancestors.

To further validate the theory at hand, I examine some additional implications. First, I study the interaction between ancestral beliefs and social organization. Since ancestral beliefs are closely linked to kinship and lineage structures (they represent the extension of the core elements of many African societies – ancestry, kinship and descent relations – to the supernatural sphere and emphasize the importance of extending one's lineage) I examine the question of whether ancestral beliefs will disappear as the lineage-based structure of societies becomes less salient. I make use of the distinction between lineage- or kin-based societies and age set societies. I focus on Kenya and use information from Moscona and Seck (2024) to distinguish between age-based and lineage-based societies

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<sup>4</sup>Similarly, in the DRC context, I find that the positive effect of patrilineality on fertility is only present when ancestor worship is practiced, which helps to better understand the positive relationship between patrilineality and fertility suggested in different contexts (BenYishay et al., 2017; Okafor et al., 2021; Fontenay et al., 2024). I provide supportive evidence that group membership (parents and children belong to the same lineage or not) is a key mechanism for understanding the relationship between kinship structure and fertility.

<sup>5</sup>Unfortunately, the DHS does not include information on siblings for men, so I use the number of men in the household aged 15-59 as a proxy for the number of men able to continue the respondent's lineage by having children.

and show that the relationship between ancestor worship and fertility is stronger in lineage-based societies. Second, I explore whether traditional beliefs in ancestors are related to lower contraceptive use. Focusing on the context of Benin, where I can control for supply-side factors related to the access to contraceptives, I show that traditional beliefs are associated with lower contraceptive use, providing additional evidence of the importance of demand rather than supply factors in explaining the low contraceptive uptake in sub-Saharan Africa (Bau et al., 2024; Dupas et al., 2024).

Overall, these results are consistent with a setting where the motive to continue the family line drives fertility upwards. I highlight the importance of taking into account how belief systems interact with important elements of the social structure in shaping economic behavior. Furthermore, this analysis suggests that future fertility decline will probably be accelerated by factors that contribute to the dissolution of clan-based institutions centered on the lineage and the extended family (e.g., increases in population density and the emergence of private property rights), which in turn impact the persistence of traditional belief systems that sustain high fertility.

This paper contributes to several strands of the literature. First, I contribute to existing analyses of the determinants of high fertility in SSA (see Church et al. (2023) for a recent review), and in particular to the literature on the relationship between cultural norms, social organization and fertility. More than 70 years ago, Lorimer (1954) noted the relationship between high and stable fertility and unilineal descent groups. Three decades later, a growing literature complemented Lorimer's analysis by linking the importance of descent in many African societies to their belief systems and certain aspects of their social organization, which focused on maintaining the family line and reverence for ancestors (Caldwell and Caldwell, 1985, 1987, 1988, 1990; Lesthaeghe, 1989; Caldwell et al., 1992). However, although the relationship between religious and cultural beliefs and fertility behavior has been extensively studied in evolutionary demography and cultural anthropology, it has been overlooked in economics. To the extent of my knowledge, I am the first to quantitatively study the influence on fertility of one of the most prevalent belief systems in the context of sub-Saharan Africa.

More recently, some papers in economics have explored whether today's fertility preferences and behaviors may be shaped by some features of social systems such as clan linkages (Bauer et al., 2006), inheritance rules (Fontenay et al., 2024; Sage, 2023), need

for old-age security due to the lack of social pensions (Chen and Roth, 2023), the occupational structure of SSA societies (Zipfel, 2022), perception of social norms (Dupas et al., 2024) or by responses in family institutions to disruptions in traditional practices and modes of production triggered by colonial institutions such as Christian missions (Guirkingner and Villar, 2022) or forced labor migration (Dupas et al., 2023). The contribution to this literature is twofold. First, I explore the influence of an alternative explanation (the importance of the traditional belief system present in many sub-Saharan African societies), and highlight how belief systems are intimately bounded to the social organization of societies in order to reward specific behaviors (i.e., high fertility rates). Second, my findings highlight the role of men's beliefs and motivations to understand reproductive patterns. In fact, it is surprising that, although it is well recognized that men in sub-Saharan Africa have higher bargaining power than their spouses and are in many cases the sole decision-makers regarding fertility, most papers examining fertility outcomes focus exclusively on the role of women's characteristics. I show in this paper that men's incentives, for example in the form of differences in the strength of the motive to continue one's lineage depending on the kinship system, are key to understand high fertility.

Second, this paper contributes to the scarce but growing literature on the relationship between religion and fertility in Sub-Sahara Africa (e.g., Ishak and Gradstein (2022), Berger and Dasré (2024), or Götmark and Turner (2023) for a review). Most often, these papers compare the fertility trajectories of major religious groups, mainly Christians and Muslims, relying on the Demographic and Health Surveys. The general consensus is that Muslims have higher fertility than Christians, while followers of African Traditional Religions and Islam have similar fertility levels. However, these papers have important limitations. First, conventional demographic surveys such as the DHS are very problematic to study these questions since they under-report the importance and the coverage of African traditional religions and beliefs (which, although more common among followers of African Traditional Religions, are transversal to religious affiliations). The under-reporting of African traditional religions is mainly a consequence of the influence of missionaries, who promoted monotheistic religions as the only socially acceptable choice (Neill, 1991; Le Rossignol et al., 2022). Therefore, people usually report being a follower of a "missionary" religion (Christianism or Islam) while holding strong tradi-

tional beliefs, which is not captured in these surveys. I circumvent these limitations by focusing on the influence of specific beliefs rather than on declared religion.<sup>6</sup>

Finally, I contribute to the broad literature on the importance of cultural characteristics for socioeconomic outcomes (e.g., Luttmer and Singhal 2011; Alesina et al. 2013; Alesina and Giuliano 2015). Regarding fertility, Fernández and Fogli (2009) study how culture influences fertility by looking at the behavior of second-generation American women. They proxy cultural beliefs in their country of ancestry by using the total fertility rates in those countries 20 years before the data they use, and show that the cultural proxies have a positive and significant effect on the number of children that second-generation American women have. In particular, this paper adds to a growing literature on the effects and consequences of traditional supernatural beliefs.<sup>7</sup> For example, Le Rossignol et al. (2022) use a series of lab-in-the-field experiments in northern DRC to show that individuals with strong traditional beliefs are treated less pro-socially. In another paper, Gershman (2016) finds a negative relationship between witchcraft beliefs and trust and other prosocial attitudes, both within Africa and globally. In the context of Benin, Stoop et al. (2019) find that voodoo adherence is associated with lower demand for health care and worse child health outcomes. Sievert (2023) shows that supernatural beliefs about the cause of illness are very common and relevant for health behavior, resulting in lower use of modern medicine, lower beliefs about the effectiveness of modern medicine, and higher stigma toward those with illness. Finally, in a recent paper, Butinda et al. (2023) argue that African traditional religions influence economic decision-making by showing that, in the context of beer sellers in the eastern Democratic Republic of the Congo, religious rituals correct overly pessimistic beliefs about the risk of theft. However, most of the work in this area has focused on witchcraft beliefs.<sup>8</sup> This paper contributes to the understanding of globally prevalent system beliefs that have received little attention, such as ancestor worship. Most of the effort dedicated to the recent study of ancestor

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<sup>6</sup>The under-reporting of African traditional religion varies by context. For example, Stoop et al. (2019) focus on Benin to examine the relationship between African traditional religion (Voodoo) and the demand for conventional healthcare. In this setting, adherence to Voodoo suffers less from the caveats presented above, since Voodoo is awarded the same status as monotheistic religions (it is mentioned explicitly in the constitution as an official religion), so traditional religion adherence is a better proxy for traditional religious beliefs.

<sup>7</sup>Indirectly, it also complements the literature on the persistence of traditional beliefs in Africa (Platteau, 2009; Nunn and Sanchez de la Sierra, 2017; Le Rossignol et al., 2022; Igboin, 2022).

<sup>8</sup>See Gershman (2022) for a review of witchcraft beliefs in economics.

worship comes from cultural anthropology. Indeed, no paper has examined the economic and social consequences and implications of beliefs in ancestors in the context of sub-Saharan Africa from a quantitative perspective, despite being recognized that these beliefs are prevalent and relevant to economic behavior.<sup>9</sup>

The remainder of the paper is organized as follows. Section 2 introduces the context and describes why there exists a close relationship between beliefs in ancestors and fertility. Section 3 presents the different datasets and variables used in the empirical analysis. Section 4 describes the different specifications and depicts the main results. Section 5 proposes a simple theoretical framework to better understand the empirical regularities. Finally, Section 6 tests some of the theoretical predictions and shows that the motive to continue one's lineage is key to explain the empirical results. Section 7 concludes.

## **2. Traditional religion and ancestorism in Africa**

An exhaustive description of African Traditional Religions (ATR) is beyond the scope of this paper (see Idowu (1973), Mbiti (1975), Opoku (1993) or, more recently, Olupona (2014) for definitions). For our purposes here, we will focus on the elements that have usually been highlighted as the main components of the structure of ATR. According to Idowu (1973), five elements go into the making of ATR: belief in God, belief in the divinities, belief in spirits, belief in the ancestors, and the practice of magic and traditional medicine. Of course, the weight of each of these components varies from society to society, ranging from very prominent in some areas to virtually absent in others. In this section I argue that of these five elements, belief in ancestors is crucial to understanding high fertility in sub-Saharan Africa.

According to Asante (2009), "ancestors are those who once lived in human society

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<sup>9</sup>There exist studies investigating the role of ancestor worship practices and fertility in other contexts, such as China. For instance, Zhang (2024) exploits a natural experiment, the Kuomintang's Retreat to Taiwan, which resettled approximately one million Chinese in Taiwan between 1945 and 1954, to show that ancestor worship (a cultural feature of the Chinese population) contributed to the transmission of son preference and high fertility rates. Hu and Tian (2018) examine the case of contemporary China. They find a positive correlation between current ancestor worship practices and childbearing and marriage outcomes. In the same context, Yang and Spencer (2022) find that the number of male siblings have a negative influence on fertility decisions, especially when individuals attribute high importance to family continuity. As the authors suggest, these findings are consistent with a free-rider behaviour in the context of a patrilineal kinship system where fertility is at least partially driven by the motive to continue a family line.

and, having fulfilled certain conditions, are now in the realm of the spirits".<sup>10</sup> Ancestor worship, or the cult of the ancestors, broadly defined as the set of rituals, practices, and beliefs designed to venerate deceased family members and perpetuate the lineage (Hu and Tian, 2018), has often been considered a crucial element of the traditional belief system of many African societies (Fortes, 1965). In contrast to cults of the dead, anthropologists have observed that the key characteristic of ancestor worship is the belief that the dead person continues to influence society (Radcliffe-Brown, 1922). This view can be summarized as follows:

"Ancestors are vested with mystical powers and authority. They retain a functional role in the world of the living, specifically in the life of their living kinsmen [...]. African kin-groups are often described as communities of both the living and the dead [...]. The African emphasis is clearly not on how the dead live but on the manner in which they affect the living (Kopytoff (1971), p.129)".

Therefore, ancestors, or the living dead, are seen as dispensers of both favor and misfortune, and their powers are most often used to ensure the survival of the lineage, for example, by contributing to the unification of families and people, or by protecting them from disease, evil, or enemies. However, ancestors are also believed to be the source of illness, misfortune, or disruption in the lives of their descendants, with their ultimate power being the curse that brings sterility and child death (Fortes, 1965; Caldwell and Caldwell, 1987; Ezenweke, 2008).

### *2.1. Prevalence of ancestral beliefs in SSA*

Before attempting to link belief in the influence of ancestors to fertility behavior, it is worth discussing why the belief in ancestors is more influential in sub-Saharan Africa than in any other region of the world. In fact, ancestor worship is not historically unique to SSA. For example, Caldwell and Caldwell (1987) hypothesized that ancestor worship may have been the original religion – understood here as a set of beliefs that structure the interaction between a community and the supernatural forces it perceives in the

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<sup>10</sup>The question of who established these criteria is not entirely clear (Igboin, 2022). However, some of these conditions, often of a moral and social nature, are common to different societies. For example, one must have lived an exemplary life by the standards of the community, respected the elders, married, and had children. In addition to having children, an additional criterion often mentioned is to have at least one son who will worship him, while having only female children may not be enough to become an ancestor. On the other hand, death should be natural and at a mature age, and proper burial rites must have taken place.

world – in many parts of the world, including Eurasia and the Americas.<sup>11</sup> Perhaps the most prominent and studied example outside SSA is South Asia, especially China, where it has been shown to influence fertility and son preference historically and even today (Ahern, 1973; Hu, 2016; Hu and Tian, 2018; Zhang, 2024).<sup>12</sup>

There are several factors that may explain the higher prevalence of beliefs in ancestors in sub-Saharan Africa compared to other regions of the world. First, the geographic isolation of SSA beyond the Sahara and the presence of several diseases prevented foreigners from gaining political power in the interior until the end of the 19<sup>th</sup> century, limiting their presence to coastal trading posts and thus preventing the spread of alternative belief systems that emphasized the relationship between individuals and external gods and undermined kinship ties (Caldwell and Caldwell, 1987; Schulz et al., 2019; Schulz, 2022).<sup>13</sup> Only the generalized use of quinine around 1850, which significantly reduced European mortality rates (Curtin, 1961), and the construction of railroads and roads facilitated the expansion of European powers into the interior of the continent, although this happened much later than in other regions (Jedwab et al., 2022).

Second, the long tradition of high land-to-man ratios in SSA has prevented the erosion of lineage and clan structures in which ancestor worship is embedded (Boserup, 1965, 1985; Platteau, 2000). In fact, ancestor worship is the extension of these relations to the supernatural sphere (Fortes, 1965; Caldwell and Caldwell, 1985). Collective farms in land-abundant societies based on production activities such as shifting cultivation are efficiently managed by extended families or lineages. As population density increases and land abundance decreases, private property rights tend to emerge and family structures shift towards the conjugal family (Platteau, 2000; Guirkingner and Platteau, 2014). Thus, agrarian societies in SSA are characterized by the presence of free peasants, free

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<sup>11</sup>This hypothesis may be traced back to Victorian anthropologist Herbert Spencer, who believed ancestorism to be the root of every religion (Grande, 2024).

<sup>12</sup>Other ancient societies, such as the Greeks, did not develop ancestor worship but rather a cult of the dead. This distinction is important. For example, Catholic societies have a cult of the saints and organize masses for the dead, but they are not considered to be ancestor worshippers either (Fortes, 1965). As already emphasized, the main difference between ancestor worship and the worship of the dead is the strong belief of the former in the influence of the ancestors in the affairs of the living. Death is a necessary but not a sufficient condition for becoming an ancestor.

<sup>13</sup>One example is the spread of Islam. As Michalopoulos et al. (2018) show, proximity to the pre-600 CE trade network is a robust predictor of Muslim adherence today, but in the case of sub-Saharan Africa these trade routes are virtually absent, and even if we look at pre-1800 CE trade networks, their development in most of SSA is also very precarious.

land, and the absence of non-working landowners, which reinforces lineage-based organization. For this reason, the *de facto* influence of beliefs in ancestors nowadays is stronger in sub-Saharan Africa than in regions where ancestor worship was traditionally widespread (e.g., Southeast Asia). As Platteau (2000) notes: "Asian societies have a strong tradition of settled agriculture, and property rights on arable lands have long been established with significant class differentiation or stratification in terms of land and other assets. Contrary to African societies which are centered around the lineage and are therefore permeated by values of extended solidarity, Asian societies, like all peasant societies (including European societies), have evolved over centuries social and family patters based on the conjugal family".

As a result of low population density and the prevalence of lineage-based systems of social organization, the intensity of kinship ties and the emphasis on descent are higher in SSA than in any other region of the world, which has hindered the dissolution of ancestral beliefs, since ancestors often symbolize the continuation of a social structure centered on lineage and descent, and therefore organized hierarchically by age, with the oldest person often occupying the position of authority (Alidou and Verpoorten, 2019).<sup>14</sup>

Finally, historical and institutional factors specific to SSA have traditionally favored the reproduction and persistence of cultural traits over time (Platteau, 2009). In particular, the existence of strong kinship ties, in which ancestral beliefs are embedded, was facilitated by the existence of weak states and the need for protection against raids during the Arab slave trade (c. 950-1950), and reinforced by the impact of colonialism and indirect rule on the social and political structure of African societies. In fact, due to the colonial administration's interest in increasing the power of local authorities, the system of indirect rule reinforced the clan-based structure of societies and increased ethnic consciousness by benefiting from the creation of uniform groups composed of individuals identified as members of a particular ethnic group and placed under the control of an African official regarded as a tribal or village chief (Ekeh, 2004; Ellis and Ter Haar, 2004;

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<sup>14</sup>In fact, age (and gender) are the two main dimensions along which African societies are stratified. Again, this is closely related to the high land-to-man ratios in SSA. In contrast, Asian village communities have well-established property rights, and (scarce) land is more subject to market exchange, which tends to concentrate ownership in the hands of a class of landowners. Moreover, lineage organization and ancestral beliefs reinforce each other because elders are often considered closest to the ancestors, as in the case of the Bakongo in the DRC or the Kaguru in Tanzania. In addition, because the hierarchy does not end with death, ancestors acquire greater authority than any living person and thus the power to intervene in the lives and affairs of their descendants.



Platteau, 2009).

Because of these contextual barriers, the new religions that spread rapidly in the 20<sup>th</sup> century have had only a limited impact on African traditional supernatural beliefs (Platteau, 2009; Igboin, 2022). Instead, modern religions have adapted to coexist with traditional supernatural beliefs, combining faith in a High God with supernatural beliefs (Platteau, 2014).<sup>15</sup> The prevalence of traditional supernatural beliefs have been noted in several recent studies, although the focus has often been put on witchcraft beliefs (van de Grijsparde et al., 2013; Leistner, 2014; Stoop and Verpoorten, 2020). With respect to beliefs in ancestors, Igboin (2022) and McCall (1995) highlight that, although ancestors remain a central feature of many (mainly rural) communities in Africa, there is little attention paid to ancestors and ancestor-related practices by scholars of African society and culture, in part due to the dominance of Western scholarship in this area of research.

For instance, data from the 2009 Pew data collection project on spiritual life (Pew Research Center, 2009), that contains information on traditional beliefs for around 25.000 respondents from 19 sub-Saharan African countries, confirms this intuition and shows that 45.6% of respondents believed in witchcraft, while 41% believed in reincarnation, and about 30% believed in ancestors.<sup>16</sup> In the case of my sample in Southern Benin, 50% of men think that their ancestors care about the continuation of the lineage (43% among those who do not report traditional religion as their main religion) or, in the same vein, Le Rossignol et al. (2022) find that in the northwestern DRC, 92% of their sample report believing either "strongly" or "very strongly" in Christianity, and at the same time, 75% of them also report believing "strongly" or "very strongly" in witchcraft.

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<sup>15</sup>The success of the Pentecostal church in SSA may be due in part to its ability to adapt to pre-existing belief systems (Auriol et al., 2020). For example, "the descent of the Holy Spirit in the Pentecostal doctrine is described as a trance, and the believers continue to think that witchcraft is powerful but are persuaded that they are now protected by a superior, divine power (Platteau (2009), p.679)".

<sup>16</sup>Beliefs were recorded as yes or no answers to the question Which, if any, of the following do you believe in? and assigned to various categories such as witchcraft, the evil eye, evil spirits, heaven, or reincarnation. Ancestral beliefs were recorded as a yes or no answer to the question Do you believe that sacrifices to spirits or ancestors can protect you from bad things happening?.

## 2.2. *Implications of Ancestor Worship on Fertility in SSA*

Why are beliefs in the ancestors closely related to fertility behavior in SSA? To answer this question, it is important to emphasize that ancestor worship belongs to the realm of kinship and lineage structures (Fortes, 1965; Gong et al., 2021). As Fortes (1965) notes, ancestor worship is a lineage cult in many African societies – that is, a cult of the basic politico-jural unit of many African societies, rather than of the domestic sphere.<sup>17</sup> Similarly, Turaki (2000) mentions that "the ancestors are the most powerful, basic and primary component of the *kinship system* of an African community". In fact, ancestor worship is the extension of the core elements of many African societies – ancestry, kinship and descent relations – to the supernatural sphere (Fortes, 1965; Caldwell and Caldwell, 1985).

The belief in the intervention of ancestral spirits in everyday life and the need for descendants to ensure the survival of the lineage is continuous with the social structure. The lineage is thus understood as "a group of descent stretching back infinitely and with an enormous spiritual investment in reaching indefinitely into the future" (Caldwell and Caldwell, 1987). Only a portion of the entire lineage is alive at any one time, and its extension into the future is the central concern of both living lineage members and ancestors. Ancestors maintain their connection to the lineage after death, and each new birth into the lineage is a way for an ancestor to return through reincarnation (Mbiti, 1975).<sup>18</sup> In fact, some authors argue that an African community consists of the unborn living (those who are about to be reincarnated), the living, and the living dead (those who are deceased but still influence the living) (Turaki, 2000). Qualitative evidence from

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<sup>17</sup>This is easily seen in matrilineal societies, where it is the mother's brother who becomes an ancestor, rather than the father himself.

<sup>18</sup>For example, among the Fon of Benin, the Yoruba of Nigeria or the Beng people of Côte d'Ivoire, new children born into the family are believed to be reincarnated when an old person has recently died, and children are often named to identify the ancestors reborn in their form (Caldwell and Caldwell, 1987; Osanyinbí and Falana, 2016). Ancestors may return in more than one child in a family. For example, (Idowu, 1973) notes for the Yoruba that "it is believed that [ancestors] reincarnate not only in one grandchild or great-grandchild, but also in several contemporary grandchildren and great-grandchildren who are brothers and sisters and cousins, aunts and nephews, uncles and nieces, *ad infinitum*". Similarly, in the context of urban South Africa, Anderson (1993), p.27, reports: "One respondent said that in 1986 she had a dream in which she saw she was pregnant. Someone took her to a big stone (probably a gravestone) on which was written the name "Isaac." The following day she enquired from an older family member, who said that Isaac was a grandfather who had died many years previously. A month later the respondent fell pregnant and a baby boy was born, whom she had to call Isaac. She then prayed and thanked the ancestors for their gift of the child. The child thereby, following traditional custom, received the "ancestor spirit" of the deceased ancestor Isaac."

first-hand data collected in Benin confirms the close link between ancestors and the importance of lineage extension: 95% of those who believe in ancestors ( $n = 1788$ ) also believe that their ancestors are deeply concerned about the continuation of their lineage. This is all the more important because in societies organized hierarchically by age, where ancestors usually have even greater authority than elders.

The emphasis on ancestry and descent and the consolidation of lineage - that is, the succession of generations - as a key element of these societies have important demographic consequences. Molnos (1973), who mentions that continuing the lineage and commemorating ancestral spirits is one of the precise reasons for wanting "as many children as possible" among societies in East Africa, states:

"The paramount objective of having children was that there should always be a living descendant to remember and honor the departed. Children meant the continuation of the lineage and the perpetuation of the family name and spirit. Descendants were needed to perform funeral ceremonies, to ensure that the parents, unlike childless people, be buried, and that the ancestral spirits be commemorated by erecting shrines, pouring out libations and offering food [...]. Among the Chaga of the north-eastern section of the United Republic of Tanzania, for instance, children were seen as a sign of approval of the parents' marriage by their forebears, and *lineages competed by producing numerous offspring for the favor of ancestors on whom their welfare was deemed to depend* (Molnos (1973), p.129)".<sup>19</sup>

These factors point at the importance of the *quantity* rather than the *quality* of children and emphasize the importance attached to the continuation of the family line. In this regard, the qualitative literature on ancestor worship does not highlight the emergence of any specific quantity-quality (QQ) trade-off in these societies (one could think, for example, that parents would invest more in those children who are believed to be reincarnated ancestors), which goes in line with the quantitative literature on the absence of a QQ trade-off in SSA (Alidou and Verpoorten, 2019).

The argument presented above can be interpreted as an extension of the well-known "old-age security motive", whereby people's needs for old-age support raise the demand for children.<sup>20</sup> In vertical systems of transmission where the continuation of the family

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<sup>19</sup>Similarly, Kuper (1961) notes that "the existence of the ancestors depends, in Swazi dogma, on the continuation of the family, and their well-being depends on family prosperity". She cites two examples of men who, "after recovering from serious illnesses, stated that they had actually returned from the dead to take wives and produce children and leave the family name on earth".

<sup>20</sup>See Lambert and Rossi (2016), Rossi and Godard (2022), or Sage (2023) for evidence of the old-age security motive for fertility in the African context.

line is of central importance due to the influence of ancestors, the question of "security" has two different facets (Goody, 1973). In addition to the standard security in old age, there is the security in the after-life which drives the demand for children in the same way as the former, and which can be obtained by making provision for the continuity of the family estate.<sup>21</sup>

Maintaining the lineage remains one's responsibility, whether in the human world or in the ancestral world. To this end, ancestors bless their progeny's fertility, provide wealth, health and children to the family, and undergo reincarnation (Grande, 2024; Olupona, 2014). It is therefore not surprising that the most common use of ancestral powers is related to lineage survival and the well-being of the kinship group. Crucially, high fertility is morally associated with joy, recognition, right living, and shows the approval of ancestors. On the contrary, low fertility or infertility are associated with misfortune, sin, mistreatment, marginalization, and shows the disapproval of ancestors.<sup>22</sup> Rituals or other reparations are usually performed to reverse these situations (Caldwell and Caldwell, 1987).<sup>23</sup>

### 3. Data

I study the impact of traditional religious and cultural beliefs in the influence of ancestors on fertility. To study this question I rely on several datasets from different contexts and time periods, some of which have never been used in economics.

#### 3.1. *Contemporary Benin*

I start by examining first-hand data that I collected in southern Benin. The sample consists of 943 households living in rural areas. The respondents were randomly selected,

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<sup>21</sup>In this regard, Olupona (2014) notes that a proverb about the Yoruba says "If the land of the ancestors is full of gold and diamonds, they will not return to the human community to solicit gifts", and, importantly, he recognizes that it emphasizes that the ancestors need their descendants to sustain themselves in the afterlife as much as the living need them in old age.

<sup>22</sup>There is no consensus about ancestral punishment. As Mekoa (2019) notes: "Although childlessness is sometimes blamed on angry ancestors, it is usually called the work of the witches because ancestors are generally interested in the growth of their own clan".

<sup>23</sup>Fear to infertility has been shown to be an important barrier to modern contraceptive methods uptake in SSA (Ochako et al., 2015), and therefore informing about the real risks and consequences of contraceptives rather than focusing on supply side frictions may be important to address the low contraceptive take-up. However, beliefs are difficult to change, and more research is needed in this area (Dupas et al., 2024).

subject to inclusion criteria (e.g., having more than 0.5 hectares of land), as part of a large project to support women in pineapple production.<sup>24</sup> The main survey was conducted in 2024 in a standard face-to-face setting. Respondents were also randomly assigned to enumerators to minimize enumerator bias. Along with a comprehensive set of socioeconomic characteristics and fertility outcomes, I directly asked about traditional beliefs in ancestors. The main variable I use to measure ancestral beliefs use the answer to the question "*do you believe that the spirits of your ancestors have an influence on the events of your life?*"<sup>25</sup><sup>26</sup>

### 3.2. *Zaire and contemporary Democratic Republic of Congo*

*West Zaire Surveys (1975-77) and Jan Vansina's Congo's Ethnography.*— I use original information from two demographic surveys conducted in the western part of the DRC between 1975 and 1977 (see Appendix E for details of these surveys).<sup>27</sup> The first contains individual-level information on 250,000 individuals in 43,000 households living in seven major cities in the DRC (in tables I refer to this survey as the urban sample). The second includes individual-level information on almost 50,000 individuals in 11,000 households distributed across four regions, and covers both rural and urban areas, excluding the large cities included in the urban survey (in tables I refer to this survey as EDOZA). Moreover, each woman over the age of 13 reported her birth calendar, including the exact dates of birth, the sex of the children, and the dates of death where applicable. Moreover, respondents' ethnicity<sup>28</sup> was also recorded and coded according to the classi-

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<sup>24</sup>Both husband and wife were supposed to be interviewed in each household. However, because the project focused on women, some households only included the woman (71/943) because she is not married or because the husband was not present at the time of the survey (29/872).

<sup>25</sup>An example was provided after each question to clarify its meaning. Regarding the first question: "For example, do you think that if you are lucky in your income-generating activities, it is because your ancestors are behind you? Regarding the second question: "For example, do your ancestors influence the health of your children or protect them from bad events?".

<sup>26</sup>This variable is rather a measure of the *strength* of individuals' traditional beliefs in ancestors. In fact, qualitative interviews revealed that it is possible to answer negatively to this questions while still believe in the *existence* (rather than influence) of ancestors. The estimates are likely to represent a lower bound of the true effect.

<sup>27</sup>Although these surveys have been used by demographers to create statistics at an aggregate level (see, for example, Tabutin (1982) or Shapiro (1996)), the use of the microdata is rather unique (they have only been used in Alvarez-Aragon et al. (2023) and Guirkingner and Villar (2022)).

<sup>28</sup>Rather than ethnic groups, ethnicity here should be understood as "tribe". In Vansina's book, several tribes may form an ethnic group, and ethnicities are grouped together in cultural regions. However, for the

fication of Vansina (1966), which allows me to match these surveys to the ethnicity-level information on beliefs in ancestors present in Vansina (1966)'s book.<sup>29</sup>

Vansina (1966) is the main ethnographic source that I use to measure the prevalence of beliefs in ancestors at the ethnicity-level in the DRC. From this book I have digitized original ethnic-level information on the practice of ancestor worship. I constructed a dummy variable that equals one if it is explicitly mentioned that an ethnic group practices ancestor worship. For example, this variable will take the value of one for the ethnic groups of the Balese-Komo region, as Vansina (1966) says:

"The Creator was often equated with the first ancestor. But a cult of the Creator is only mentioned among the Balese [...]. Ancestor worship was present everywhere. Chickens were sacrificed and offerings made [...]. A few protective charms, which ensured human fertility, were common (Vansina (1966), p.100)".

This ethnographic source has two clear advantages. First, it contains explicit information on the main variable of interest so no proxy is needed. Second, it provides a much more detailed description of the ethnographic landscape of the DRC than traditional datasets such as the Ethnographic Atlas. For example, while the EA records 60 ethnic groups in the territory of the present-day DRC, and about 400 for the entire SSA, this number increases to about 250 tribes in Vansina (1966). Of these, 66% practice ancestor worship. The matching procedure used here has also the advantage of taking migration into account, compared to matching techniques using purely geographical characteristics, and will allow me to exploit variations in ethnicity-level beliefs in ancestors among migrants of the same origin and destination, thus isolating beliefs from local economic conditions and institutions. Overall, I am able to match about 60,000 women over the age of 13 with the information provided by Vansina (1966).<sup>30</sup>

*DRC Demographic and Health Surveys (DHS).*— Finally, I complement the analysis with

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purpose of this paper, where I use the characteristics of these "tribes", I use the most granular information possible, and will use ethnic group instead of tribe for the sake of expôosition clarity (as I will use as well information from the Ethnographic Atlas (Murdock, 1967) and Berezkin's Folklore and Mythology Catalog (Berezkin, 2015), which use ethnic groups as their unit of analysis).

<sup>29</sup>Cogneau and Dupraz (2015) wonder to what extent the classifications of ethnic groups made by anthropologists have influenced the classifications used by the surveys. In the case of the demographic surveys used here, their codebook explicitly mentions that they adapted the questionnaires to record ethnicity as classified by Vansina (1966).

<sup>30</sup>This figure represents about 70% of the total number of women over the age of 13 included in the demographic surveys. Of the non-compliant subsample, 30% are individuals born in a country other than the DRC.

the two available rounds of the Demographic and Health Surveys available for the Democratic Republic of the Congo (2007-2013), a nationally representative survey that provides detailed information on education, literacy, occupation, religion, fertility preferences, and contraceptive methods. I use data from both the men's and women's questionnaires. In total, there is information on about 40,000 individuals living in 785 clusters. I follow the methodology of previous studies (i.e., Lowes, 2022) and use location to match the DHS data with the ethnographic information.<sup>31</sup> To do this, I combine the location of each DHS cluster with a digitized version of the map of ethnicities from Vansina (1966). The matching procedure is shown in Appendix B.

### 3.3. *Sub-Saharan Africa: Pew Research Center's Forum on Religion and Public Life*

In an attempt to examine the external validity of the previous analysis performed in two specific contexts, I use individual-level information on contemporary supernatural beliefs from 25,000 respondents in 19 SSA countries. This information comes from a survey conducted between December 2008 and April 2009 by the Pew Research Center's Forum on Religion and Public Life (Pew Research Center, 2009). It includes individual-level information on religious beliefs, practices, knowledge, and attitudes toward other faiths for about 25,000 respondents in 19 sub-Saharan African countries. I measure the strength of ancestral beliefs by using the responses to the following question: "*Do you ever participate in traditional African ceremonies or perform special acts to honor or celebrate your ancestors?*".<sup>32</sup> Finally, this survey also contains information on socio-demographic characteristics such as age, gender, education, marital status, religion, number of children, region of residence within the country, or self-reported income status.<sup>33</sup> Appendix

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<sup>31</sup>Using self-reported ethnicity to match the DHS data with Vansina's information on ancestor worship is less satisfactory because the information on ethnicity in the DHS for the DRC is very limited, and the match of the DRC ethnicities with the Ethnographic Atlas is quite difficult. In fact, only 7 different ethnicities (or groups of ethnicities) are recorded in the DHS (compared to about 130 after matching with Vansina).

<sup>32</sup>I show that my results do not change when using different measures of ancestral beliefs. For example, I construct a dummy variable that equals one if the respondent answers "yes" to the question "*Do you believe that sacrificing to spirits or ancestors can protect you from bad things happening?*". Another alternative is to construct a dummy variable equal to one if the respondent answers "yes" to the following question: "*Which, if any, of the following do you believe in? Reincarnation?*".

<sup>33</sup>As with the first-hand data from Benin, this survey measures ancestral beliefs at the individual level, and therefore does not consider ethnicity as the only vector of cultural transmission, since the analysis here does not rely on ancestral ethnic group's characteristics.

C.1 shows the spatial distribution of the main explanatory variables.

### 3.4. *Berezkin's Folklore and Mythology Catalogue and Ethnographic Atlas*

Beliefs in ancestral influence may have been widespread in preindustrial societies. To show that this relationship is not driven by other factors affecting the post-independence fertility trajectories of societies, I explore whether the relationship between beliefs in ancestors and fertility exists in the oral traditions of pre-industrial societies across the globe, when ancestral beliefs were likely prevalent worldwide. The most widely used data set in economics for the study of pre-industrial societies is the Ethnographic Atlas (EA), which collects detailed ethnographic information on 1,265 ethnic groups spread across the globe (Murdock, 1967). However, the EA does not include information on fertility or ancestor worship. Therefore, I supplement the EA with data on more than 2,500 folkloric motifs in 958 oral traditions from Berezkin's Folklore and Mythology Catalogue (Berezkin, 2015). Michalopoulos and Xue (2021) digitized the information in Berezkin's catalog and created a comprehensive dataset that includes the proportion of folkloric motifs in the oral tradition of ethnic groups related to different concepts.<sup>34</sup> These motifs are considered to be the building blocks of oral traditions, representing their characteristics, important experiences, events and images (Galor et al., 2023). This information can be particularly valuable because culture is transmitted from generation to generation through oral traditions such as myths, folklore, stories, or proverbs.<sup>35</sup> I measure ancestral beliefs in pre-industrial societies by looking at the proportion of folkloric motifs in an ethnic group's oral tradition related to the word *ancestor*. Similarly, I proxy the importance of fertility in pre-industrial societies by looking at the proportion of motifs related to the word *birth* in an ethnic group's oral tradition.<sup>36</sup>

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<sup>34</sup>They also validate the content of the catalog and link the groups in Berezkin's collection to the EA.

<sup>35</sup>Even in recent times, the presence of ancestors in the oral and written traditions of African societies is widespread. For example, Appendix A shows a poem written by the Senegalese writer Birago Diop (1906-1989) in 1960.

<sup>36</sup>I also use alternative measures. For example, I show that my results remain unchanged if I use the word *ancestral*, the combination of the words *ancestor* and *worship*, or the word *fertility* instead of *birth*.



## 4. The influence of ancestral beliefs on fertility

### 4.1. Evidence from Contemporary Benin

I start by examining the relationship between beliefs in ancestors and fertility using first-hand data collected among a sample of (mostly married) rural households ( $n = 943$ ) in southern Benin. This is a valuable context that allows me to disentangle alternative mechanisms: almost everyone's main activity is agriculture (98% of men and 90% of women have agricultural fields), fertility is still very high (the average number of children ever born is close to 8 for men and above 5 for women, and 42% of married households are polygamous), and the clan-based structure of the society is pervasive (45% of men and 41% of women have received pressure from their extended family or clan to increase their number of children). Moreover, I can accurately measure ancestral beliefs, which are widespread: 43% (31%) of men (women) believe that ancestors have an influence on their lives (out of which 96% think that their ancestors are deeply concerned about the survival of their extended family and the continuation of their lineage).

The baseline specification, estimated through ordinary least squares, takes the following form:

$$Y_i = \alpha + \beta_1 \text{AncestralBeliefs}_i + X_i' \Phi + \epsilon_i \quad (1)$$

Where  $Y_i$  is the total number of children ever born and  $\text{AncestralBeliefs}_i$  is a dummy variable that equals one if the respondent answers *yes* to the question *Do you believe that the spirits of your ancestors influence the events of your life?*. The vector  $X_i'$  includes an extensive set of control variables. As baseline controls, I always include age, age squared, education, and whether the household has access to electricity. Second, I add additional control variables that help me to disentangle alternative mechanisms: total number of agricultural fields, whether the household has a TV, and a dummy variable that takes the value 1 if respondent have experienced pressure from their extended family or clan to increase their number of children. Finally, I always include religion fixed effects (five main categories: Catholic, Celestial Church of Christ, Evangelical, Vodoun, and Other).

Qualitative evidence suggests that men are the main decision-makers regarding the number of children in this context. Therefore, I focus on two samples. First, I estimate equation 1 for men. Second, I focus on women and examine the effect of both husband

and wife's beliefs. <sup>37</sup>

Table 1 presents the estimates for men while sequentially including different vectors of control variables. Columns (1)-(3) do not include religion fixed effects, while column (4)-(6) do. Columns (1) and (4) do not include control variables, while columns (2) and (5) include baseline controls (age, education, and access to electricity), and columns (3) and (6) include the full list of controls. Robust standard errors are reported in parenthesis. Across all specifications, I find a positive, stable and statistically significant relationship between beliefs in ancestors and fertility: holding strong traditional beliefs in the influence of ancestors is associated with more than one additional child. These effects are quantitatively large. The coefficient is similar to the effect of having attended school, or to the difference in the total fertility rate of SSA and other low-income regions.

Table 1: Beliefs in ancestors and male fertility in contemporary Benin

|                   | Number of children ever born |                     |                     |                     |                     |                     |
|-------------------|------------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                   | (1)                          | (2)                 | (3)                 | (4)                 | (5)                 | (6)                 |
| Ancestral beliefs | 1.848***<br>(0.379)          | 1.667***<br>(0.345) | 1.188***<br>(0.298) | 1.460***<br>(0.388) | 1.267***<br>(0.350) | 1.152***<br>(0.355) |
| Baseline controls | No                           | Yes                 | Yes                 | No                  | Yes                 | Yes                 |
| Extended controls | No                           | No                  | Yes                 | No                  | No                  | Yes                 |
| Religion FE       | No                           | No                  | No                  | Yes                 | Yes                 | Yes                 |
| Mean Y            | 7.995                        | 8.016               | 8.016               | 8.012               | 8.029               | 8.029               |
| N                 | 846                          | 832                 | 832                 | 841                 | 830                 | 830                 |

NOTE. Data: First-hand data collected in southern Benin. The table reports OLS estimates. The outcome variable is the total number of children ever born. "Ancestral beliefs" is a dummy variable that equals one if the respondent answers *yes* to the question *Do you believe that the spirits of your ancestors influence the events of your life?*. Baseline controls include age, age squared, whether the household has electricity and education. Extended controls include the number of agricultural fields, whether the household owns a TV, whether the household has been part of a program to encourage the production of pineapple, and whether the respondent has received pressure from the extended family to have children. Religion fixed effects includes five categories: Vodoun, Roman Catholic, Evangelic, Celeste and other. Robust standard errors in parenthesis. \*\*\* for  $p < 0.01$ , \*\* for  $p < 0.05$ , \* for  $p < 0.1$ .

Table 2 presents the estimates for women. Now, the explanatory variable of interest is a dummy that equals one if both husband and wife hold strong beliefs in ancestors. The same set of control variables as in Table 1 are included. Overall, these results strongly suggest that traditional beliefs in ancestors are still an important factor determining fertility outcomes.

<sup>37</sup>Appendix F.4 replicates these results using information from women only, as well as looking at men but using the beliefs of both spouses as explanatory variable. Interestingly, the size of the coefficients is even higher when both spouses hold strong beliefs in the influence of ancestors, rather than the husband alone. These effects suggest some degree of complementarity between spouses and room for non-negligible importance of women's decision making.

Table 2: Beliefs in ancestors and female fertility in Benin

|                     | Number of children ever born |                     |                     |                     |                     |                     |
|---------------------|------------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                     | (1)                          | (2)                 | (3)                 | (4)                 | (5)                 | (6)                 |
| Men + Women beliefs | 0.750***<br>(0.178)          | 0.641***<br>(0.161) | 0.614***<br>(0.160) | 0.825***<br>(0.198) | 0.696***<br>(0.180) | 0.704***<br>(0.178) |
| Baseline controls   | No                           | Yes                 | Yes                 | No                  | Yes                 | Yes                 |
| Extended controls   | No                           | No                  | Yes                 | No                  | No                  | Yes                 |
| Religion FE         | No                           | No                  | No                  | Yes                 | Yes                 | Yes                 |
| Mean Y              | 5.365                        | 5.352               | 5.352               | 5.365               | 5.352               | 5.352               |
| N                   | 846                          | 840                 | 840                 | 846                 | 840                 | 840                 |

NOTE. Data: First-hand data collected in southern Benin. The sample is restricted to women. The table reports OLS estimates. The outcome variable is the total number of children ever born. "Men + Women beliefs" is a dummy variable that equals one if both the husband and the wife answer *yes* to the question *Do you believe that the spirits of your ancestors influence the events of your life?*. Baseline controls include age, age squared, whether the household has electricity and education. Extended controls include the number of agricultural fields, whether the household owns a TV, whether the household has been part of a program to encourage the production of pineapple, and whether the respondent has received pressure from the extended family to have children. Religion fixed effects includes five categories: Vodoun, Roman Catholic, Evangelic, Celeste and other. Robust standard errors in parenthesis.

\*\*\* for  $p < 0.01$ , \*\* for  $p < 0.05$ , \* for  $p < 0.1$ .

**Robustness.-** Appendix F provides several robustness tests of the relationship between beliefs in ancestors and fertility. First, I show that the results do not change when I estimate an exponential regression model using a pseudo-Poisson maximum likelihood estimator, or when I use alternative definitions of the explanatory and dependent variables.<sup>38</sup> Second, I show in specification curves that these results are robust to the inclusion of additional control variables, such as polygamy, overall religiosity (measured as an indicator if the respondent donates money to the church when COVID happened), household size, or having livestock. I also show that the results are not driven by similar dynamics in fertility and ancestral beliefs over the life cycle or over time (same results when including age fixed effects). Third, I show that the results are robust to the use of the total number of *alive* children as dependent variable, to reduce concerns about differential mortality. Finally, I rule out that the effects are driven by differences in the inheritance rules of ethnic groups (Fontenay et al., 2024), or by differences in old-age security that could be correlated with ancestral beliefs (Rossi and Rouanet, 2015; Rossi and Godard, 2022; Sage, 2023).<sup>39</sup>

<sup>38</sup>Alternative measures of beliefs in ancestors are: dummy variable that equals one if the respondent answers *yes* to the question *Do you think that your ancestors' spirits care about your extended family and the continuation of the family line?*, or when I combine both measures and generate a third variable that equals one when at least one of the two variables takes the value one.

<sup>39</sup>An additional concern in this setting could be reverse causality. In fact, having more kids could

#### 4.2. Epidemiological approach: Congo

I then examine the same relationship using unique individual-level data from women born during the colonial period in the Democratic Republic of the Congo (DRC). Examining these data is interesting for several reasons. First, because the digitization of Vansina (1966) allows me to directly identify the existence of ancestor worship at a granular ethnic group level. Second, because the DRC is one of the most prominent examples of delay in the fertility transition in Tropical Africa. In fact, fertility in the DRC increased during the second half of the 20<sup>th</sup> century (Romaniuk, 2011). Third, the nature of the 1970s Demographic Urban sample, composed mainly of post-independence migrants from rural areas to seven major cities of the DRC, allows me to replicate the epidemiological approach introduced by Fernández and Fogli (2009). I exploit the heterogeneity in ancestral beliefs while holding constant the environment in which individuals grew up by comparing migrants who were born in the same territory and who live in the same city at the time of the survey, but whose beliefs in ancestors differ. This strategy is useful to rule out confounding factors related to the environment of individuals (either economic conditions or institutions).

The equation I estimate now takes the following form (with small variations depending on the specification). Let  $i$  denote individuals,  $e$  denote ethnicity,  $c$  denote city of residence, and  $r$  denote territory (or city) of birth.<sup>40</sup>

$$Y_{iect} = \beta_0 + \beta_1 AncestralBeliefs_e + X_i' \Phi + \alpha_{cr} + \epsilon_{iect} \quad (2)$$

Where  $Y_{iect}$  is a fertility outcome,  $AncestralBeliefs_e$  is a dummy variable that equals one if the respondent belongs to an ethnic group that traditionally practiced ancestor worship. The vector  $X_i'$  includes an extensive set of individual-level control variables, such as age, age squared, whether the place of birth was rural or urban, education, whether the respondent is a migrant, whether respondent's father is alive at the time

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reinforce people's thinking about lineage and heritage, and make people feel blessed and thankful to ancestors. However, I think that this is unlikely to explain my results for several reasons. First, beliefs in ancestors are deeply-rooted cultural beliefs, whose origin is uncertain, and unlikely to emerge at the individual level. Second, the results with the historical data from the DRC alleviate this concern (see Section 4.2), since ancestral beliefs are measured at the ethnic group level, using information on the practice of ancestor worship before colonization from Vansina (1966).

<sup>40</sup>The territory is the second administrative unit in the Democratic Republic of the Congo, smaller than the province. The 25 provinces of DR Congo are divided into 145 territories and 32 cities.

of the survey, employment status, whether the respondent works in agriculture (only available for the urban sample), and year of installation in the current city. Finally,  $\alpha_{cr}$  denotes city of residence  $\times$  territory of birth fixed effects.  $\epsilon$  is an error term, and standard errors are clustered at the ethnic group level, since the treatment is constructed at that level. The presence of city of residency  $\times$  territory of birth fixed effects means that, when estimating equation 2, the effect of ancestor worship is identified by comparing respondents from different ethnicities with different reliance on ancestor worship, living in the same city at the time of the survey and born in the same territory.<sup>41</sup>

The key underlying assumption when including origin-destination fixed effects is that culture is easier to carry across space than local institutions and economic conditions.<sup>42</sup> The empirical strategy employed here differs in several aspects from the seminar paper of Fernández and Fogli (2009). First, it is more demanding since it uses within-country variation in cultural beliefs. Moreover, I compare individuals born close to each other (but from different ethnicity and with different beliefs in ancestors) instead of in different countries, holding constant both economic conditions and institutions also at the place of origin. Third, I carefully control for the year of installation in the city of residence at the time of the survey, which allows me to eliminate concerns related to a potential correlation between time since migration and the prevalence of ancestral beliefs, which could confound identification and is often ignored in the literature (Bertoli et al., 2024). Finally, I contribute to the existing discussion by bringing a new explanation for the origin of different fertility preferences across individuals and social groups.

Table 3 shows the results. I find a positive relationship between ancestor worship and the total number of children ever born in the different samples. For each survey, I report the results on the total number of women over the age of 13 (columns 1, 3, 5 and 7) and restricting on those over the age of 30 (columns 2, 4, 6 and 8), since most of childless women are really young (75% of women younger than 20 are childless). First,

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<sup>41</sup>The variation in the number of ethnicities by territory of birth is surprisingly high: on average, people from about 27 different ethnic groups (tribes) are born in a territory (out of 220 in my sample). This ranges from only one ethnicity in 17% of territories, to 119 ethnicities in Kinshasa. See appendix G.1 for details.

<sup>42</sup>This strategy builds on the work of Fernández and Fogli (2009), who study how culture influences fertility by examining the behaviour of second-generation American women. They use past values of total fertility rates in their country of ancestry to proxy for cultural beliefs, and show that they still have a significant explanatory power for their number of children. Then, by comparing second-generation American women, they hold market and institutions constant while they may differ culturally as their parents' country of origin is different.

belonging to an ethnic group with ancestor worship is associated with an increase in the number of children of about 20% in the full sample. The difference between urban and rural areas is not very large: the effect of ancestor worship on fertility in the EDOZA (rural) sample of women over 30 represents 22% of the mean, while it represents 16% in the urban sample (note that, when using the urban sample, I use city of residence FE instead of region of residence FE).<sup>43</sup> The urban sample allows me to include city of residence  $\times$  territory of birth fixed effects (columns 7 and 8). The results remain robust and highly significant, although the magnitude of the coefficient is reduced, from about 19% of the urban sample mean when using city of residence fixed effects to about 8% when including the interaction with territory of birth. Overall, these results suggest that there is still a positive relationship between ancestor worship and fertility in the context of urban post-independence Congo.

Table 3: Ancestor worship and fertility in the DRC

|                            | Number of children ever born |                     |                    |                     |                      |                     |                     |                     |
|----------------------------|------------------------------|---------------------|--------------------|---------------------|----------------------|---------------------|---------------------|---------------------|
|                            | Full sample                  |                     | EDOZA              |                     | Urban sample         |                     |                     |                     |
|                            | All                          | +30                 | All                | +30                 | All                  | +30                 | All                 | +30                 |
| Ancestral beliefs          | 0.433***<br>(0.0967)         | 0.861***<br>(0.205) | 0.564**<br>(0.246) | 1.052***<br>(0.387) | 0.363***<br>(0.0559) | 0.833***<br>(0.173) | 0.125**<br>(0.0480) | 0.349***<br>(0.128) |
| Controls                   | Yes                          | Yes                 | Yes                | Yes                 | Yes                  | Yes                 | Yes                 | Yes                 |
| FE                         | Yes                          | Yes                 | Yes                | Yes                 | Yes                  | Yes                 | Yes                 | Yes                 |
| City $\times$ Territory FE | No                           | No                  | No                 | No                  | No                   | No                  | Yes                 | Yes                 |
| Mean Y                     | 2.883                        | 5.322               | 3.546              | 4.866               | 2.703                | 5.542               | 2.700               | 5.537               |
| R-squared                  | 0.528                        | 0.246               | 0.407              | 0.267               | 0.576                | 0.252               | 0.587               | 0.281               |
| N                          | 60064                        | 23194               | 12825              | 7551                | 47239                | 15643               | 47055               | 15491               |

NOTE. Data: Demographic Survey of 1970s and EDOZA. Columns 1 and 2 combine both the Urban Demographic survey and EDOZA. Columns 3 and 4 use only the EDOZA sample. Columns 5-8 use only the urban sample. In columns 2, 4, 6 and 8, the sample is restricted to women older than 30 years old. The table reports OLS estimates. The outcome variable is the total number of children ever born. "Ancestor worship" is a dummy variable that equals one if the ethnic group  $e$  practice ancestor worship. Controls include age, age squared, whether the place of birth was urban or rural, whether the respondent has primary education, dummy equal to one if migrant, dummy equal to one if the father's respondent is alive, dummy equal to one if the respondent is working, dummy equal to one if the respondent is a farmer (only available in the urban sample), number of household members, and year of installation in the current city. City/region fixed effects include city of residence in the case of the urban sample and region of residence in the case of EDOZA. Standard errors clustered at the Vansina's ethnic group-level in parenthesis. \*\*\* for  $p < 0.01$ , \*\* for  $p < 0.05$ , \* for  $p < 0.1$ .

**Robustness.-** Although these results are robust both to the inclusion of origin and destination fixed effects and to the inclusion of an extensive set of control variables, the main

<sup>43</sup>Interestingly, the fertility rates of women over the age of 30 are higher in the urban sample than in EDOZA. This is consistent with earlier works examining the fertility rates in big cities like Kinshasa (Shapiro, 1996).

remaining concern is that of omitted variables. In this setting, as noted by Fernández and Fogli (2009), unobserved differences in human capital (once woman's education is controlled for) may bias the results if the unobserved component that depends on parental human capital varies with ethnicity in a way that is correlated with ancestor worship. For example, societies in which ancestor worship is important may have been less receptive to missionaries, which could have lowered the human capital of the ethnic network, broadly defined, relative to neighboring groups, affecting the probability of marriage or obtaining a job. To address this concern, I examine whether ancestor worship is related to wages. Indeed, if unobserved human capital accounts for part of the results, it should be reflected in wages. To do this, I resort to an expenditure survey which includes 1/50 of the total number of households identified in the same seven cities of the 1970s Urban Demographic Survey.<sup>44</sup> Table G4 in Appendix G reports the results from the regression of wages on ancestor worship and a vector of control variables, including education. Ancestor worship remains insignificant in every specification, suggesting that unobserved human capital is not responsible for the results.

An important limitation of the 1970s datasets is that they do not contain any information on religious beliefs. More importantly, the rapid expansion of Christianity in the context of the DRC took place hand in hand with missionaries' activities, which may have weakened family and ethnic lineages, and emphasized universal over family/ethnicity-centered moral values (Platteau, 2009; Reybrouck, 2014; Bergeron, 2023).<sup>45</sup> At the same time, exposure to missionary presence in colonial Congo had important effects on education (Alvarez-Aragon et al., 2023) and fertility outcomes (Guirkingner and Villar, 2022). In Appendix G, I construct a measure of exposure to missionary presence at birth for all individuals born before 1948 in the DRC using both the EDOZA sample and the urban demographic survey, and show that the positive influence of ancestor worship on fertility is not affected when I control for exposure to Christian missions.

Finally, I show that my results hold when I use age-specific fertility rates instead of total number of children ever born (Table ??G1 G1d when looking at both the extensive

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<sup>44</sup>See Appendix E for details about this survey.

<sup>45</sup>The relationship between missionaries and the practice of ancestor worship is controversial. While some authors suggest that Christian and Islamic doctrines prohibited ancestor worship, other authors suggest that ancestor worship is often allowed to continue because it is considered a cultural rather than a religious practice (Caldwell and Caldwell, 1990; Olupona, 2014). In fact, this is what I observe in Benin: there is no correlation between being Catholic and beliefs in the influence of ancestors.

and the intensive margin, and are robust to the use of a pseudo-poisson maximum-likelihood estimator (see subsection G.3). This is important, as the dependent variable contains a large proportion of zeros (37% of women over the age of 13 in my sample). This is a well know fact in certain areas of Central Africa, known as the "infertility belt", specially before the 1990s (Bongaarts et al., 1984; Larsen, 2003), where infertility rates among married women could reach 25%.

#### 4.3. *Evidence from contemporary sub-Saharan Africa*

Next, I zoom out and study the wider context of sub-Saharan Africa. Using data from the PEW research center, which collected detailed information of different types of contemporary supernatural beliefs, I estimate the follwing equation:

$$Y_{ic} = \alpha + \beta AncestralBeliefs_i + X_i' \Phi + \phi_c + \epsilon_{ic} \quad (3)$$

Let  $i$  denote individuals and  $c$  denote countries. Then  $Y_{ic}$  is the fertility outcome of individual  $i$  living in country  $c$ , defined either as the total number of children ever born or as a dummy variable equal to one if the respondent has more than 4 children.  $AncestralBeliefs_i$  is the main explanatory variable and is constructed as a binary variable equal to one if the respondent answers "Yes" to the question "*Do you ever participate in traditional African ceremonies or perform special acts to honor or celebrate your ancestors?*". The vector  $X_i'$  contains a set of individual-level covariates that vary according to the specification, and may include age, age squared, gender, whether the respondent lives in a urban or rural area, whether the respondent is Christian/Muslim or from another religion, a dummy variable that equals one if the respondent has completed primary education, a dummy variable that equals one if the respondent is married, and a dummy variable that equals one if the respondent finds himself/herself in a good economic situation. Finally, the analysis exploits within-country variation by including  $\phi_c$ , which represents country fixed effects, to account for any country-level time-invariant characteristics that may be correlated with both fertility levels and traditional beliefs. Robust standard errors are included in parenthesis.

The results are shown in Table 4. I find a positive and robust relationship between ancestral beliefs and fertility outcomes. In terms of magnitude, the coefficient ranges from 0.5 (25% of the outcome mean) more children in column (1) to about 0.2 more



children (8% of the outcome mean) in column (4), once we include the full set of control variables. This effect is similar to the coefficient associated with living in a rural area (0.2) and almost the same as the influence of having less than primary education (0.3). Interestingly, ancestral beliefs are also associated with an increase in the probability of having more than four children of about one percentage point (10% of the outcome mean).

Table 4: Ancestor worship and fertility in sub-Saharan Africa

|                     | Number of children   |                      |                      |                      | P( $\geq 5$ children)  |                        |                        |                       |
|---------------------|----------------------|----------------------|----------------------|----------------------|------------------------|------------------------|------------------------|-----------------------|
|                     | (1)                  | (2)                  | (3)                  | (4)                  | (5)                    | (6)                    | (7)                    | (8)                   |
| Ancestral beliefs   | 0.507***<br>(0.0378) | 0.248***<br>(0.0302) | 0.233***<br>(0.0301) | 0.172***<br>(0.0292) | 0.0364***<br>(0.00493) | 0.0157***<br>(0.00456) | 0.0139***<br>(0.00452) | 0.0101**<br>(0.00459) |
| Country FE          | Yes                  | Yes                  | Yes                  | Yes                  | Yes                    | Yes                    | Yes                    | Yes                   |
| Controls            | No                   | Yes                  | Yes                  | Yes                  | No                     | Yes                    | Yes                    | Yes                   |
| Extended Controls 1 | No                   | No                   | Yes                  | Yes                  | No                     | No                     | Yes                    | Yes                   |
| Extended Controls 2 | No                   | No                   | No                   | Yes                  | No                     | No                     | No                     | Yes                   |
| Mean Y              | 2.092                | 2.086                | 2.086                | 2.096                | 0.0978                 | 0.0975                 | 0.0975                 | 0.0975                |
| R-squared           | 0.0635               | 0.417                | 0.424                | 0.480                | 0.0418                 | 0.174                  | 0.179                  | 0.187                 |
| N                   | 22926                | 22791                | 22791                | 21997                | 22926                  | 22791                  | 22791                  | 21997                 |

NOTE. Data: PEW research forum 2008-2009 Survey. The outcome variables are defined as follows: in columns (1)-(4), it is the total number of children, and in columns (5)-(8) it is the probability of having at least 5 children. The explanatory variable is an indicator equal to one if the respondent answers "yes" to the question "Do you ever participate in traditional African ceremonies or perform special acts to honor or celebrate your ancestors?". Basic controls include age, age<sup>2</sup> and sex. Extended controls 1 also include whether the individual lives in a urban or rural area and whether he/she is christian/muslim or from other religion. Finally, extended controls 2 also include a dummy variable that equals one if the respondent has completed primary education, a dummy that equals one if the respondent is married, a dummy variable that equals one if the respondent finds himself/herself in a good economic situation, and a dummy equal to one if the respondent did not have money at some point during the last year to buy food for his/her family. Robust standard errors in parenthesis. \*\*\* for  $p < 0.01$ , \*\* for  $p < 0.05$ , \* for  $p < 0.1$ .

**Robustness.-** In Appendix H, I show that these results are robust to alternative definitions of the explanatory variable, to the inclusion of region (instead of country) fixed effects, or to alternative clustering of the standard errors.<sup>46</sup>

However, the main concern here is that the measures of beliefs in ancestors may be capturing something related to traditional religious beliefs that may not be necessarily related to ancestors. I show that my results are not driven by potential confounders such as overall religiosity or the importance of alternative traditional magico-religious beliefs. Indeed, religiosity has been shown to be positively related to fertility (Hayford

<sup>46</sup>Since the sample design in Pew Research Center (2009)'s survey was a stratified random sample at the regional level (province, region or district depending on the country), an alternative is to cluster the standard errors at the regional level. Table H5 in Appendix H shows that this choice does not matter for my results, even if clustered standard errors at this level may be inflated, since the number of clusters in the sample equals the number of clusters in the population (Abadie et al., 2022).

and Morgan, 2008; Herzer, 2019). Therefore, even though religion is controlled for in all specifications, the results would still be biased if the measure of ancestral beliefs is correlated with residual religiosity (not captured by religious affiliation). Table H6 in Appendix H shows that the coefficient on ancestor worship remains unchanged after controlling for the frequency of religious service attendance (Herzer, 2019) and the importance of religion in respondents' lives (Hayford and Morgan, 2008), which are the two most common ways to measure religiosity in the literature. A second potential confounder relates to the importance of alternative traditional magical-religious beliefs, which have been shown to have an important influence on economic life (Platteau, 2014; Le Rossignol et al., 2022). Table H7 in Appendix H shows that alternative measures of traditional supernatural beliefs, such as the use of traditional religious healers, belief in miracles, participation in initiation rituals, beliefs in witchcraft, or knowledge about ancestral, tribal, animist or other traditional African religions, do not affect the results, suggesting that my proxies effectively capture something related specifically to the belief in ancestors.

#### 4.4. *Traditional ancestral beliefs: scope and relevance*

Ancestral beliefs have possibly been widespread before the spread of modern religions and alternative belief systems.<sup>47</sup> In this section I draw on information from the oral tradition of ethnic groups (Berezkin, 2015; Michalopoulos and Xue, 2021), which likely represents a period well before the European colonization of Africa, to examine the extent and importance of ancestor worship at both continental and global scales.<sup>48</sup>

To understand the scope and importance of ancestral beliefs at a large scale, I explore

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<sup>47</sup>Figure C2 in Appendix C.2 plots the distribution of the share of motifs in an ethnic group's oral tradition related to ancestors. As it is clearly visible, the presence of ancestors in the folklore of ethnic groups is not specific of sub-Saharan Africa. We find extensive presence of ancestors in the oral tradition of societies in America and South Asia and the Pacific.

<sup>48</sup>The term "precolonial" used in many empirical papers that rely on data from the Ethnographic Atlas is sometimes confusing. In fact, the EA is based on ethnographic data collected mostly in the early 20<sup>th</sup> century, although anthropologists have tried to represent the situation as it was before European influence (i.e., for SSA, only 5% of ethnic groups have information collected before 1890, while 73% of ethnicities have information collected between 1900 and 1930, and 20% of ethnicities have information collected between 1930 and 1960). Thus, the EA approximates 19<sup>th</sup> century institutions, and the results of analyses using ethnographic information at any other point in time would likely be different. It is therefore important not to consider these variables as time-invariant, "deep-rooted" factors, thus neglecting the massive process of cultural change that has always occurred in Africa across time and space. In contrast, Michalopoulos and Xue (2021) argue that the data from Berezkin's catalog likely represent a period long before that recorded in the EA.

the relationship between the share of motifs related to fertility or birth and the share of motifs related to ancestors in a regression framework.<sup>49</sup> In their paper analyzing the Folklore data, Michalopoulos and Xue (2021) transform their dependent variables using  $\log(0.01+Y)$  to account for the skewed nature of concept intensity across oral traditions. However, these "log-like" transformations have well known problems (Chen and Roth, 2023). Instead, I follow the seminal paper of Silva and Tenreyro (2006) and use a pseudo-poisson maximum-likelihood estimator, which is well behaved when the proportion of zeros in the dependent variable is large (Gourieroux et al., 1984a,b; Santos Silva and Tenreyro, 2011).<sup>50</sup> I estimate the following model:

$$Y_e = \exp(\alpha + \beta \text{AncestralBeliefs}_e + X_e' \Phi) \epsilon_e \quad (4)$$

Table 5 shows the conditional correlations. The vector  $X_e'$  includes ethnicity-level controls divided in three categories. The folklore controls include the total number of motifs in an ethnic group's oral tradition, the number of publishers of the sources used to identify those motifs, and the earliest year of publication in the group's oral tradition. The vector of ethnographic controls includes domestic organization around extended families and the presence of segmented communities and localized clans (according to Enke (2019), these are two key characteristics that reflect strong extended family networks, and therefore help me to disentangle the influence of an extended supportive kinship network from the importance of ancestral beliefs), whether patrilineality is the main descent type, whether the group has partible or impartible inheritance, political complexity, the prevalence of monogamy, the importance of pastoralism, the use of the plow, and the proportion of motifs related to the word "supernatural" (to disentangle the influence of ancestor worship from the importance of alternative supernatural beliefs). Finally, I also control for a number of geographic covariates, including tropical climate, precipitation, ruggedness, land quality (population-weighted), and agricultural suitability. Continent

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<sup>49</sup>One concern may be related to whether oral traditions are a good proxy of the actual relevance of these concepts. Here, I assume that, if the share of motifs related to ancestors in an ethnic group's oral tradition is large, then the beliefs in ancestors are likely to be strong. Although beliefs in ancestors and fertility change over time and therefore these measures are difficult to validate, Michalopoulos and Xue (2021) show that episodes in folklore accurately reflect the physical environment of ethnic groups (i.e., proximity to earthquake zones or the intensity of lightning strikes), which increases my confidence in using the motifs as proxies for the importance of fertility and ancestor worship.

<sup>50</sup>Table J4 in Appendix J shows that the results do not change when I estimate a linear regression model by ordinary least squares.

fixed effects are always included when I focus on the global sample. The appendix D provides a detailed description of how the variables were constructed.

I find a positive relationship between the share of motifs related to ancestors and the share of motifs related to birth at both the global and African levels, suggesting that ancestral beliefs may have contributed to historically high fertility levels worldwide (Caldwell and Caldwell, 1987). One percentage point increase in the share of motifs related to ancestors increase the share of motifs related to birth by about 5%. These results are robust to the inclusion of an extensive set of control variables and to alternative measures of both fertility and ancestral beliefs (see Appendix J).

Table 5: Ancestor worship and fertility in Folklore

|                       | Share of motifs related to <i>birth</i> |                       |                       |                       |                        |                        |                        |                        |
|-----------------------|-----------------------------------------|-----------------------|-----------------------|-----------------------|------------------------|------------------------|------------------------|------------------------|
|                       | SSA Sample                              |                       |                       |                       | Global Sample          |                        |                        |                        |
|                       | (1)                                     | (2)                   | (3)                   | (4)                   | (5)                    | (6)                    | (7)                    | (8)                    |
| Ancestral beliefs     | 0.0517***<br>(0.00324)                  | 0.0484***<br>(0.0109) | 0.0389***<br>(0.0136) | 0.0376***<br>(0.0124) | 0.0499***<br>(0.00360) | 0.0472***<br>(0.00468) | 0.0472***<br>(0.00625) | 0.0460***<br>(0.00655) |
| SSA Sample            | Yes                                     | Yes                   | Yes                   | Yes                   | No                     | No                     | No                     | No                     |
| Folklore controls     | No                                      | Yes                   | Yes                   | Yes                   | No                     | Yes                    | Yes                    | Yes                    |
| Ethnographic controls | No                                      | No                    | Yes                   | Yes                   | No                     | No                     | Yes                    | Yes                    |
| Geographic controls   | No                                      | No                    | No                    | Yes                   | No                     | No                     | No                     | Yes                    |
| Mean Y                | 4.177                                   | 4.177                 | 3.895                 | 3.908                 | 4.086                  | 4.086                  | 3.877                  | 3.848                  |
| N                     | 407                                     | 407                   | 307                   | 306                   | 1245                   | 1245                   | 951                    | 862                    |

NOTE. Data: Ethnographic Atlas and Folklore. In columns (1)-(4), the sample is restricted to SSA. The table reports Pseudo-Poisson Maximum Likelihood (PPML) estimators. The outcome variable is the share of motifs related to "birth" in the oral tradition of an ethnic group. "Ancestor worship" is the share of motifs related to "ancestors" in an ethnic group's oral tradition. Folklore controls include the total number of motifs in an ethnic group's oral tradition, the number of publishers of the sources in the group's oral tradition, and the earliest year of publication in the group's oral tradition. Ethnographic controls include whether the domestic organization is around independent nuclear families, whether people are part of localized clans that live as segmented communities, whether the ethnic group is patrilineal, impartible inheritance, political complexity, whether monogamy is dominant, whether the group practices pastoralism, use of historical plough, historical economic development, practice of intensive agriculture, and the share of motifs in an ethnic group related to "supernatural". Geographic controls include tropical climate, precipitations, ruggedness, land quality (population weighted), and agricultural suitability. Robust standard errors in parenthesis. \*\*\* for  $p < 0.01$ , \*\* for  $p < 0.05$ , \* for  $p < 0.1$ .

## 5. Conceptual Framework: Implications

In the previous section I have shown that ancestral beliefs contribute to higher fertility rates in sub-Saharan Africa. I have argued that, because of the central role of descent and lineage in societies with strong beliefs in ancestors, where both living and dead lineage members work together to extend the lineage into the future, the motive to continue the family line becomes pivotal. In this section, I provide suggestive evidence that the motive to continue the family line is what drives the positive relationship between ancestral

beliefs and fertility, rather than alternative explanations. I develop a simple model of fertility where the total number of children belonging to one's lineage is a public good. This model provides specific predictions that I will test in the data and that would be consistent with the motive to continue one's lineage driving fertility upwards, but that would be difficult to explain by other competing hypothesis.

In the model, the couple head maximizes the utility of the couple in a unitary framework. Since most societies in sub-Saharan Africa are strongly patriarchal, I assume that the couple head is always male. I therefore assume that the husband has the final say in fertility decisions and that they are not made by other members of the extended family (e.g., the maternal uncle in the case of matrilineal societies). This assumption seems reasonable in the sense that husbands are likely to have more control over their wives than other family members, and it is standard in the context of SSA (see Dupas et al. (2023) or Fontenay et al. (2024)). In Appendix K, I discuss the case where the woman contributes to the decision on the number of children, and the additional implications in this situation.<sup>51</sup> The agent derives utility from three components: 1) consumption ( $c_i$ ), the net monetary benefit of having children ( $n_i$ ), and the total number of children belonging to his lineage ( $N$ ). Additive separability is assumed to represent preferences:

$$U_i = u(c_i) + b(n_i) + \beta \ell(N)$$

Where the functions  $u(\cdot)$ ,  $b(\cdot)$ , and  $\ell(\cdot)$  satisfy the standard assumptions that  $u' > 0$ ,  $b' > 0$ ,  $\ell' > 0$ ,  $u'' < 0$ ,  $b'' < 0$ ,  $\ell'' < 0$ . The extra term ( $\ell$ ) represents the utility that the individual derives from extending his lineage into the future, or in other words, it represents the motive to continue the family line. In fact, in societies with strong ancestral beliefs, the extension of the lineage into the future and the importance attached to the succession of generations is the primary concern of both the living members of the lineage and the ancestors. The parameter  $\beta$  takes the value 0 if ancestral beliefs are absent in the society and 1 if ancestral beliefs are prevalent.<sup>52</sup> The composition of the

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<sup>51</sup>The assumption of men being the main decision-takers regarding fertility is not crucial for the results. In fact, I would get the same main predictions if I assume that both spouses have the same bargaining power, since their preferences are aligned in patrilineal societies (both spouses belong to the same kin group and therefore expand their lineage by having children) but are non-aligned in matrilineal societies (the husband belongs to a different lineage than his wife and children, and therefore only the mother expands her lineage by having children).

<sup>52</sup>The parameter  $\beta$  could take values between 0 and 1 and be interpreted as a factor representing the intensity of ancestral beliefs. For example, it could be the subjective probability that the ancestors will be

total number of children belonging to the lineage ( $N$ ) depends on the structure of the kinship system.<sup>53</sup> In patrilineal societies, the father and his own children belong to the same lineage, as do the children of his male siblings. On the other hand, in matrilineal societies, the father and his children belong to different lineages, and only the children of his sisters continue his lineage. Therefore, the total number of children belonging to the lineage of the (male) head of the couple is:

$$N = \begin{cases} n_i + \sum_{j \neq i} n_j & \text{if patrilineal, } j \text{ indexes male siblings} \\ \sum_{k \neq i} n_k & \text{if matrilineal, } k \text{ indexes female siblings} \end{cases}$$

Therefore, the maximization problem becomes:

$$\begin{aligned} \max_{c_i, n_i} \quad & U_i = u(c_i) + b(n_i) + \beta \ell(N) \\ \text{s.t.} \quad & c_i + n_i \leq y_i \\ & N = \begin{cases} n_i + \sum_{j \neq i} n_j & \text{if patrilineal} \\ \sum_{k \neq i} n_k & \text{if matrilineal} \end{cases} \end{aligned}$$

### 5.1. Matrilineal case

I first study the solution under matrilineality. In matrilineal societies, the lineage of the couple head is expanded through the children of his female siblings. Thus, the maximization problem becomes:

$$\begin{aligned} \max_{c_i, n_i} \quad & U_i = u(c_i) + b(n_i) + \beta \ell\left(\sum_{k \neq i} n_k\right) \\ \text{s.t.} \quad & c_i + n_i \leq y_i \end{aligned}$$

The first-order conditions for an interior solution gives the following optimality condition:

$$\frac{\partial U_i(c_i, n_i, N)}{\partial c_i} = u'(c_i) = b'(n_i) = \frac{\partial U_i(c_i, n_i, N)}{\partial n_i} \quad (5)$$

This condition says that the marginal utility of private consumption must be equal to the marginal utility of having children. Note that since men cannot continue their family line with their own children, the third term of the utility function disappears, and we

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reincarnated in each of the new children belonging to the lineage.

<sup>53</sup>See Appendix L for a simple introduction to kinship systems.

are back to the case where the motive to continue the family line does not play a role in fertility behavior.<sup>54</sup>

### 5.2. Patrilineal case

Let's now turn to the patrilineal case. Now, the lineage of the household head is expanded through his own children as well as through the children of his male siblings. The maximization problem then becomes:

$$\begin{aligned} \max_{c_i, n_i} \quad & U_i = u(c_i) + b(n_i) + \beta \ell(n_i + \sum_{j \neq i} n_j) \\ \text{s.t.} \quad & c_i + n_i \leq y_i \end{aligned}$$

The first-order conditions for an interior solution gives the following optimality condition:

$$u'(c_i) = b'(n_i) + \beta \ell'(n_i + \sum_{j \neq i} n_j) \quad (6)$$

Note that since the term  $\beta \ell'(n_i + \sum_{j \neq i} n_j)$  is always positive, the optimal number of children is now higher than in the matrilineal case. In this case, an individual's utility depends not only on his own number of children, but also on the total number of children of his male siblings (i.e., the number of newborns who continue the lineage). Therefore, we need to better understand the interaction between the two. We can define the best response function as:

$$r_i(n_j) = \arg\max_{n_i} U_i(c_i, n_i, N)$$

Substituting it in the optimality condition of equation K.1, we have that for interior solutions:

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<sup>54</sup>This result follows from the simplifying assumption that men are the sole decision-makers regarding the number of children. If, instead, women also participate in the decision to have children, the motive to continue the family line would be present even in matrilineal societies, where the mother and her children belong to the same lineage. However, it would be less important than in patrilineal societies, where the children of the couple continue the lineage of both the mother and the father, since the wife is now part of her husband's lineage. An additional implication is that the difference in fertility preferences between patrilineal and matrilineal societies due to the motive to continue the family line will be more important for men than for women. This is because men have an additional motive for fertility only in patrilineal societies (continuation of the family line), while women have this motive in both patrilineal and matrilineal societies.

$$F = u'(c_i) - b'(r_i(n_j)) - \beta \ell'(r_i(n_j) + \sum_{j \neq i} n_j) = 0$$

Using the implicit function theorem:

$$\frac{dr_i(n_j)}{dn_j} = -\frac{\frac{\partial F}{\partial n_j}}{\frac{\partial F}{\partial r_i(n_j)}} = -\frac{\frac{\partial^2 U_i(c_i, n_i, n_i + \sum n_j)}{\partial n_i \partial n_j}}{\frac{\partial^2 U_i(c_i, n_i, n_i + \sum n_j)}{\partial n_i^2}} < 0 \quad (7)$$

The expression K.1 means that if  $n_i$  and  $n_j$  are both positive, each time one increases, the other will decrease to maintain the equilibrium between the marginal utility of consumption and children. In other words, there is a negative relationship between the number of children of the agent's male siblings and the agent's own number of children. This is because the two are strategic substitutes: when the number of children of the agent's male siblings is high, he does not "need" to have children to continue his family line. On the contrary, when the number of children of the agent's male siblings is very low, he increases his number of children to extend his lineage. This result mirrors the standard free-rider problem, where each consumer has an incentive to enjoy the benefits of the public good provided by others while providing an inadequate amount of it herself.<sup>55</sup>

So far, the analysis I have made assumes that the husband exerts a major influence on reproductive decisions. However, specially in matrilineal societies, women may have some degree of autonomy in decision making (Lowes, 2022; Tène, 2024). Interestingly, if we solve the problem from the woman's perspective (see Appendix K for details), we get an additional prediction. Following the same logic, we now find that, in matrilineal societies with strong beliefs in ancestors, there is a negative relationship between female fertility and the number of children of the woman's sisters. This is because in matrilineal societies, the woman sisters' children belong to the woman's lineage, while the children of the woman's brothers belong to the lineage of their wives.

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<sup>55</sup>Note that this result cannot be fully explained from an evolutionary perspective. In evolutionary terms, inclusive fitness (the ability of an individual to pass on its genes-both through its offspring and through the offspring of close relatives with whom it shares genes) is increased when siblings have children, and thus the pressure to do so from one's own offspring is reduced. The difference with an evolutionary explanation is that it is not just the total number of siblings that matters, but also their sex composition. In patrilineal societies with ancestor worship, we expect a negative relationship between the number of *male* siblings and one's own fertility.



This simple model is therefore consistent with the following predictions:

- *Proposition 1*: On average, fertility is higher in societies with ancestral beliefs.
- *Proposition 2*: The positive influence of ancestral beliefs on fertility is driven by patrilineal ethnic groups, where both parents and the children belong to the same lineage.
- *Proposition 3*: In patrilineal societies with beliefs in ancestors, there is a negative relationship between one's own fertility and the number of male siblings' children.
- *Proposition 4*: In matrilineal societies with beliefs in ancestors, there is a negative relationship between female fertility and the number of women sisters' children.

In the next section, I provide empirical evidence to support these predictions.

## **6. Test of the theory: Implications**

I have shown that ancestral beliefs contribute to higher fertility rates in sub-Saharan Africa. I have argued that because of the central role of descent and lineage in societies with strong beliefs in ancestors, where both living and dead lineage members work together to extend the lineage into the future, the motive to continue the family line plays an important role in these societies. Moreover, high fertility is socially rewarded and morally associated with joy, recognition, and right living, while low fertility is associated with misfortune and sin. However, although the results shown in Section 4 hold in different samples and time periods, and are robust to the inclusion of many different control variables, the possible association between ancestral beliefs and fertility could still be capturing unobserved confounders.

In this section, I show that these results are specifically due to the motive to continue one's lineage. To do this, I test the very specific predictions provided by the simple model of fertility decisions of Section 5. The validation of these predictions in the data is consistent with a setting where the motive to continue the family line drives fertility upwards, and are difficult to explain by alternative hypothesis.

### 6.1. *Kinship structure, ancestral beliefs and fertility behavior*

I start by investigating the relationship between ancestral beliefs and fertility by kinship system. I test the the second prediction of the model, namely that the positive influence of ancestral beliefs on fertility is driven by patrilineal ethnic groups, in two different settings. First, focusing on the Democratic Republic of Congo, I combine the ethnicity-level information on the practice of ancestor worship from Vansina (1966) with the type of kinship system from Murdock (1967), and match this information to the DRC DHS clusters using ethnic group boundaries.<sup>56</sup> About half of the sample lives in a DHS cluster belonging to the ancestral territory of a patrilineal ethnic group, and 13% live in clusters belonging to the ancestral territory of an ethnic group that did not practice ancestor worship. In all specifications I include a vector of individual-level control variables such as age, age squared, gender, whether the respondent is Catholic, single years of education, whether the respondent lives in a urban or rural area, whether the respondent belongs to the top 40% of the wealth distribution, and dummies for provinces. Finally, since the variable measuring ancestral beliefs is defined at the DHS cluster level, I cluster the standard errors at that level. Second, I use ethnicity-level information at a global scale from Berezkin (2015)s catalog and the Ethnographic Atlas (Murdock, 1967).

Tables 6 and 7 reports the results. Consistent with the theoretical framework, I find that, in both cases, the positive influence of ancestral beliefs on fertility is driven by patrilineal ethnic groups, even at the global level during precolonial times. In the DRC, individuals from ethnic groups who practiced ancestor worship have higher fertility levels, larger ideal family sizes, and want more children in the future, but only if they belong to a patrilineal ethnic group. Interestingly, spouses' preferences are substantially affected (column 4 in table 6). In patrilineal societies with strong beliefs in ancestors, husbands want more children than their wives. However, in matrilineal societies with beliefs in ancestors, this relationship is reversed, and now wives want more children than their husbands. This is because in matrilineal societies children only belong to their mother's line, and therefore the motive to continue one's lineage is absent for men. Overall, these results go in line with the first prediction of the simple framework depicted in Section 5 – that is, the relationship between ancestral beliefs and fertility is

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<sup>56</sup>The matching procedure between DHS clusters and Vansina (1966)' information is described in Section 3.

stronger in patrilineal societies, where both parents and the children belong to the same lineage.

Table 6: Ancestor worship, fertility, and kinship structures

|                                | (1)<br>Nb of children | (2)<br>Ideal Nb of children | (3)<br>Want more children | (4)<br>Husband > wife |
|--------------------------------|-----------------------|-----------------------------|---------------------------|-----------------------|
| Ancestor worship               | -0.165<br>(0.104)     | -0.0988<br>(0.228)          | -0.0346<br>(0.0268)       | -0.102**<br>(0.0503)  |
| Ancestor worship x Patrilineal | 0.263**<br>(0.126)    | 0.422*<br>(0.254)           | 0.0593**<br>(0.0293)      | 0.144**<br>(0.0561)   |
| Province FE                    | Yes                   | Yes                         | Yes                       | Yes                   |
| Controls                       | Yes                   | Yes                         | Yes                       | Yes                   |
| Mean Y                         | 3.086                 | 6.445                       | 0.760                     | 0.474                 |
| R-squared                      | 0.622                 | 0.241                       | 0.221                     | 0.0631                |
| N                              | 35891                 | 33188                       | 29568                     | 10793                 |

NOTE. Data: Demographic and Health Surveys of DRC (2007 and 2013-2014). The table reports OLS estimates. The sample is restricted to women in column 4. The outcome variable in column (1) is the total number of children ever born. In column (2), it is the ideal number of children. In column (3) it is a dummy variable that equals one if the respondent wants to have an additional child. Finally, in column (4), a dummy variable that equals one if the husband wants more children than his wife. "Ancestor Worship" is a dummy variable that equals one if the DHS cluster  $c$  of individual  $i$  belongs to the ethnic homeland of an ethnic group that traditionally practice ancestor worship as reported in Vansina (1966). "Patrilineality" is a dummy variable that equals one if the DHS cluster  $c$  of individual  $i$  belongs to the ethnic homeland of a patrilineal ethnic group as reported in Murdock (1967). Both interaction terms are included as controls. Controls also include age, age squared, gender, a dummy variable that equals one if the respondent is catholic, single years of education, whether the DHS cluster is urban, whether the respondent works, whether the respondent works in agriculture and a dummy variable that equals one if the respondent is in the top 40% of the wealth distribution. Standard errors clustered at the DHS cluster-level in parenthesis. \*\*\* for  $p < 0.01$ , \*\* for  $p < 0.05$ , \* for  $p < 0.1$ .

Table 7: Ancestor worship, fertility and kinship in Folklore

|                                 | Share of motifs related to <i>birth</i> |                       |                      |                      | Share of motifs related to <i>fertility</i> |                      |                      |                      |
|---------------------------------|-----------------------------------------|-----------------------|----------------------|----------------------|---------------------------------------------|----------------------|----------------------|----------------------|
|                                 | (1)                                     | (2)                   | (3)                  | (4)                  | (5)                                         | (6)                  | (7)                  | (8)                  |
| Ancestral beliefs               | 0.00983<br>(0.0150)                     | 0.00648<br>(0.0160)   | 0.00182<br>(0.0204)  | -0.0143<br>(0.0265)  | 0.0160<br>(0.0318)                          | -0.00549<br>(0.0311) | -0.00815<br>(0.0349) | -0.00188<br>(0.0400) |
| Ancestral beliefs x Patrilineal | 0.0430***<br>(0.0151)                   | 0.0441***<br>(0.0153) | 0.0498**<br>(0.0200) | 0.0649**<br>(0.0259) | 0.124***<br>(0.0320)                        | 0.131***<br>(0.0315) | 0.124***<br>(0.0355) | 0.121***<br>(0.0406) |
| Folklore controls               | No                                      | Yes                   | Yes                  | Yes                  | No                                          | Yes                  | Yes                  | Yes                  |
| Ethnographic controls           | No                                      | No                    | Yes                  | Yes                  | No                                          | No                   | Yes                  | Yes                  |
| Geographic controls             | No                                      | No                    | No                   | Yes                  | No                                          | No                   | No                   | Yes                  |
| Mean Y                          | 4.102                                   | 4.102                 | 3.877                | 3.848                | 0.736                                       | 0.736                | 0.687                | 0.660                |
| N                               | 1228                                    | 1228                  | 951                  | 862                  | 1228                                        | 1228                 | 951                  | 862                  |

NOTE. Data: Ethnographic Atlas and Folklore. In columns (1)-(4), the outcome variable is the share of motifs related to "birth" in the oral tradition of an ethnic group, while in columns (5)-(9), it is the share of motifs related to "fertility". The table reports Pseudo-Poisson Maximum Likelihood (PPML) estimators. "Ancestor worship" is the share of motifs related to "ancestors" in an ethnic group's oral tradition. "Patrilineality" equals one if the major descent type of the ethnic group is patrilineal descent. Both interaction terms are included as controls. Folklore controls include the total number of motifs in an ethnic group's oral tradition, the number of publishers of the sources in the group's oral tradition, and the earliest year of publication in the group's oral tradition. Ethnographic controls include whether the domestic organization is around independent nuclear families, whether people are part of localized clans that live as segmented communities, whether the ethnic group is patrilineal, impartible inheritance, political complexity, whether monogamy is dominant, whether the group practices pastoralism, use of historical plough, historical economic development, practice of intensive agriculture, and the share of motifs in an ethnic group related to "supernatural". Geographic controls include tropical climate, precipitations, ruggedness, land quality (population weighted), and agricultural suitability. Robust standard errors in parenthesis. \*\*\* for  $p < 0.01$ , \*\* for  $p < 0.05$ , \* for  $p < 0.1$ .

These results are not driven by inherent differences between patrilineal and matrilineal societies different than beliefs in ancestors, such as differences in women's decision-making and empowerment in favor of women in matrilineal societies (Tene, 2020; Lowes, 2022), or to higher economic returns of children and the need of having sons for old age support (Rossi and Godard, 2022). I show in Appendix M that the relationship between patrilineality and fertility only holds when ancestral beliefs are prevalent, highlighting the importance of kin membership in explaining these fertility differences.<sup>57</sup>

**Kinship systems and fertility behavior.-** These results contribute to our understanding of how kinship systems influence individuals' behavior. Although the influence of kinship systems on developmental outcomes has received considerable attention in economics in recent years (e.g., Moscona et al., 2020; Lowes, 2021, 2022; Bau and Fernández, 2021), the influence of descent structure on fertility has not been extensively studied, despite Lorimer (1954)'s efforts to link resistance to fertility decline to unilineal descent groups as early as 70 years ago.<sup>58</sup> A promising avenue for future research concerns the systematic study of the relationship between patrilineality and fertility, especially in the context of sub-Saharan Africa. This paper contributes to this line of inquiry by highlighting the different incentives than people face in patrilineal as compared to matrilineal societies when they care about the continuation of their lineage. In Appendix M, I dig deeper into the general relationship between patrilineal descent and fertility preferences.

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<sup>57</sup>Moreover, although some papers suggest that women's decision-making and material conditions are better in matrilineal societies, it is incorrect to compare matrilineality to matriarchy or to a utopian feminist society. For example, Coontz and Henderson (1986) argues that it is patrilocal – that is, the system in which women move to their husband's kin group upon marriage – that has enabled men to use and appropriate women's labor and products. Because of the emphasis on the importance of residence rules over unilineal descent, it is argued that matrilineal societies without matrilocality – that is, societies in which the wife must still live with her husband's kin group – are just as restrictive for women as patrilineal and patrilocal societies, despite the fact that inheritance and kinship membership are traced through female lines. In my sample, none of the matrilineal ethnic groups are matrilocal. In this regard, Sage (2023) shows that, in matrilineal (and patrilocal) societies, the old-age security motive also drives fertility upwards. Finally, as Mizinga (2000) notes for the matrilineal Tonga society in southern Zambia, authority is still concentrated in men who exercise control over marriage. What is changing is the composition of those who concentrate power. Instead of the husband and his lineage, power is concentrated in the men of the matrikin, especially the maternal uncle. Moreover, marriage is virilocal and bridewealth is practiced, so women are not entrusted with control of property as they leave their households upon marriage.

<sup>58</sup>In unilineal descent systems, kin is determined only through one gender. In the case of patrilineal societies, kin is determined through the father's side. On the contrary, kin is determined through the mother's side in matrilineal societies.

## 6.2. *Free-rider behavior*

The second set of predictions generated by the model states that there could be a negative relationship between one's own fertility and the number of family members able to continue one's lineage with their own children.<sup>59</sup>

I use data from the DRC Demographic and Health Surveys to explore this question in two steps. First, I examine whether there exists a negative relationship between fertility and the number of husband's brothers in patrilineal societies with strong beliefs in ancestors for both men and women. Unfortunately, the DHS does not include information on siblings' gender for men. However, I do have some information on household composition. For example, I know the total number of eligible men (those aged 15-59) in the household. I use this measure as a proxy for the presence of family members capable of extending the family line.<sup>60</sup> Second, I focus on women and examine the relationship between female fertility and the number of sisters. For women, the DHS provides information of the gender and number of siblings, so I can accurately test the model's prediction.

Tables 8 shows the results regarding the relationship between one's fertility and the number of family male members able to continue one's lineage. Consistent with the intuition in section 5, I find that the number of eligible men in the household is associated with lower fertility, but only in patrilineal ethnic groups. Moreover, the negative effect of the number of eligible men on fertility is driven by ethnic groups with strong ancestral beliefs, since the trade-off between one's own fertility and the number of brothers is particularly salient when the motive to continue the family line plays an important role in fertility behavior.

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<sup>59</sup>This result helps to rationalize some empirical findings in other contexts where ancestor worship still exists to some degree, such as China. For example, Yang and Spencer (2022) find that "while the number of siblings of husbands and wives is of little or no consequence, the number of brothers of husbands matters: couples in which husbands have more brothers have fewer children. This is consistent with free-riding in a patrilineal kinship system, where fertility is at least partly driven by the motive to continue a family line.

<sup>60</sup>There are similar proportions of eligible men in the household by kinship system (52% of male respondents live in a household with at least one other eligible man in patrilineal societies, versus 45% in matrilineal societies). The main limitation of this analysis is that I only have information on the type of relationship to the household head for respondents, not for other household members, and therefore there may be compositional differences. For example, the probability that the other eligible male is a brother may be higher in patrilineal (and patrilocal) ethnic groups, and therefore my measure of eligible men in the household may be a better proxy for the number of male siblings in patrilineal ethnic groups.

Table 8: Kinship network, ancestor worship and fertility

|                              | Total number of children even born |                       |                       |                     |
|------------------------------|------------------------------------|-----------------------|-----------------------|---------------------|
|                              | (1)                                | (2)                   | Ancestral beliefs=1   | Ancestral beliefs=0 |
|                              |                                    |                       | (3)                   | (4)                 |
| Number of eligible men       | 0.0136<br>(0.0270)                 | 0.0362<br>(0.0268)    | 0.0326<br>(0.0270)    | 0.0133<br>(0.169)   |
| Patrilineal x N Eligible men | -0.114***<br>(0.0376)              | -0.0932**<br>(0.0376) | -0.111***<br>(0.0417) | -0.0319<br>(0.172)  |
| Province FE                  | Yes                                | Yes                   | Yes                   | Yes                 |
| Basic controls               | Yes                                | Yes                   | Yes                   | Yes                 |
| All controls                 | No                                 | Yes                   | Yes                   | Yes                 |
| Mean Y                       | 3.111                              | 3.099                 | 3.054                 | 3.432               |
| R-squared                    | 0.635                              | 0.641                 | 0.655                 | 0.578               |
| N                            | 12233                              | 12049                 | 9706                  | 1515                |

NOTE. Data: Demographic and Health Surveys of DRC (2007 and 2013-2014). The table reports OLS estimates. The outcome variable is the total number of children even born. The sample is restricted to men. The sample is restricted to ethnic groups with ancestor worship in column (3) and to ethnic groups without ancestor worship in column (4). "Patrilineal" is a dummy variable that equals one if the ethnic group  $e$  has patrilineal descent. "Nb of eligible men" is the number of men aged 15-59 in the household. Basic controls include age, age squared and gender. All controls also include a dummy variable that equals one if the respondent is catholic, single years of education, whether the DHS cluster is urban, whether the respondent works, whether the respondent works in agriculture and a dummy variable that equals one if the respondent is in the top 40% of the wealth distribution. Standard errors clustered at the DHS cluster-level in parenthesis. \*\*\* for  $p < 0.01$ , \*\* for  $p < 0.05$ , \* for  $p < 0.1$ .

Finally, Table 9 shows the results on the relationship between women's fertility and the number of sisters. I find that there is a negative relationship between the number of sisters and women's fertility in matrilineal societies, but only if beliefs in ancestors are widespread. Moreover, I do not find any negative relationship between the number of brothers and women's fertility in matrilineal societies with strong beliefs in ancestors, since the brothers' children belong to the lineage of their wives, and therefore do not continue one's family line. Overall, the combination of these results are consistent with the hypothesis that the motive to continue the family line is strong in societies where beliefs in the influence of ancestors are important, which contributes to sustain high fertility.

Table 9: Kinship network, ancestor worship and fertility

|                          | Total number of children even born |                       |                      |                       |
|--------------------------|------------------------------------|-----------------------|----------------------|-----------------------|
|                          | Ancestral beliefs=1                |                       | Ancestral beliefs=0  | Ancestral beliefs=1   |
|                          | (1)                                | (2)                   | (3)                  | (4)                   |
| Number of sisters        | 0.0520***<br>(0.00898)             | 0.0454***<br>(0.0102) | 0.0595**<br>(0.0230) |                       |
| Matrilineal x N Sisters  | -0.0310**<br>(0.0142)              | -0.0270*<br>(0.0155)  | -0.00360<br>(0.0534) |                       |
| Number of brothers       |                                    |                       |                      | 0.0380***<br>(0.0109) |
| Matrilineal x N Brothers |                                    |                       |                      | -0.00581<br>(0.0155)  |
| Province FE              | Yes                                | Yes                   | Yes                  | Yes                   |
| Controls                 | Yes                                | Yes                   | Yes                  | Yes                   |
| Mean Y                   | 3.074                              | 3.048                 | 3.277                | 3.048                 |
| R-squared                | 0.614                              | 0.627                 | 0.549                | 0.627                 |
| N                        | 26445                              | 21437                 | 3233                 | 21437                 |

NOTE. Data: Demographic and Health Surveys of DRC (2007 and 2013-2014). The table reports OLS estimates. The outcome variable is the total number of children even born. The sample is restricted to women. The sample is restricted to ethnic groups with ancestor worship in columns (2) and (4) and to ethnic groups without ancestor worship in column (3). "Matrilineal" is a dummy variable that equals one if the ethnic group  $e$  has matrilineal descent. "Nb sisters" is the number of sisters that the respondent has ever had. "Nb brothers" is the number of brothers that the respondent has ever had. Basic controls include age, age squared and gender. All controls also include a dummy variable that equals one if the respondent is catholic, single years of education, whether the DHS cluster is urban, whether the respondent works, whether the respondent works in agriculture and a dummy variable that equals one if the respondent is in the top 40% of the wealth distribution. Standard errors clustered at the DHS cluster-level in parenthesis. \*\*\* for  $p < 0.01$ , \*\* for  $p < 0.05$ , \* for  $p < 0.1$ .

### 6.3. Additional results: ancestral beliefs and social organization

Underlying the relationship between patrilineality, ancestors, and fertility, there is the more fundamental institution of lineage playing a pivotal role in these societies. In fact, ancestor worship belongs to the realm of kinship and lineage structures, since the whole system of beliefs, rituals and practices is centered around the clan or the extended family, and ultimately represents the extension of these aspects of the social organization to the supernatural sphere (Fortes, 1965). One interesting question then arises: will the traditional belief system of African societies, with a primary emphasis on the influence of ancestors on the everyday life of people, erode as the clan-based structure of societies dilutes?

To provide some evidence on this question, I make use of the distinction between lineage-based and age-based societies to examine whether the effect of beliefs in ancestors on fertility is stronger in lineage-based societies (Moscona and Seck, 2024). In

kin-based or lineage-based societies, the main social group is the extended family, and social loyalty is within the kin group. In age-based societies, on the other hand, the main social group is the age group, i.e., a group of individuals of similar age who enter adulthood at the same time.<sup>61</sup> As one of the primary goals of both of living members and ancestors is the extension of the lineage into the future, I expect ancestor worship to play little role in age-based societies (where "horizontal" organization is key) as compared to lineage-based societies (where "vertical" organization is key). I follow the classification of Moscona and Seck (2024) to identify in the contemporary PEW data which ethnic groups are organized along age lines and which ethnic groups are dominated by lineage organization. Although they focus on both Kenya and Uganda, I focus only on respondents from Kenya because I have almost no variation in Uganda (96% of respondents from Uganda belong to an ethnic group where lineage structure dominates, versus 27% in Kenya). Table 10 shows the results. Although the estimates are less precise due to the reduced sample size, I find a positive relationship between ancestral beliefs and fertility in kinship-based societies, while this relationship is weaker in age-based societies and becomes null when the full set of controls is included in the specifications (columns 2, 4, and 6).

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<sup>61</sup>One question that follows from the description of age-set societies and the earlier interpretation of the role of ancestors and their direct connection to kinship structure is whether ancestor spirits should exist at all in societies that are not organized around kinship. First, it is important to note that, as Moscona and Seck (2024) mention, it is possible for an ethnic group to have both kinship and age-set structures. Although many groups have either a strong lineage structure or a strong age structure, there is a continuum of possibilities in between. From this perspective, it is not inconceivable to have variations in the strength of beliefs in ancestors within both lineage and age-based societies. Second, there may be different sets of ancestral spirits. For example, Kenyatta (1965) notes for the Gukuyu (an age-based society from central Kenya) that in addition to the spirits of the father and mother and the clan spirits, there are age-group spirits. The key argument here is that these spirits will play a different role in the society, and the motive of lineage continuation for fertility will be less important.



Table 10: Ancestor worship, fertility and age-based social organization

|                               | Number of children |                   |                    |                    |                     |                   |
|-------------------------------|--------------------|-------------------|--------------------|--------------------|---------------------|-------------------|
|                               | (1)                | (2)               | (3)                | (4)                | (5)                 | (6)               |
| Ancestral beliefs             | 0.741**<br>(0.354) | 0.328<br>(0.248)  | 0.861**<br>(0.389) | 0.720**<br>(0.330) | 0.895***<br>(0.306) | 0.538*<br>(0.308) |
| Ancestral beliefs x Age-based | -0.113<br>(0.404)  | -0.243<br>(0.283) | -0.341<br>(0.455)  | -0.674*<br>(0.379) | -0.792**<br>(0.370) | -0.505<br>(0.355) |
| Country FE                    | Yes                | Yes               | Yes                | Yes                | Yes                 | Yes               |
| Controls                      | No                 | Yes               | No                 | Yes                | No                  | Yes               |
| Mean Y                        | 1.654              | 1.667             | 2.379              | 2.378              | 2.963               | 2.964             |
| R-squared                     | 0.0146             | 0.565             | 0.0120             | 0.462              | 0.0105              | 0.326             |
| N                             | 1198               | 1177              | 775                | 767                | 669                 | 662               |

NOTE. Data: PEW research forum 2008-2009 Survey (Kenya). The sample is restricted to women over the age of 25 in columns 3 and 4, and to women with at least one children in columns 5 and 6. The outcome variable is the total number of children. *Age-based* is an indicator equal to one if the respondent belongs to an aged-based society. The explanatory variable is an indicator equal to one if the respondent answers "yes" to the question "Do you ever participate in traditional African ceremonies or perform special acts to honor or celebrate your ancestors?". Controls include age, age<sup>2</sup>, sex, urban/rural place of residence, religion, education, a dummy that equals one if the respondent is married, a dummy variable that equals one if the respondent finds himself/herself in a good economic situation, and a dummy equal to one if the respondent did not have money at some point during the last year to buy food for his/her family. Robust standard errors in parenthesis. \*\*\* for  $p < 0.01$ , \*\* for  $p < 0.05$ , \* for  $p < 0.1$ .

#### 6.4. Family planning and contraceptive use

A critical issue that is directly related to the analysis presented above concerns the effectiveness of family planning policies and contraceptive uptake. If the motive to continue the family line is very strong due to the prevalence of beliefs in ancestors, the demand for contraceptives may be very low. In this regard, supply-side constraints are sometimes not relevant in explaining high fertility rates in SSA, pointing at the importance of demand-side factors. For example, Dupas et al. (2024) conducted a large-scale randomized controlled trial in Burkina Faso offering free contraceptives to women, and found no significant effect on fertility, suggesting that financial constraints or limited access to contraceptives were not important drivers of high fertility rates.<sup>62</sup>

Traditional beliefs centered on the influence of ancestors on everyday life can be closely linked to contraceptive uptake for several reasons. In addition to higher demand for children in societies with ancestral beliefs, low fertility in these societies is associated with sin, misfortune and the disapproval of ancestors, who care for the fate of the lineage,

<sup>62</sup>In fact, the little effect of contraception on fertility is probably because contraception is not employed to restrict fertility, but rather as a substitute for traditional mechanisms to avoid pregnancy, such as sexual abstinence (Caldwell and Caldwell, 1985).

while men and women who continue to parent children are persons in good moral and social standing, who get their ancestors' approval. Thus, the fear of sterilization is easily understood (Caldwell and Caldwell, 1988). In fact, fear to infertility has been shown to be an important barrier to modern contraceptive methods uptake in SSA (Ochako et al., 2015; Bertrand, 2023; Bau et al., 2024). On the other hand, if bearing children is influenced by the desire and concerns of ancestors, people may think that effective methods to act on the number of children should include contact with the spiritual world through divination or rituals, rather than using modern contraceptives.

I provide below suggestive evidence that holding traditional beliefs prevents contraceptive use. I use data from all available rounds of the DHS for Benin.<sup>63</sup> I mainly rely on Benin DHS because of several reasons. First, there is detailed information on women's contraceptive use (and on the reasons for not using), as well as information on the place of residence of respondents. This information will allow me to look at contraceptive use while holding supply factors constant, as I will be able to compare people living in the same villages or neighbourhoods within cities.<sup>64</sup> Second, Benin is a unique context where adherence to African traditional religion is arguably a good proxy for traditional beliefs, and Benin's main traditional religion (Voodoo) is well known to have strong beliefs in ancestral spirits.<sup>65</sup> Finally, history provides plausibly exogenous variation in adherence to traditional religion that helps with identification.<sup>66</sup>

Table 11 shows the results. I find that, once supply-side factors and other important determinants of contraceptive uptake (including age, education, wealth, employment, marital status, or polygamy) are accounted for, traditional beliefs are associated with lower contraceptive use and intention to use (columns 1-4 and 5, respectively). Note that these results are likely to be a lower bound, as self-reported African traditional religion adherence in Benin is not a perfect proxy for ancestral beliefs, and the control

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<sup>63</sup>In Appendix N.1, I show the same results for the DRC, using the merged information from the DHS, Vansina, and the Ethnographic Atlas.

<sup>64</sup>The merged data of the DRC DHS, Vansina, and Murdock do not allow me to implement this strategy, since all people within a cluster are assigned to the same ethnic group in Vansina's map.

<sup>65</sup>Voodoo is explicitly mentioned in the constitution as an official religion, acquiring the same status as "missionary religions". Since Voodoo is not socially nor politically marginalized, concerns about underreporting of traditional religion in this contexts are limited.

<sup>66</sup>In Appendix N.2, I verify that there is a relationship between demand for children and traditional beliefs in these data as well. I show a positive relationship that holds with DHS cluster fixed effects, and is robust to the same instrumental variable strategy that I will implement below.

group likely includes respondents with ancestral beliefs. Moreover, this relationship seems to be specific to fertility-related outcomes, suggesting that the belief in ancestral spirits may be important to understand these findings. For example, I do not find any relationship between traditional beliefs and the belief in witchcraft (column 7, measured as a dummy variable that equals one if the respondent thinks that HIV/AIDS can be caused by witchcraft). Overall, these results suggest that the demand for contraception is lower among followers of African traditional religions, where the influence of ancestors on people's life and fertility is central.

Table 11: Traditional beliefs and contraceptive use

|                     | (1)<br>Currently        | (2)<br>Traditional    | (3)<br>Modern           | (4)<br>Ever             | (5)<br>Intention       | (6)<br>Side-effects  | (7)<br>AIDS witchcraft |
|---------------------|-------------------------|-----------------------|-------------------------|-------------------------|------------------------|----------------------|------------------------|
| Traditional beliefs | -0.0268***<br>(0.00760) | -0.00366<br>(0.00378) | -0.0218***<br>(0.00676) | -0.0261***<br>(0.00976) | -0.0342***<br>(0.0120) | -0.0268*<br>(0.0138) | 0.0205<br>(0.0156)     |
| Controls            | Yes                     | Yes                   | Yes                     | Yes                     | Yes                    | Yes                  | Yes                    |
| Round FE            | Yes                     | Yes                   | Yes                     | Yes                     | Yes                    | Yes                  | Yes                    |
| Cluster FE          | Yes                     | Yes                   | Yes                     | Yes                     | Yes                    | Yes                  | Yes                    |
| Mean Y              | 0.166                   | 0.0592                | 0.104                   | 0.562                   | 0.382                  | 0.158                | 0.472                  |
| N                   | 43662                   | 43662                 | 43662                   | 43662                   | 36399                  | 14508                | 27237                  |

NOTE. Data: Demographic and Health Surveys of Benin (1996, 2001, 2006, 2011 and 2017). The table reports OLS estimates. The sample is restricted to women. The outcome variable in column (1) is a dummy that equals one if the respondent was using contraceptives at the time of the survey. In column (2) a dummy that equals one if the respondent was using traditional methods. In column (3) a dummy that equals one if the respondent was using modern methods. In column (4) a dummy that equals one if the respondent has ever used contraceptives. In column (5) a dummy that equals one if the respondent intends to use contraceptives in the future. In column (6) a dummy that equals one if the respondent does not use contraceptive because of fear to side effects. In column (7) a dummy that equals one if the respondent thinks that aids may be caused by witchcraft. "Traditional beliefs" is a dummy variable that equals one if the respondent declares African traditional religion as his/her main religion. Controls also include age, education, whether the respondent works, marital status, polygamy, and wealth quintiles. Standard errors clustered at the DHS cluster-level in parenthesis. \*\*\* for  $p < 0.01$ , \*\* for  $p < 0.05$ , \* for  $p < 0.1$ .

**Instrumental variable.-** I follow the strategy developed by Stoop et al. (2019) and instrument traditional religion with a dummy variable that equals one if the respondent's ethnicity is Adja and lives within the historical boundaries of the ancient Dahomey Kingdom. This instrument exploits the fact that religious affiliation is, to some extent, determined by history and tradition. The Adja people were the founders of the Dahomey Kingdom, birth place of the Voodoo religion, whose followers believe in the influence of ancestors on everyday life, and who showed fierce resistance against evangelization. The exclusion restriction implies that, conditional on observables, being Adja influences contraceptive use only through its influence on traditional beliefs. As I show in Table N4, socioeconomic characteristics are fairly similar across ethnic groups, while there are stark differences in their belief systems (and, in turn, in the number of children). These

differences are likely to persist due to different historical and institutional paths.

The results and details on the choice of the instrument can be found in N.3. The results point in the same direction, and are larger in terms of magnitude.<sup>67</sup> In particular, holding strong traditional beliefs is closely associated to fear to side-effects, which is code for fear of jeopardizing one's future fertility (Bertrand, 2023; Bau et al., 2024).

Overall, these results suggest informing about the real risks and consequences of contraceptives rather than focusing on supply side frictions may be important to address the low contraceptive take-up. However, beliefs rooted in deep cultural practices and social organization are difficult to change, and more research is needed in this area (Dupas et al., 2024).

### 6.5. *Heterogeneity and dynamics*

In this section, I discuss the correlates of ancestral beliefs as well as the factors that contribute to reduce its influence on fertility. The main results are shown in Appendix I. Identifying when and where the influence of ancestor worship is stronger may help us to better understand the dynamics of this belief system, which in turn would increase our understanding of current and future trends in fertility rates. I begin by examining which variables are correlated with ancestral beliefs in the first-hand data from Benin (see subsection I.1), where I can accurately measure ancestral beliefs. Second, I test for heterogeneous effects along different socioeconomic dimensions using the multi-country data from the PEW surveys (see subsection I.2).

Two main conclusions arise from this analysis. First, holding beliefs in ancestors is weakly correlated with socioeconomic characteristics, such as education, age, having a TV at home, polygyny, the number of fields for agriculture, or having livestock. Interestingly, a sharp distinction emerges between "missionary" religions and "New" Christian churches: while being a member of a "missionary religion" (i.e., Catholic) is not correlated with holding beliefs in ancestors, being a member of an Evangelic church is the strongest negative correlate of ancestral beliefs.<sup>68</sup> This raises the question of the demo-

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<sup>67</sup>This is not surprising, since the OLS estimates were likely a lower bound (although Voodoo is a better proxy of traditional religion than in other countries or regions, it is by no means a perfect proxy, and people who report other religion as their main religion may still hold traditional beliefs).

<sup>68</sup>This result is consistent with historical accounts highlighting the strong opposition of new Christian churches to African traditional beliefs. For example, Caldwell and Caldwell (1987) note: "Increasingly, the

graphic consequences associated with the massive conversion movement towards new Christian churches that sub-Saharan Africa is undergoing since a few decades.

Second, distance to Cotonou (the economic capital of the country) is positively associated with ancestral beliefs, suggesting that urbanization may contribute to the dissolution of this belief system. The cross-country analysis with the PEW data confirms that urbanization is the strongest factor reducing the influence of ancestral beliefs on fertility, and that other socioeconomic characteristics may not contribute to attenuate this influence.<sup>69</sup> These results are not surprising if we take into account that ancestor worship is closely linked to lineage and kinship structures, and that urbanization (and the associated process of land scarcity and privatization of property rates) contribute to undermining strong kinship ties and the lineage-based structure of societies

There are conflicting views in the literature about the impact of urbanization on traditional practices. For example, Hunter (1961) argues that urbanization contributes to the decline of ancestor worship by eroding traditional practices and clan structures. On the contrary, Theron (1996) and Coertze (2004) state that African traditional religions and worldviews still have a strong influence on people's lives despite urbanization. In a more recent paper, Bae (2007) asserts that in the South African context, the importance of ancestor worship has not diminished in urban areas. In fact, beliefs in ancestors play a role in maintaining local identities in the face of colonization, westernization, and urbanization. This hypothesis is consistent with the economic theory of cultural persistence among minority groups advanced by Bisin and Verdier (2000), in which the sense of identity is an important component of the individual's utility, so that brides and grooms prefer to be matched with a person of the same culture because this gives them a better chance of transmitting their culture to their children. Finally, in the specific context of the Democratic Republic of the Congo and with a specific mention of ancestor worship,

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cutting edge of change in African Christianity has been not the imported, mainstream missionary churches but the locally generated sects known as African or Ethiopian Churches [...]. The fact that they are very much a grass-roots phenomenon and do not share the apprehension of African cultural manifestations that has often marked the missionary churches has tended to hide their greater willingness to challenge those aspects of the traditional belief system that they consider unacceptable".

<sup>69</sup>Particularly interesting is the null effect of education on the importance of ancestor worship. In fact, if ancestor worship contributes to the stalls in fertility observed in many sub-Saharan African countries, these results suggest that education alone may have a limited impact on the reduction of fertility. These results are compatible with recent papers suggesting that slow education progress is not among the predominant factors in explaining fertility stalls in SSA (Schoumaker and Sánchez-Páez, 2024).

Romaniuk (2011), p.15, asserts that "ethnicity still plays a role in interpersonal networks of socialization and in the wheeling and dealing of political patronage, but many of the customs associated with ethnicity tend to disappear [...]. Ancestral memory, which has traditionally supported high fertility in African societies, is waning as people are increasingly cut off from their ancestral lands," and Shapiro (2010), p. 14, argues for contemporary Kinshasa that "ethnicity, once the most important factor associated with differential fertility, has become largely irrelevant to fertility. Here, I provide evidence that urbanization may contribute to the decline in fertility rates by reducing the importance of kinship and the centrality of extended families and clan-based organization, in addition to its effects through increased education, better access to services such as healthcare or contraceptives, or better infrastructure (de la Croix and Gobbi, 2017).

## **7. Conclusion**

The fertility decline associated with the demographic transition has been a key factor in moving from stagnation to sustained economic growth (Galor et al., 2023). Moreover, high fertility rates combined with reductions in infant mortality lead to high child dependency ratios, are detrimental to maternal and child health, and are associated with low educational attainment and poor living standards (Stover and Ross, 2010; African Union, 2007). Understanding the origins of the demographic transition is therefore critical for development.

This paper revisits the literature that stressed the importance of deep-seated cultural norms, rooted in traditional beliefs, in explaining resistance to fertility decline in sub-Saharan Africa (Lesthaeghe, 1989; Caldwell and Caldwell, 1987; Caldwell et al., 1992). In particular, I examine the role played by the belief in the influence of ancestors, one of the (historically) most prominent belief systems, both globally and within sub-Saharan Africa. I document and test the hypothesis that ancestral beliefs, which stresses the importance of ancestry and descent, and therefore attaches great importance to the continuation of the family line, contributes to sustain high fertility rates. I find a positive association between beliefs in ancestors and fertility in different context and time periods, that holds across ethnic groups, individuals within countries and regions, and across migrants who live in the same city and were born in similar places, but differ in the practice of ancestor worship.

Second, I test whether the relationship between ancestral beliefs and fertility is specifically due to the importance attributed to the continuation of one's lineage. First, I show that the influence of ancestral beliefs on fertility depends on the kinship system, since the motive to continue one's lineage differs. I find that the positive effect of ancestral beliefs on fertility is driven by patrilineal societies, where both parents and children belong to the same lineage, and thus having children is always a way to continue one's family line. On the contrary, the father does not belong to his wife's lineage in matrilineal societies, and children belong exclusively to the wife's lineage, and thus the motive to continue the family line with one's own offspring is absent for men. Second, consistent with free-rider behavior in a setting where the motive to continue one's lineage is a public good, I find a negative relationship between fertility and the number of family members able to continue one's lineage.

Overall, these findings reinforce Casterline (2017)'s conclusion, notably the hypothesis that an important basis for resistance to fertility decline is both cultural and social, partly because beliefs and social norms have proven to be difficult to shift even in experimental settings (Dupas et al., 2024). Moreover, I show some of the demographic implications of the existing traditional belief system (e.g., lower use of modern contraceptives due to high fear to infertility), and highlight the importance of demand (e.g., information campaigns about contraceptives' side effects) rather than supply factors to influence reproductive decisions. Finally, I provide suggestive evidence that both the process of urbanization (that contributes to erode lineage institutions and the clan-based structure of societies) and the emergence of endogenous African religions who strongly oppose traditional beliefs (such as Evangelical churches in southern Benin) may contribute to the potential future decline of fertility levels. An important avenue for future research is to understand which factors (e.g., son preference or ethnic traits) increase or decrease in importance as fertility declines, or to understand what are the causes and the social and economic consequences of those institutions that challenge some traditional beliefs and aspects of the social structure that they consider unacceptable (i.e., evangelical churches).

## References

- Abadie, A., Athey, S., Imbens, G. W., and Wooldridge, J. M. (2022). When Should You Adjust Standard Errors for Clustering?\*. *The Quarterly Journal of Economics*, 138(1):1–35. 29
- African Union (2007). Status report on maternal newborn child and adolescent health: Focusing on unfinished business in africa. 50
- Ahern, E. (1973). *The cult of the dead in a Chinese village*. Stanford: Stanford University Press. 11
- Alesina, A. and Giuliano, P. (2015). Culture and institutions. *Journal of Economic Literature*, 53(4):898–944. 8
- Alesina, A., Giuliano, P., and Nunn, N. (2013). On the origins of gender roles: Women and the plough. *Quarterly Journal of Economics*, 128(2):469–530. 8, 4, 5
- Alidou, S. and Verpoorten, M. (2019). Family size and schooling in sub-saharan africa: testing the quantity-quality trade-off. *J Popul Econ*, 32:13531399. 12, 15
- Alvarez-Aragon, P., Guirking, C., and Villar, P. (2023). Legacy of Colonial Education: Unveiling Persistence Mechanisms in the D.R. Congo. *DeFiPP Working Papers 2305*, University of Namur, Development Finance and Public Policies. 17, 27, 7, 24
- Anderson, A. (1993). African pentecostalism and the ancestor cult : Confrontation or compromise. *Missionalia : Southern African Journal of Mission Studies*, 21(1):26–39. 14
- Angrist, J. D. and Pischke, J.-S. (2009). *Mostly Harmless Econometrics: An Empiricists Companion*. Princeton: Princeton University Press. 12
- Asante, M. K. (2009). Ancestors. In Asante, M. K. and Mazama, A., editors, *Encyclopedia of African Religion*. Volume 1. SAGE Publications, Inc. 9
- Auriol, E., Lassébie, J., Panin, A., Raiber, E., and Seabright, P. (2020). God Insures those Who Pay? Formal Insurance and Religious Offerings in Ghana. *The Quarterly Journal of Economics*, 135(4):1799–1848. 13
- Bae, C. S. (2007). *Ancestor Worship and the Challenges it poses to the Christian Mission and Ministry*. PhD Dissertation: University of Pretoria. 49
- Bau, N. and Fernández, R. (2021). Culture and the family. Working Paper 28918, National Bureau of Economic Research. 40
- Bau, N., Henning, D. J., Low, C., and Steinberg, B. (2024). Family planning, now and later: Infertility fears and contraceptive take-up. Working Paper 32735, National Bureau of Economic Research. 6, 46, 48
- Bauer, M., Chytilová, J., and Streblov, P. (2006). Effects of education on determinants of high desired fertility evidence from ugandan villages. *IES Working Paper 239/2006*, IES FSV. Charles University. 6
- BenYishay, A., Grosjean, P., and Vecchi, J. (2017). The fish is the friend of matriliney: Reef density and matrilineal inheritance. *Journal of Development Economics*, 127:234–249. 5, 39
- Berezkin, Y. (2015). Folklore and mythology catalogue: Its lay-out and potential for research. *The Retrospective Methods Network Newsletter*, 10:5870. 18, 20, 30, 38
- Berger, H. and Dasré, A. (2024). Religious affiliation, education, and fertility in sub-saharan africa. *World Development*, 184:106723. 7
- Bergeron, A. (2023). Missions, Social Ties, and Moral Universalism: Evidence from the Belgian Congo. *Work in progress*. 27
- Berggreen, S. and Gokmen, G. (2023). Traditional norms and parental investment in human capital. *Working Paper*. 4, 37, 38
- Bertoli, S., Clerc, M., Loper, J., and Roca-Fernández, . (2024). Migration and the epidemiological approach: time and self-selection into foreign ancestries matter. *Working paper*. 25
- Bertrand, J. T. (2023). *The Dogged Pursuit of Progress: 50 years of Family Planning in the DR Congo*. Forthcoming. 46, 48



- Bisin, A. and Verdier, T. (2000). Beyond the Melting Pot: Cultural Transmission, Marriage, and the Evolution of Ethnic and Religious Traits\*. *The Quarterly Journal of Economics*, 115(3):955–988. 49
- Bongaarts, J. (2017). Africa's unique fertility transition. *Population and Development Review*, 43:39–58. 2
- Bongaarts, J. and Casterline, J. (2013). Fertility transition: Is sub-saharan africa different? *Popul Dev Rev*, 38:153–168. 2
- Bongaarts, J., Frank, O., and Lesthaeghe, R. (1984). The proximate determinants of fertility in sub-saharan africa. *Population and Development Review*, 10:511. 28
- Boserup, E. (1965). *The Conditions of Agricultural Growth: The Economics of Agrarian Change Under Population Pressure*. Routledge. 11
- Boserup, E. (1985). Economic and demographic interrelationships in sub-saharan africa. *Population and Development Review*, 11(3):383–397. 11
- Butinda, L. D., Lameke, A. A., Nunn, N., Posch, M., and Sánchez de la Sierra, R. (2023). On the importance of african traditional religion for economic behavior. Working Paper 31430, National Bureau of Economic Research. 2, 8
- Caldwell, J. and Caldwell, P. (1987). The cultural context of high fertility in sub-saharan africa. *Popul Dev Rev*, (13):409. 2, 6, 10, 11, 14, 16, 32, 48, 50, 31
- Caldwell, J. C. and Caldwell, P. (1985). Cultural forces tending to sustain high fertility in tropical africa. *Population, Health and Nutrition Department*. World Bank. 6, 11, 14, 45
- Caldwell, J. C. and Caldwell, P. (1988). Is the asian family planning program model suited to africa? *Studies in Family Planning*, 19(1):19–28. 2, 6, 46
- Caldwell, J. C. and Caldwell, P. (1990). High fertility in sub-saharan africa. *Scientific American*, (262):118125. 2, 6, 27
- Caldwell, J. C., Caldwell, P., and Orubuyole, I. (1992). Fertility decline in africa: A new type of transition? *Popul Dev Rev*, (2):211–242. 6, 50
- Casterline, J. B. (2017). Prospects for fertility decline in africa. *Population and Development Review*, 43:3–18. 2, 51
- Chen, J. and Roth, J. (2023). Logs with Zeros? Some Problems and Solutions\*. *The Quarterly Journal of Economics*, page qjad054. 7, 31
- Chu, C. Y. C. (1991). Primogeniture. *Journal of Political Economy*, 99(1):78–99. 15, 16
- Church, A. C., Ibitoye, M., Chettri, S., and Casterline, J. B. (2023). Traditional supports and contemporary disrupters of high fertility desires in sub-saharan africa: a scoping review. *Reprod Health*, (20). 6, 38
- Coertze, R. D. (2004). *Etnografiese Studies in Suidelike Africa*. Pretoria: VanSchaik. 49
- Cogneau, D. and Dupraz, Y. (2015). institutions historiques et développement économique en afrique: Une revue sélective et critique de travaux récents. *Histoire mesure*, pages 103–134. 18
- Coontz, S. and Henderson, P. (1986). *Women's Work, Men's Property: The Origins of Gender and Class*. London: Verso, pp. 220. 40
- Curtin, P. D. (1961). "the white man's grave:" image and reality, 1780-1850. *Journal of British Studies*, 1(1):94–110. 11
- de la Croix, D. and Gobbi, P. E. (2017). Population density, fertility, and demographic convergence in developing countries. *Journal of Development Economics*, 127:13–24. 50
- Dupas, P., Falezan, C., Mabeu, M. C., and Rossi, P. (2023). Long-run impacts of forced labor migration on fertility behaviors: Evidence from colonial west africa. Working Paper 31993, National Bureau of Economic Research. 7, 33
- Dupas, P., Jayachandran, S., Lleras-Muney, A., and Rossi, P. (2024). Drivers of fertility: Experimental evidence from burkina faso. Technical report. 2, 6, 7, 16, 45, 48, 51
- Ekeh, P. (2004). Individuals basic security needs and the limits of democracy in Africa. In Berman, B., Eyoh, D., and Kymlicka, W., editors, *Ethnicity and Democracy in Africa*. Ohio University Press, Oxford: James Currey and Athens. 12

- Ellis, S. and Ter Haar, G. (2004). *Worlds of Power. Religious Thought and Political Practice in Africa*. Hurst Company, London. 12
- Enke, B. (2019). Kinship, Cooperation, and the Evolution of Moral Systems\*. *The Quarterly Journal of Economics*, 134(2):953–1019. 31, 4
- Ezenweke, E. (2008). The cult of ancestors: A focal point for prayers in african traditional communities. *UNIZIK Journal of Religion and Human Relations*, (1). 10
- Fenske, J. (2013). Does land abundance explain african institutions? *The Economic Journal*, 123(573):1363–1390. 5
- Fernández, R. and Fogli, A. (2009). Culture: An empirical investigation of beliefs, work, and fertility. *American Economic Journal: Macroeconomics*, 1(1):146–77. 8, 24, 25, 27
- Fontenay, S., Gobbi, P. E., and Goñi, M. (2024). Fertility in Sub-Saharan Africa: the Role of Inheritance. Working Papers ECARES 2024-01, ULB – Université Libre de Bruxelles. 5, 6, 23, 33, 15, 39
- Fortes, M. (1965). Some Reflections on Ancestor Worship in Africa. In Fortes, M. and Dietorlen, G., editors, *African systems of thought*. London: Oxford University Press. 2, 10, 11, 14, 43
- Fox, R. (1983). *Kinship and Marriage: An Anthropological Perspective*. Cambridge University Press. 4
- Galor, O., Klemp, M., and Wainstock, D. C. (2023). The impact of the prehistoric out-of-africa migration on cultural diversity. *Working Paper, Brown University*. 20, 50
- Gay, V., Gobbi, P., and Goñi, M. (2023). Revolutionary Transition: Inheritance Change and Fertility Decline. *CEPR Discussion Paper No. 18607*. CEPR Press, Paris London. 15
- Gershman, B. (2016). Witchcraft beliefs and the erosion of social capital: Evidence from sub-saharan africa and beyond. *Journal of Development Economics*, 120:182–208. 8
- Gershman, B. (2022). Witchcraft beliefs, social relations, and development. Available at SSRN: <https://ssrn.com/abstract=3896078> or <http://dx.doi.org/10.2139/ssrn.3896078>. 8
- Gong, W., Zhu, M., Gürel, B., and Xie, T. (2021). The lineage theory of the regional variation of individualism/collectivism in china. *Frontiers in Psychology*, 11. 14
- Goody, J. (1973). Strategies of heirship. *Comparative Studies in Society and History*, 15(1):3–20. 16, 12, 17
- Gourieroux, C., Monfort, A., and Trognon, A. (1984a). Pseudo maximum likelihood methods: Applications to poisson models. *Econometrica*, 52(3):701–20. 31
- Gourieroux, C., Monfort, A., and Trognon, A. (1984b). Pseudo maximum likelihood methods: Theory. *Econometrica*, 52(3):681–700. 31
- Grande, L. (2024). *The Evolution of Religions: A History of Related Traditions*. ISBN:9780231216517. Columbia University Press. 3, 11, 16
- Guirkinger, C. and Platteau, J.-P. (2014). The effect of land scarcity on farm structure: Empirical evidence from mali. *Economic Development and Cultural Change*, 62(2):195–238. 11
- Guirkinger, C. and Villar, P. (2022). Pro-birth policies, missions and fertility: historical evidence from Congo. *DeFiPP Working Paper 4, Université de Namur*. 7, 17, 27, 24
- Götmark, F. and Turner, N. (2023). Human fertility and religions in sub-saharan africa: A comprehensive review of publications and data, 2010-2020. *Afr J Reprod Health*, 27(1):119–171. 7
- Hayford, S. R. and Morgan, S. P. (2008). Religiosity and fertility in the united states: The role of fertility intentions. *Social forces; a scientific medium of social study and interpretation*, 86(3):1163–1188. 29, 30
- Herzer, D. (2019). A note on the effect of religiosity on fertility. *Demography*, 56(3):991–998. 30
- Holden, C. J., Sear, R., and Mace, R. (2003). Matriliney as daughter-biased investment. *Evolution and Human Behavior*, 24(2):99–112. 39
- Hu, A. (2016). Ancestor worship in contemporary china: An empirical investigation.

- China Review*, 16(1):169–186. 11
- Hu, A. and Tian, F. (2018). Still under the ancestors' shadow? Ancestor worship and family formation in contemporary China. *Demographic Research*, 38(1):1–26. 9, 10, 11
- Hunter, M. (1961). *Reaction to conquest: Effects of contact with Europeans on the Pondo of South Africa*. London: Oxford University Press. 49
- Idowu, E. B. (1973). *African Traditional Religion: A Definition*. London: SCM Press Ltd. 9, 14
- Igboin, B. O. (2022). *Beliefs and Veneration of Ancestors*, pages 107–118. Springer International Publishing, Cham. 8, 10, 13
- Ishak, P. W. and Gradstein, M. (2022). Losing my religion (or maybe not): Religion and fertility patterns in africa. *Working Paper*. 7
- Jedwab, R., Meier zu Selhausen, F., and Moradi, A. (2022). The economics of missionary expansion: evidence from africa and implications for development. *Journal of British Studies*, 27:149–192. 11
- Kenyatta, J. (1965). *Facing Mount Kenya*. Knopf Doubleday Publishing Group. 3, 44, 12
- Kopytoff, I. (1971). Ancestors as elders in africa. *Africa: Journal of the International African Institute*, 41(2):129–142. 10
- Kuper, H. (1961). *An African Aristocracy: Rank Among the Swazi*. Routledge. 15
- Lambert, S. and Rossi, P. (2016). Sons as widowhood insurance: Evidence from senegal. *Journal of Development Economics*, 120:113–127. 15, 17
- Larsen, U. (2003). Infertility in central africa. *Tropical Medicine & International Health*, 8(4):354–367. 28
- Le Rossignol, E. and Lowes, S. (2022). Ancestral livelihoods and moral universalism: Evidence from transhumant pastoralist societies. Working Paper 30259, National Bureau of Economic Research. 4
- Le Rossignol, E., Lowes, S., and Nunn, N. (2022). Traditional supernatural beliefs and prosocial behavior. *NBER working paper* 29695. 7, 8, 13, 30
- Leistner, E. (2014). Witchcraft and African Development. *African Security Review*, 23(1):53–77. 13
- Lesthaeghe, R. J. (1989). *Reproduction and Social Organization in Sub-Saharan Africa*. Berkeley. University of California Press. 6, 50
- Lorimer, F. (1954). Culture and human fertility. *Paris: UNESCO*. 6, 40, 37
- Lowes, S. (2021). Kinship structure, stress, and the gender gap in competition. *Journal of Economic Behavior Organization*, 192:36–57. 40
- Lowes, S. (2022). Kinship structure and the family: Evidence from the matrilineal belt. Working Paper 30509, National Bureau of Economic Research. 19, 36, 40
- Luttmer, E. F. P. and Singhal, M. (2011). Culture, context, and the taste for redistribution. *American Economic Journal: Economic Policy*, 3(1):157–79. 8
- Mbiti, J. S. (1975). *Introduction to African religion*. London: Heinemann. 9, 14
- McCall, J. C. (1995). Rethinking ancestors in africa. *Africa*, 65(2):256–270. 13
- Mekoa, I. (2019). The living-dead/ancestors as guardians of morality in african traditional religious thought. *Glob J Arch Anthropol*, 10(5). 16
- Michalopoulos, S., Naghavi, A., and Prarolo, G. (2018). Trade and geography in the spread of islam. *The Economic Journal*, 128(616):3210–3241. 11
- Michalopoulos, S. and Xue, M. M. (2021). Folklore. *Quarterly Journal of Economics*, 136(4):1993–2046. 20, 30, 31, 3
- Mizinga, F. M. (2000). Marriage and bridewealth in a matrilineal society: The case of the tonga of southern zambia: 1900–1996. *African Economic History*, (28):53–87. 40
- Molnos, A. (1973). *Cultural Source Material for Population Planning in East Africa*. Bibliography, institute of African Studies, University of Nairobi, East African Publishing House, Nairobi. 15
- Moscona, J., Nunn, N., and Robinson, J. A. (2020). Segmentary lineage organization and

- conflict in sub-saharan africa. *Econometrica*, 88(5):1999–2036. 40
- Moscona, J. and Seck, A. A. (2024). Age set versus kin: Culture and financial ties in east africa. *American Economic Review*, 114(9):274891. 5, 43, 44
- Murdock, G. P. (1967). Ethnographic atlas: a summary. *Ethnology*, 6(2):109236. 5, 18, 20, 38, 39, 3, 41
- National Research Council (1993). Factors affecting contraceptive use in sub-saharan africa. 2
- Neill, S. (1991). *A History of Christian Missions*. 2 edition. ed. Penguin Books, London; New York. 7
- Nunn, N. and Puga, D. (2012). Ruggedness: The Blessing of Bad Geography in Africa. *The Review of Economics and Statistics*, 94(1):20–36. 5
- Nunn, N. and Sanchez de la Sierra, R. (2017). Why being wrong can be right: Magical warfare technologies and the persistence of false beliefs. *American Economic Review*, 107(5):582–87. 8
- Ochako, R., Mbondo, M., Aloo, S., Kaimenyi, S., Thompson, R., Temmerman, M., and Kays, M. (2015). Barriers to modern contraceptive methods uptake among young women in kenya: A qualitative study global health. *BMC Public Health*, 15. 16, 46
- Okafor, S., Onu, J. C., and Nwaeze, V. C. (2021). Rural small-scale women farmers and preference for family size in south-east nigeria. *Comparative Population Studies*, 46. 5, 38, 43
- Olupona, J. K. (2014). *African Religions: A Very Short Introduction*. Oxford University Press. 9, 16, 27
- Opoku, K. A. (1993). *African traditional religion: An enduring heritage*, pages 67–82. De Gruyter, Berlin, Boston. 9
- Osanyinbí, . and Falana, K. (2016). An evaluation of the akure yorùbá traditional belief in reincarnation. *Open Journal of Philosophy*, 6:59–67. 14
- Oster, E. (2019). Unobservable selection and coefficient stability: Theory and evidence. *Journal of Business & Economic Statistics*, 37(2):187–204. 18
- Pew Research Center (2009). Tolerance and tension: Islam and christianity in sub-saharan africa. 4, 13, 19, 29
- Platteau, J.-P. (2000). *Institutions, social norms and economic development*. Harwood Academic. 11, 12
- Platteau, J.-P. (2009). Institutional obstacles to african economic development: State, ethnicity, and custom. *Journal of Economic Behavior Organization*, 71(3):669–689. 8, 12, 13, 27
- Platteau, J.-P. (2014). *Redistributive Pressures in Sub-Saharan Africa: Causes, Consequences, and Coping Strategies*, page 153207. Cambridge University Press. 13, 30
- Radcliffe-Brown, A. R. (1922). *The andaman islanders*. Cambridge University Press. 2, 10
- Radcliffe-Brown, A. R. and Forde, D. (1950). *African Systems of Kinship and Marriage*. Oxford University Press. 37
- Reybrouck, D. V. (2014). *Congo: The Epic History of a People*. HarperCollins. 27
- Romaniuk, A. (2011). Persistence of high fertility in tropical africa: the case of the democratic republic of the congo. *Population and development review*, 37(1):128. 24, 50
- Rossi, P. and Godard, M. (2022). The old-age security motive for fertility: Evidence from the extension of social pensions in namibia. *American Economic Journal: Economic Policy*, 14(4):488–518. 15, 23, 40, 17
- Rossi, P. and Rouanet, L. (2015). Gender preferences in africa: A comparative analysis of fertility choices. *World Development*, 72:326–345. 23
- Sage, M. (2023). Children are a poor women’s wealth: How inheritance rights affect fertility. *Working Paper*. 6, 15, 23, 40, 17
- Santos Silva, J. and Tenreiro, S. (2011). Further simulation evidence on the performance of the poisson pseudo-maximum likelihood estimator. *Economics Letters*, 112(2):220–222. 31

- Schneidman, M. (1990). Mortality and fertility trends in zaire. *Internal Discussion Paper, World Bank*. 7
- Schoumaker, B. and Sánchez-Páez, D. (2024). Disruptions in educational progress and fertility dynamics by educational level: Unraveling the link between education and fertility stalls in sub-saharan africa. *Population and Development Review*. 2, 49
- Schulz, J. F. (2022). Kin Networks and Institutional Development. *The Economic Journal*, 132(647):2578–2613. 11
- Schulz, J. F., Bahrami-Rad, D., Beauchamp, J. P., and Henrich, J. (2019). The church, intensive kinship, and global psychological variation. *Science*, 366(6466):eaau5141. 11
- Shapiro, D. (1996). Fertility decline in Kinshasa. *Population Studies*, 50:89103. 17, 26, 7
- Shapiro, D. (2010). Ongoing fertility transition in kinshasa: Evidence from the 2007 dhs. *paper presented at the Annual Meeting of the Population Association of America, Dallas, TX*. 50
- Shapiro, D. and Hinde, A. (2017). On the pace of fertility decline in sub-Saharan Africa. *Demographic Research*, 37(40):1327–1338. 2
- Shapiro, D. and Tambashe, B. (2001). Fertility in the democratic republic of congo. *Pennsylvania State University, Department of Economics, Working Papers*. 7
- Sievert, C. (2023). Supernatural beliefs about illness and modern medicine use: Evidence from the drc. *Working Paper*. 8
- Silva, J. M. C. S. and Tenreyro, S. (2006). The Log of Gravity. *The Review of Economics and Statistics*, 88(4):641–658. 31
- Stoop, N. and Verpoorten, M. (2020). Risk, envy and magic in the artisanal mining sector of south kivu, democratic republic of congo. *Development and Change*, 51(5):1199–1224. 13
- Stoop, N., Verpoorten, M., and Deconinck, K. (2019). Voodoo, vaccines, and bed nets. *Economic Development and Cultural Change*, 67(3):493–535. 8, 47
- Stover, J. and Ross, J. (2010). How increased contraceptive use has reduced maternal mortality. *Maternal and Child Health Journal*, 5(14):687695. 50
- Tabutin, D. (1982). Evolution régionale de la fécondité dans louest du zaïre. *Population (french edition)*, page 2950. 17, 7
- Tene, E. (2020). On the Historical Roots of Gender Norms: Evidence from Matrilineal Societies in Sub-Saharan Africa . *Working Paper*. 40
- Theron, P. F. (1996). *African Traditional Cultures and the Church*. Pretoria: Institute for Missiological and Ecumenical Research. 49
- Turaki, Y. (2000). *Christianity and African Gods A method in Theology*. Potchefstroomse: Universiteit vir Christelike Hoer Onderwys. 14
- Tène, E. (2024). On the historical roots of gender norms: Evidence from matrilineal societies in sub-saharan africa. *Working Paper*. 36
- van de Grijsparde, H., Voors, M., Bulte, E., and Richards, P. (2013). Who Believes in Witches? Institutional Flux in Sierra Leone. *African Affairs*, 112(446):2247. 13
- Vansina, J. (1966). Introduction a lethnographie du congo. *Centre de Recherche et d'Information Socio-Politiques*. 4, 5, 18, 19, 24, 38, 39, 2, 20, 44
- World Bank (1986). Population growth and policies in sub-saharan africa. 2
- Yang, W. and Spencer, B. G. (2022). Kinship and fertility: Brother and sibling effects on births in a patrilineal system. *Journal of Economic Behavior Organization*, 195:158–170. 9, 41
- Zhang, Y. (2024). The persistence of son preference: Cultural transmission of chinese migrants in taiwan. *Toulouse School of Economics*. 9, 11
- Zipfel, C. (2022). The demand side of africa's demographic transition: desired fertility, wealth, and jobs. *STICERD - Economic Organisation and Public Policy Discussion Papers Series 71*. 2, 7

## **Online Appendix**

### **ANCESTRAL BELIEFS AND FERTILITY IN SUB-SAHARAN AFRICA**

November 20, 2024

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**A. (Extract of) The Breath of the Ancestors (Birago Diop, 1960)**

*A.1. English*

Listen more often  
to things than to beings  
the voice of the fire  
the water that speaks  
the voice of the wind  
the bush that weeps  
this is the ancestors breathing

The breath of the ancestors  
who are not gone  
who are not underground  
who are not dead  
those who have passed are not gone  
they are in a womans breast  
in a childs wailing song  
in the coals that wont rest

*A.1. Original*

Écoute plus souvent  
les choses que les êtres  
La voix du feu sétend  
entends la voix de leau  
Écoute dans le vent  
le buisson en sanglot  
Cest le souffle des ancêtres

Le souffle des ancêtres morts  
qui ne sont pas partis  
Qui ne sont pas sous terre  
qui ne sont pas morts  
Ceux qui sont morts ne sont jamais partis  
Ils sont dans le sein de la femme  
Ils sont dans lenfant qui vagit  
Et dans le tison qui senflamme

## B. Matching procedure: Vansina and DRC DHS

Figure B1: Ethnic groups in Vansina (1966) and DHS clusters



(a) Digitized version of Vansina (1966)

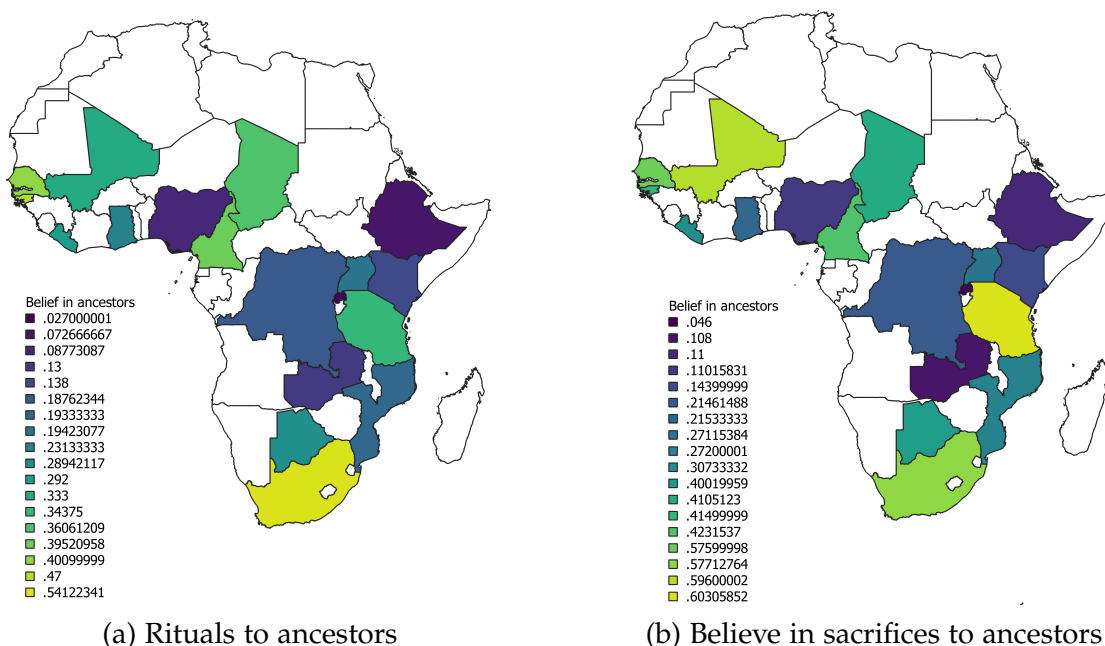
(b) Ancestor worship and DHS clusters

Notes: Panel B1a displays ethnic group boundaries for the DRC digitized from Vansina (1966). Panel B1b assigns to each DHS cluster the value taken by the variable ancestor worship in the ethnic homeland where the DHS cluster is located.

## C. Geographical distribution of variables: PEW and Folklore

### C.1. PEW Variables

Figure C1: Share of respondent believing in Ancestors



(a) Rituals to ancestors

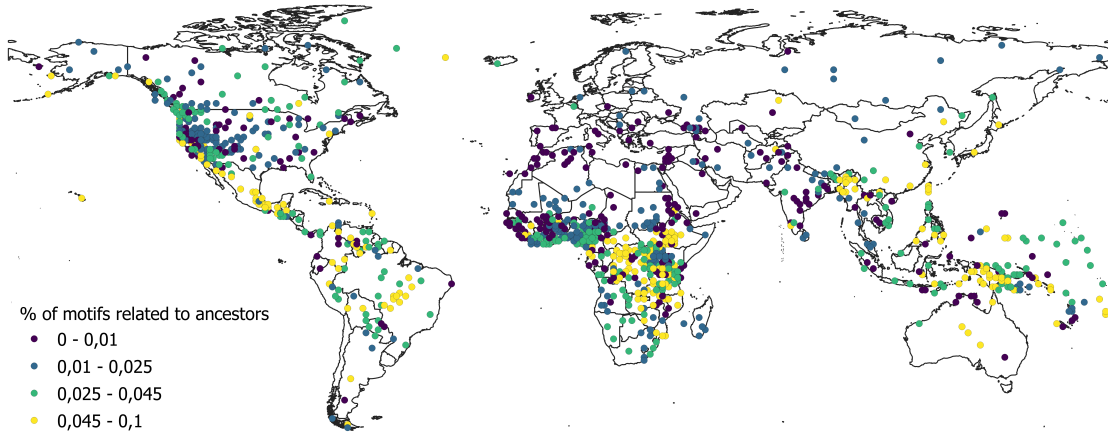
(b) Believe in sacrifices to ancestors

Notes: The figure displays the share of individuals in each country of the PEW sample believing in ancestors. Panel C1a shows the geographic distribution of the share of respondents that answered "yes" to the question "Do you ever participate in traditional African ceremonies or perform special acts to honor or celebrate your ancestors?". Panel C1b shows the geographic distribution of the share of respondents that answered "yes" to the question "Do you believe that sacrificing to spirits or ancestors can protect you from bad things happening?".



## C.2. Distribution of motifs related to ancestors

Figure C2: Share of Motifs related to ancestors in the oral tradition of EA's ethnic groups



Note: The figure displays the location of the 1.265 groups in the Ethnographic Atlas (Murdock, 1967), and the distribution of the share of motifs related to the word "ancestor".

## D. Variable definitions

### D.1. Ethnicity-level variables

#### D.1.1. Folklore:

**Number of motifs:** Total number of motifs recorded in each Berezkin's group. Data comes from Michalopoulos and Xue (2021).

**Number of publishers:** Total number of publishers of the sources in the group's oral tradition. Data comes from Michalopoulos and Xue (2021).

**Earliest year of publication:** Earliest year of publication in the group's oral tradition. Data comes from Michalopoulos and Xue (2021).

**Motifs related to "supernatural":** Share of motifs tagged by supernatural related terms in the oral tradition of an ethnic group. Data comes from Michalopoulos and Xue (2021).

#### D.1.2. Ethnographic Atlas:

**Domestic organization around extended families:** is based on variable V8 of the Ethnographic Atlas (domestic organization). It is constructed as a dummy variable that is zero if domestic organization is "Independent polyandrous families", "Polygynous: unusual

co-wives pattern", "Polygynous: usual co-wives pattern", "Minimal (stem) extended families", "Small extended families", and "Large extended families". It equals one if domestic organization is "Independent nuclear family, monogamous" or "Independent nuclear family, occasional polygyny". This measure is based on Enke (2019).

**Segmented communities and localized clans:** is based on variable V15 of the Ethnographic Atlas (community marriage organization). It is constructed as a dummy variable that is zero if community organization is "Demes, not segregated into clan barrios", "Agamous communities", "Exogamous communities, not clans". It takes on a value of one if community organization is "Segmented communities without local exogamy", "Segmented communities, localized clans, local exogamy", or "Clan communities, or clan barrios". This measure is based on Enke (2019).

**Patrilineal:** is based on variable V43 of the Ethnographic Atlas (descent: major type). It is a dummy variable that takes the value one if a groups major mode of descent is patrilineal as opposed to any other mode of descent.

**Political complexity:** is based on variable V33 of the Ethnographic Atlas (jurisdictional hierarchy beyond local communities). This is a categorical variable ranging from no levels beyond local community to four levels beyond local community. I use it as it appears in the Ethnographic Atlas (ranging from 0 to 4).

**Monogamy:** is based on variable V9 of the Ethnographic Atlas (marital composition: monogamy and polygamy). It is constructed as a dummy variable that takes on a value of one if an ethnic group's dominant form of marital composition is monogamous.

**Pastoralism:** is based on variables V4 (animal husbandry) and V40 (predominant type of animal husbandry) of the Ethnographic Atlas. To construct this variable, I follow Le Rossignol and Lowes (2022). From variable v4, I create a dummy variable that equals one if the predominant type of animal raised is a herding animal such as cattle, sheep, or camelids. Then, pastoralism is measured by multiplying this new dummy variable by variable V40.

**Historical use of the plough:** is based on variable V39 of the Ethnographic Atlas (animals and plow cultivation). I construct an indicator variable for traditional plough agriculture that equals one if the plough was present (whether aboriginal or not) and zero otherwise. This measure is similar to Alesina et al. (2013).

**Historical economic development:** is based on variable V30 of the Ethnographic Atlas

(settlement patterns). I use this variable as in Alesina et al. (2013). Each ethnic group is categorized into one of the following categories describing their pattern of settlement: "nomadic or fully migratory", "semi-nomadic", "semi-sedentary", "compact but temporary settlements", "neighborhoods of dispersed family homes", "separated hamlets forming a single community", "compact and relatively permanent", "complex settlements". The variable takes on the values of 1 to 8, with 1 indicating fully nomadic groups and 8 groups with complex settlement.

**Practice of intensive agriculture:** is based on variable V28 of the Ethnographic Atlas (intensity of agriculture). I follow Alesina et al. (2013) and construct an indicator that equals one if the society belongs to the categories "intensive agriculture" or "intensive irrigated agriculture", and zero otherwise.

#### *D.1.3. Geographic variables:*

**Tropical climate:** is taken from Alesina et al. (2013). It is computed as the proportion of land within a 200 kilometer radius of an ethnic groups centroid that is classified as being either tropical or subtropical. The classification of thermal climates comes from the GAEZ 2002 database.

**Precipitation:** is taken from Fenske (2013). It is average annual precipitation (mm). The data comes from the International Institute for Applied Systems Analysis.

**Ruggedness:** is taken from Nunn and Puga (2012). It is constructed as the average across points on a grid 1 kilometer apart within a country of an index of terrain ruggedness. For details about how the index is constructed, see Nunn and Puga (2012).

**Land quality:** is taken from Fenske (2013). It is measured as a non-additive combination of climate constraints, soil constraints and terrain slope constraints, coming from the Food and Agriculture Organizations Global Agro-Ecological Zones (FAO-GAEZ) project.

**Agricultural suitability:** is taken from Alesina et al. (2013). Using information on global geo-climatic conditions for crop cultivation from the FAOs Global Agro-Ecological Zones (GAEZ) v3.0 database, they calculate the fraction of this land that is suitable for the cultivation of barley, wheat, rye, sorghum, foxtail millet, or pearl millet. They use this measure to construct the average suitability of the land (within 200 kilometers of the centroid of each ethnic group) historically inhabited by a locations ancestors.

## **E. West Zaire Surveys (1975-77)**

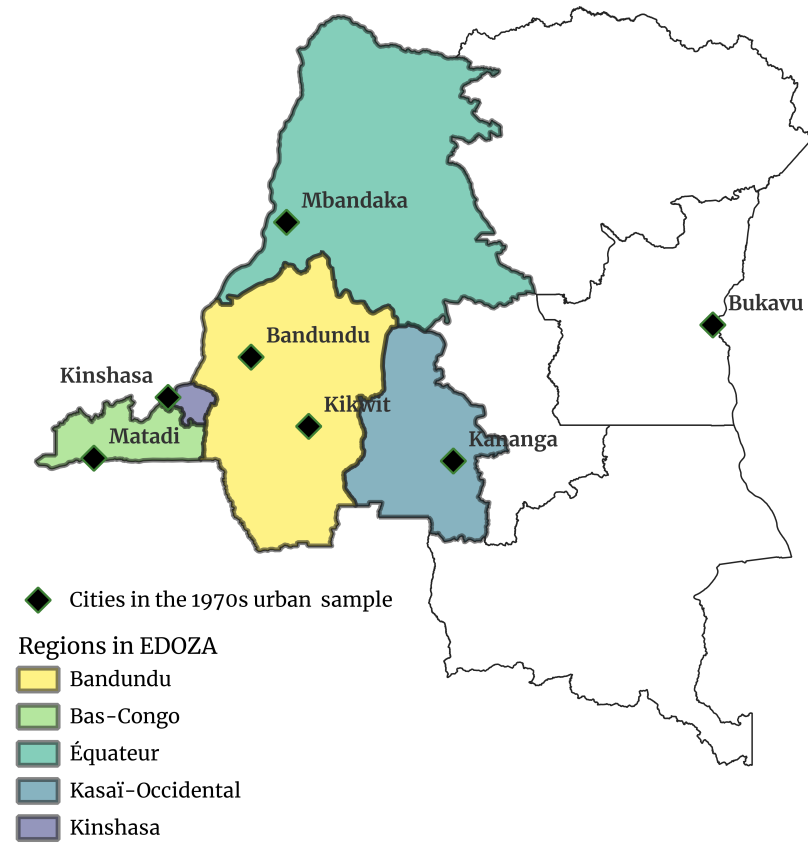
These surveys were part of a large research project led by the Université Catholique de Louvain in collaboration with the Congolese Official Institute of Research and Statistics. The surveys were conducted between 1975 and 1977 and aimed at collecting basic demographic information for the Democratic Republic of the Congo, due to the paucity of this type of data. These surveys are the outcome of two parallel studies (see Figure E1 for the sampled territories):

**Urban Demographic and Budgetary Survey.**—A demographic and budgetary survey were conducted in the six largest cities of western Zaire: Kinshasa, Matadi, Bandundu, Kikwit, Mbandaka and Kananga. This survey was led by Joseph Houyoux, professor at the Sociology department of the Université Catholique de Louvain, in Belgium. On the one hand, the demographic survey contains individual-level information on about 250,000 individuals in 43,000 households. After a census conducted in each of these cities to count and identify all households, 1/10 of the total number of households was randomly selected to be included in the sample. Demographic information was collected on age, sex, ethnicity, place of birth, and migration status, as well as socioeconomic characteristics such as years of education and occupation. Moreover, for each woman over the age of 13, births calendars are also recorded, including the month and year of each birth, the sex of the children, and the dates of death, if applicable. On the other hand, the budgetary survey contains information on household expenditures and transactions for 1/50 of the households identified in each city (two daily passages during one month).

**Enquête Démographique de l'Ouest du Zaïre (EDOZA).**—A demographic survey conducted in the rest of urban areas (excluding the large cities included in the previous survey) and in rural areas of four western regions: Bas-Congo, Bandundu, Kasai Occidental and Equateur (only in the Equateur and Tshuapa districts). This survey was led by the Demography department of the Université Catholique de Louvain. The original survey includes around 210,000 individuals in about 45,000 households. However, contrary to the Urban Demographic survey where I have the full sample, I have been able to locate 50,000 individuals in 11,000 households so far. To the extent of my knowledge, the rest of the EDOZA sample is lost.

One potential concern relates to the quality of these surveys. Although these surveys were conducted during Mobutu's dictatorship, both the design and implementation of the survey and data collection were managed jointly by a team of demographers based at the Congolese Institute for Research and Statistics and the Université Catholique de Louvain in Belgium. Data cleaning and statistical programming were carried out in Belgium. These data are considered high quality by demographers who rely on them to study demographic dynamics in the Congo in well-published academic papers (Tabutin, 1982; Shapiro, 1996; Shapiro and Tambashe, 2001; Schneidman, 1990). Finally, these surveys used the same methodology to construct variables (i.e., birth calendars) as current benchmark surveys, such as the Demographic and Health Surveys. Although the data from these surveys have been used by demographers, it has only been used in an aggregated way to study fertility and mortality levels and trends across regions and time. Recently, some papers in economics have used the microdata of the Urban Demographic survey (Guirkingner and Villar, 2022; Alvarez-Aragon et al., 2023). To the best of my knowledge, no paper has yet exploited the microdata from EDOZA.

Figure E1: Cities included in the urban survey and regions included in EDOZA



*Note:* Own elaboration using QGIS.

## F. Ancestral beliefs and fertility in southern Benin

### F.1. Poisson model

Table F1: Beliefs in ancestors and fertility in contemporary Benin

|                   | Number of children ever born |                      |                      |                      |                      |                      |
|-------------------|------------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
|                   | (1)                          | (2)                  | (3)                  | (4)                  | (5)                  | (6)                  |
| Ancestral beliefs | 0.229***<br>(0.0451)         | 0.209***<br>(0.0407) | 0.191***<br>(0.0414) | 0.184***<br>(0.0474) | 0.159***<br>(0.0422) | 0.146***<br>(0.0431) |
| Baseline controls | No                           | Yes                  | Yes                  | No                   | Yes                  | Yes                  |
| Extended controls | No                           | No                   | Yes                  | No                   | No                   | Yes                  |
| Religion FE       | No                           | No                   | No                   | Yes                  | Yes                  | Yes                  |
| Mean Y            | 7.995                        | 8.016                | 8.016                | 8.012                | 8.029                | 8.029                |
| N                 | 846                          | 832                  | 832                  | 841                  | 830                  | 830                  |

NOTE. Data: First-hand data collected in southern Benin. The table reports Pseudo-Poisson Maximum Likelihood (PPML) estimators. The outcome variable is the total number of children ever born. "Ancestral beliefs" is a dummy variable that equals one if the respondent answers *yes* to the question *Do you believe that the spirits of your ancestors influence the events of your life?*. Baseline controls include age, age squared, whether the household has electricity and education. Extended controls include the number of agricultural fields, whether the household owns a TV, whether the household has been part of a program to encourage the production of pineapple, and whether the respondent has received pressure from the extended family to have children. Religion fixed effects includes five categories: Vodoun, Roman Catholic, Evangelic, Celeste and other. Robust standard errors in parenthesis. \*\*\* for  $p < 0.01$ , \*\* for  $p < 0.05$ , \* for  $p < 0.1$ .

### F.2. Alternative explanatory variables

Table F2: Beliefs in ancestors and fertility in contemporary Benin

|                   | Number of children ever born |                     |                     |                     |                     |                     |
|-------------------|------------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                   | (1)                          | (2)                 | (3)                 | (4)                 | (5)                 | (6)                 |
| Ancestral beliefs | 1.905***<br>(0.358)          | 1.648***<br>(0.326) | 1.513***<br>(0.317) | 1.590***<br>(0.378) | 1.264***<br>(0.340) | 1.172***<br>(0.335) |
| Baseline controls | No                           | Yes                 | Yes                 | No                  | Yes                 | Yes                 |
| Extended controls | No                           | No                  | Yes                 | No                  | No                  | Yes                 |
| Religion FE       | No                           | No                  | No                  | Yes                 | Yes                 | Yes                 |
| Mean Y            | 7.995                        | 8.016               | 8.016               | 8.012               | 8.029               | 8.029               |
| N                 | 846                          | 832                 | 832                 | 841                 | 830                 | 830                 |

NOTE. Data: First-hand data collected in southern Benin. The table reports OLS estimates. The outcome variable is the total number of children ever born. "Ancestral beliefs" is a dummy variable that equals one if the respondent answers *yes* to the question *Do you think that your ancestors' spirits care about your extended family and the continuation of the family line?*. Baseline controls include age, age squared, whether the household has electricity and education. Extended controls include the number of agricultural fields, whether the household owns a TV, whether the household has been part of a program to encourage the production of pineapple, and whether the respondent has received pressure from the extended family to have children. Religion fixed effects includes five categories: Vodoun, Roman Catholic, Evangelic, Celeste and other. Robust standard errors in parenthesis. \*\*\* for  $p < 0.01$ , \*\* for  $p < 0.05$ , \* for  $p < 0.1$ .

Table F3: Beliefs in ancestors and fertility in contemporary Benin

|                   | Number of children ever born |                     |                     |                     |                     |                     |
|-------------------|------------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                   | (1)                          | (2)                 | (3)                 | (4)                 | (5)                 | (6)                 |
| Ancestral beliefs | 1.818***<br>(0.354)          | 1.577***<br>(0.321) | 1.472***<br>(0.312) | 1.503***<br>(0.369) | 1.205***<br>(0.333) | 1.138***<br>(0.327) |
| Baseline controls | No                           | Yes                 | Yes                 | No                  | Yes                 | Yes                 |
| Extended controls | No                           | No                  | Yes                 | No                  | No                  | Yes                 |
| Religion FE       | No                           | No                  | No                  | Yes                 | Yes                 | Yes                 |
| Mean Y            | 7.995                        | 8.016               | 8.016               | 8.012               | 8.029               | 8.029               |
| N                 | 846                          | 832                 | 832                 | 841                 | 830                 | 830                 |

NOTE. Data: First-hand data collected in southern Benin. The table reports OLS estimates. The outcome variable is the total number of children ever born. "Ancestral beliefs" is a dummy variable that equals one if the respondent answers *yes* to the question *Do you believe that the spirits of your ancestors influence the events of your life?* or to the question *Do you think that your ancestors' spirits care about your extended family and the continuation of the family line?*. Baseline controls include age, age squared, whether the household has electricity and education. Extended controls include the number of agricultural fields, whether the household owns a TV, whether the household has been part of a program to encourage the production of pineapple, and whether the respondent has received pressure from the extended family to have children. Religion fixed effects includes five categories: Vodoun, Roman Catholic, Evangelic, Celeste and other. Robust standard errors in parenthesis. \*\*\* for  $p < 0.01$ , \*\* for  $p < 0.05$ , \* for  $p < 0.1$ .

### F.3. Number of alive children

Table F4: Beliefs in ancestors and fertility in contemporary Benin

|                   | Number of children alive |                     |                     |                     |                     |                     |
|-------------------|--------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                   | (1)                      | (2)                 | (3)                 | (4)                 | (5)                 | (6)                 |
| Ancestral beliefs | 1.415***<br>(0.331)      | 1.285***<br>(0.306) | 0.858***<br>(0.261) | 1.105***<br>(0.341) | 0.953***<br>(0.313) | 0.854***<br>(0.315) |
| Baseline controls | No                       | Yes                 | Yes                 | No                  | Yes                 | Yes                 |
| Extended controls | No                       | No                  | Yes                 | No                  | No                  | Yes                 |
| Religion FE       | No                       | No                  | No                  | Yes                 | Yes                 | Yes                 |
| Mean Y            | 7.147                    | 7.179               | 7.179               | 7.165               | 7.190               | 7.190               |
| N                 | 842                      | 828                 | 828                 | 837                 | 826                 | 826                 |

NOTE. Data: First-hand data collected in southern Benin. The table reports OLS estimates. The outcome variable is the total number of children alive at the time of the survey. "Ancestral beliefs" is a dummy variable that equals one if the respondent answers *yes* to the question *Do you believe that the spirits of your ancestors influence the events of your life?*. Baseline controls include age, age squared, whether the household has electricity and education. Extended controls include the number of agricultural fields, whether the household owns a TV, whether the household has been part of a program to encourage the production of pineapple, and whether the respondent has received pressure from the extended family to have children. Religion fixed effects includes five categories: Vodoun, Roman Catholic, Evangelic, Celeste and other. Robust standard errors in parenthesis. \*\*\* for  $p < 0.01$ , \*\* for  $p < 0.05$ , \* for  $p < 0.1$ .



#### F.4. Effects on women and complementarity between spouses

Table F5: Beliefs in ancestors and female fertility in Benin

|                   | Number of children ever born |                     |                     |                     |                     |                     |
|-------------------|------------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                   | (1)                          | (2)                 | (3)                 | (4)                 | (5)                 | (6)                 |
| Ancestral beliefs | 0.570***<br>(0.155)          | 0.423***<br>(0.139) | 0.369***<br>(0.140) | 0.630***<br>(0.166) | 0.460***<br>(0.150) | 0.427***<br>(0.149) |
| Baseline controls | No                           | Yes                 | Yes                 | No                  | Yes                 | Yes                 |
| Extended controls | No                           | No                  | Yes                 | No                  | No                  | Yes                 |
| Religion FE       | No                           | No                  | No                  | Yes                 | Yes                 | Yes                 |
| Mean Y            | 5.345                        | 5.333               | 5.333               | 5.345               | 5.333               | 5.333               |
| N                 | 943                          | 937                 | 937                 | 943                 | 937                 | 937                 |

NOTE. Data: First-hand data collected in southern Benin. The sample is restricted to women. The table reports OLS estimates. The outcome variable is the total number of children ever born. "Ancestral beliefs" is a dummy variable that equals one if the respondent answers *yes* to the question *Do you believe that the spirits of your ancestors influence the events of your life?*. Baseline controls include age, age squared, whether the household has electricity and education. Extended controls include the number of agricultural fields, whether the household owns a TV, whether the household has been part of a program to encourage the production of pineapple, and whether the respondent has received pressure from the extended family to have children. Religion fixed effects includes five categories: Vodoun, Roman Catholic, Evangelic, Celeste and other. Robust standard errors in parenthesis. \*\*\* for  $p < 0.01$ , \*\* for  $p < 0.05$ , \* for  $p < 0.1$ .

Table F6: Beliefs in ancestors and fertility in contemporary Benin

|                   | Number of children ever born |                     |                     |                     |                     |                     |
|-------------------|------------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                   | (1)                          | (2)                 | (3)                 | (4)                 | (5)                 | (6)                 |
| Ancestral beliefs | 2.695***<br>(0.519)          | 2.310***<br>(0.478) | 2.066***<br>(0.473) | 2.235***<br>(0.533) | 1.825***<br>(0.493) | 1.626***<br>(0.488) |
| Baseline controls | No                           | Yes                 | Yes                 | No                  | Yes                 | Yes                 |
| Extended controls | No                           | No                  | Yes                 | No                  | No                  | Yes                 |
| Religion FE       | No                           | No                  | No                  | Yes                 | Yes                 | Yes                 |
| Mean Y            | 7.995                        | 8.016               | 8.016               | 8.012               | 8.029               | 8.029               |
| N                 | 846                          | 832                 | 832                 | 841                 | 830                 | 830                 |

NOTE. Data: First-hand data collected in southern Benin. The table reports OLS estimates. The outcome variable is the total number of children ever born. "Ancestral beliefs" is a dummy variable that equals one if both the husband and the wife answer *yes* to the question *Do you believe that the spirits of your ancestors influence the events of your life?*. Baseline controls include age, age squared, whether the household has electricity and education. Extended controls include the number of agricultural fields, whether the household owns a TV, whether the household has been part of a program to encourage the production of pineapple, and whether the respondent has received pressure from the extended family to have children. Religion fixed effects includes five categories: Vodoun, Roman Catholic, Evangelic, Celeste and other. Robust standard errors in parenthesis. \*\*\* for  $p < 0.01$ , \*\* for  $p < 0.05$ , \* for  $p < 0.1$ .

#### *F.5. Specification curves: robustness to additional controls*

In this section I show that the results are robust to the inclusion of alternative control variables and their different combinations. In addition to the baseline and extended controls (age, education, access to electricity, TV ownership, quantity of agricultural land, pressure from the extended family to have children and whether the respondent is participating in a program to increase pineapple production), I include:

*Polygyny:* Since I am focusing on men's ever-born children, polygyny is an important factor to consider. In fact, about 60% of men who report having 10 or more children are polygynous. I have not included this in the main specification because it may be a "bad control" (Angrist and Pischke, 2009). For example, Goody (1973) notes that one can gain security in the afterlife (i.e., continuity of the family line) directly by having children or indirectly by adding wives. Similarly, Kenyatta (1965) mentions in the context of the Gikuyu people of central Kenya that:

"If a man dies without a male child his family group comes to an end. This is one thing that the Gikuyu fear dreadfully, and it can be said to be one of the factors behind the polygamous system of marriage. There is no doubt that perpetuation of family or kinship group is the main principal of every Gikuyu marriage. For the extinction of a kinship groups means cutting off the ancestral spirits from visiting the earth, because there is no one left to communicate with them (Kenyatta (1965), p.13)".

*Father practices Vodoun:* To proxy for parental preferences for children and aspects related to the traditional practices of the family, I include a dummy variable that equals one if the father's main religion is Vodoun. This variable is highly correlated to the respondent's religion: when the respondent declares Vodoun as his main religion, his father practices Vodoun in the 85% of cases.

*Household size:* This variable takes into account the structure of the household. Households with more members can more easily externalize the costs of having children, and more members may be the result of stronger preferences towards children in previous generations.

*Money to church to get protection against COVID:* I try to account for strong religiosity (beyond the inclusion of religion fixed effects) by looking at whether the respondent has donated some money to his church to get protected against COVID (average is 14%).

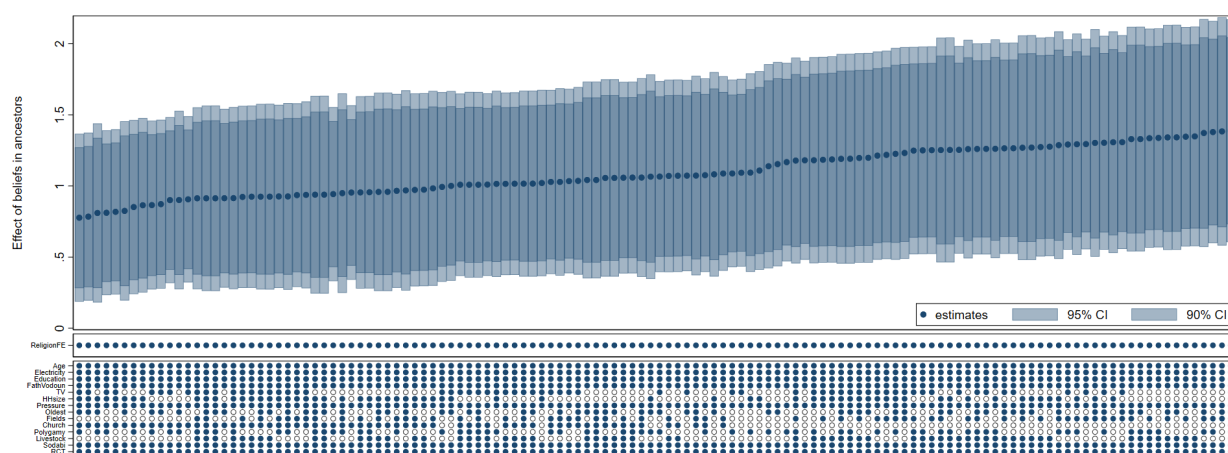
*Drinking sodabi to get protection against COVID:* I proxy for the strength of other traditional practices (not related to beliefs in supernatural powers) by measuring the subjec-

tive effectiveness of traditional medicine. I include an indicator that equals one if the respondent drunk a traditional alcoholic drink (sodabi) to get protection against COVID.

*Livestock ownership:* As an additional measure of wealth, I include a dummy variable that takes the value one if the respondent owns livestock.

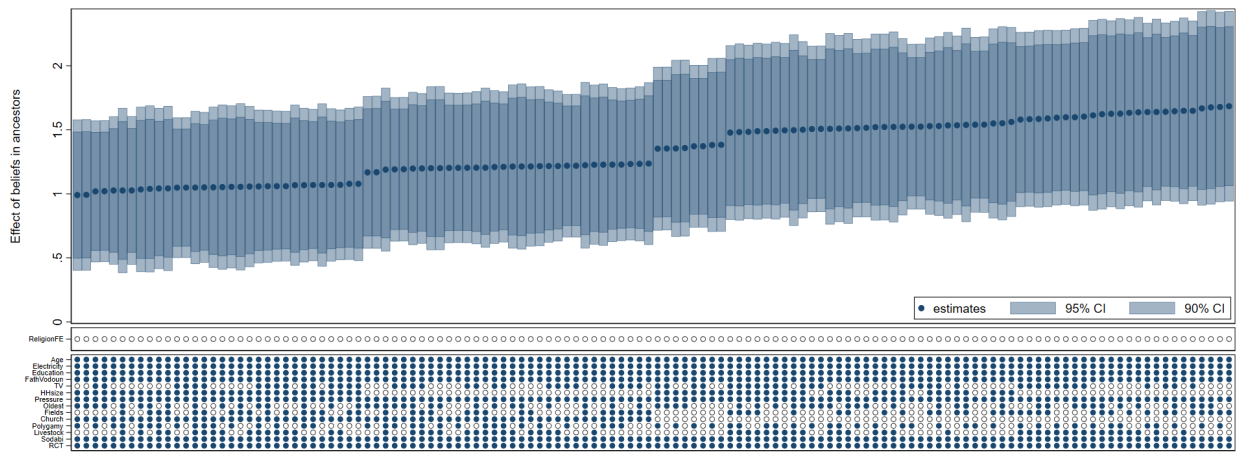
*Oldest brother:* Finally, since siblings may compete for limited resources in their household and first-born children are often prioritized in accessing resources, or in their marriage prospects, I include an indicator that takes value one if the respondent is the oldest sibling (among those from the same mother).

Figure F1: Specification curve: additional controls and religion fixed effects



*Note:* The figure shows the specification curves for the effect of beliefs in the influence of ancestors. Each dot is a coefficient from Eq. 1 with a different set of control variables. The vertical bars, from darkest to lightest, denote the 95% and 90% confidence intervals. All specifications include religion fixed effects. *HHsize* is the total number of people living in the household. *FatherVodoun* equals one if the (main) religion of the respondent's father is Vodoun and zero otherwise. *Oldest* takes the value one if the respondent is the oldest brother and zero otherwise. *Church* equals one if the respondent donated to his church to get protected from COVID. *Polygamy* equals one if the respondent is in a polygamous union and zero otherwise. *Livestock* is an indicator that takes the value one if the respondent owns livestock. *Sodabi* takes the value one if the respondent drank sodabi (a traditional alcoholic drink) to get protected against COVID.

Figure F2: Specification curve: additional controls

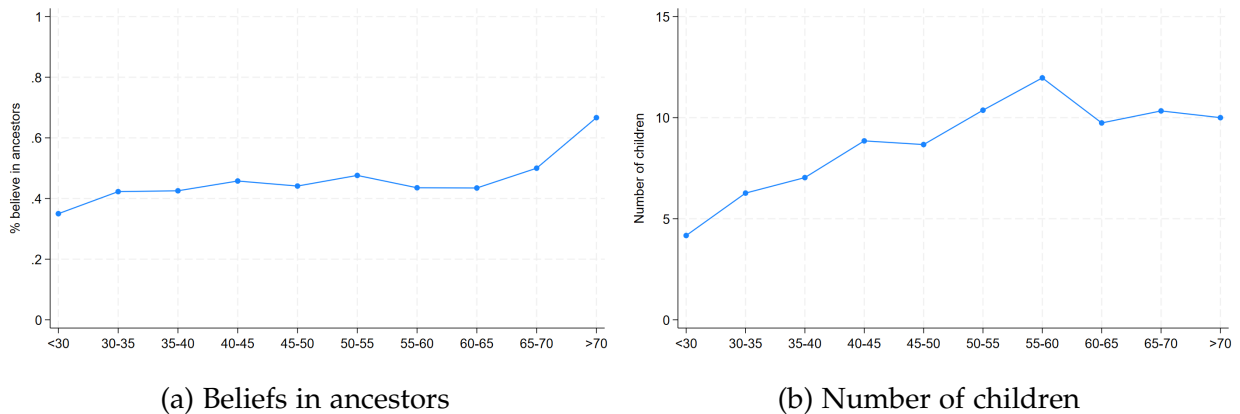


Note: The figure shows the specification curves for the effect of beliefs in the influence of ancestors. Each dot is a coefficient from Eq. 1 with a different set of control variables. The vertical bars, from darkest to lightest, denote the 95% and 90% confidence intervals. Religion fixed effects are not included. *HHsize* is the total number of people living in the household. *FatherVodoun* equals one if the (main) religion of the respondent's father is Vodoun and zero otherwise. *Oldest* takes the value one if the respondent is the oldest brother and zero otherwise. *Church* equals one if the respondent donated to his church to get protected from COVID. *Polygamy* equals one if the respondent is in a polygamous union and zero otherwise. *Livestock* is an indicator that takes the value one if the respondent owns livestock. *Sodabi* takes the value one if the respondent drank sodabi (a traditional alcoholic drink) to get protected against COVID.

#### F.6. Age dynamics: age fixed effects

One potential confounder could be related to similar dynamics in the evolution of beliefs and the number of children over the life cycle or over time, that are difficult to capture in a lineal model. However, this does not seem to be the case (see Figure F3). Although the number of children is an increasing function of age (although at decreasing pace), the share of people believing in ancestors is rather constant. In Table F7 I show that adding age fixed effects does not change the results.

Figure F3: Dynamics of beliefs in ancestors and children by age group



Notes: Panel (a) plots the share of people believing in the influence of ancestors on their lives (n=846) by age group. Panel (b) plots the average number of children by age group.

Table F7: Beliefs in ancestors and fertility in contemporary Benin

|                   | Number of children ever born |                     |                     |                     |                     |                     |
|-------------------|------------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                   | (1)                          | (2)                 | (3)                 | (4)                 | (5)                 | (6)                 |
| Ancestral beliefs | 1.848***<br>(0.379)          | 1.743***<br>(0.358) | 1.249***<br>(0.311) | 1.460***<br>(0.388) | 1.369***<br>(0.365) | 1.251***<br>(0.371) |
| Baseline controls | No                           | Yes                 | Yes                 | No                  | Yes                 | Yes                 |
| Extended controls | No                           | No                  | Yes                 | No                  | No                  | Yes                 |
| Religion FE       | No                           | No                  | No                  | Yes                 | Yes                 | Yes                 |
| Age FE            | No                           | Yes                 | Yes                 | No                  | Yes                 | Yes                 |
| Mean Y            | 7.995                        | 8.016               | 8.016               | 8.012               | 8.029               | 8.029               |
| N                 | 846                          | 832                 | 832                 | 841                 | 830                 | 830                 |

NOTE. Data: First-hand data collected in southern Benin. The table reports OLS estimates. The outcome variable is the total number of children ever born. "Ancestral beliefs" is a dummy variable that equals one if the respondent answers *yes* to the question *Do you believe that the spirits of your ancestors influence the events of your life?*. Baseline controls include whether the household has electricity and education. Extended controls include the number of agricultural fields, whether the household owns a TV, whether the household has been part of a program to encourage the production of pineapple, and whether the respondent has received pressure from the extended family to have children. Age fixed effects included. Religion fixed effects includes five categories: Vodoun, Roman Catholic, Evangelic, Celeste and other. Robust standard errors in parenthesis. \*\*\* for  $p < 0.01$ , \*\* for  $p < 0.05$ , \* for  $p < 0.1$ .

#### F.7. Partible vs Impartible inheritance

Differences in inheritance rules may have an influence on fertility behavior. For example, impartible inheritance (where a single heir receives the full amount of the family's land) has been shown to increase fertility rates (when comparing with partible inheritance rules) both in Europe (Gay et al., 2023) and in sub-Saharan Africa (Fontenay et al., 2024). In Benin, there are ethnic groups with both partible (Yoruba) and impartible inheritance (Fon and Adja), so it is an interesting setting to test whether the positive relationship between ancestral beliefs and fertility is affected by differences in inheritance rules. In fact, impartible inheritance may be correlated with ancestral beliefs. For example, Chu (1991) has shown that impartible inheritance (e.g., primogeniture) may emerge in traditional societies as the family's head optimal policy when the objective function is the continuation of the family line, which as I have shown is particularly important in societies with strong ancestral beliefs. Therefore, impartible inheritance could be a mediator in the relationship between ancestral beliefs and fertility behaviour.

I conduct two different tests to rule out this possibility. First, as impartible inheritance is defined at the ethnic group level, I show in Table N4 that my results remain unchanged when I include ethnicity fixed effects. Second, I test the correlation between ancestral beliefs, impartible inheritance and fertility (Table F9). I show that there is positive cor-

relation between ancestral beliefs and impartible inheritance (column 1), pointing in the direction of Chu (1991)'s hypothesis, namely that impartible inheritance may emerge in places where the motive to continue the family line plays an central role. However, there is not significant relationship between impartible inheritance and fertility (see column 2 without any control variables and column 3 after including some control variables). Finally, when I include both ancestral beliefs and impartible inheritance, only ancestral beliefs matter for fertility behavior. Overall, these results suggest that, although impartible inheritance and the motive to continue the family line may be related, the positive influence on fertility is mostly due to the later.

Table F8: Beliefs in ancestors and fertility in contemporary Benin

|                   | Number of children ever born |                     |                     |                     |                     |                     |
|-------------------|------------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                   | (1)                          | (2)                 | (3)                 | (4)                 | (5)                 | (6)                 |
| Ancestral beliefs | 1.460***<br>(0.388)          | 1.267***<br>(0.350) | 1.152***<br>(0.355) | 1.326***<br>(0.394) | 1.117***<br>(0.355) | 1.043***<br>(0.359) |
| Baseline controls | No                           | Yes                 | Yes                 | No                  | Yes                 | Yes                 |
| Extended controls | No                           | No                  | Yes                 | No                  | No                  | Yes                 |
| Religion FE       | Yes                          | Yes                 | Yes                 | Yes                 | Yes                 | Yes                 |
| Ethnicity FE      | No                           | No                  | No                  | Yes                 | Yes                 | Yes                 |
| Mean Y            | 8.012                        | 8.029               | 8.029               | 8.012               | 8.029               | 8.029               |
| N                 | 841                          | 830                 | 830                 | 841                 | 830                 | 830                 |

NOTE. Data: First-hand data collected in southern Benin. The table reports OLS estimates. The outcome variable is the total number of children ever born. "Ancestral beliefs" is a dummy variable that equals one if the respondent answers *yes* to the question *Do you believe that the spirits of your ancestors influence the events of your life?*. Baseline controls include age, age squared, whether the household has electricity and education. Extended controls include the number of agricultural fields, whether the household owns a TV, whether the household has been part of a program to encourage the production of pineapple, and whether the respondent has received pressure from the extended family to have children. Religion fixed effects includes five categories: Vodoun, Roman Catholic, Evangelic, Celeste and other. Robust standard errors in parenthesis. \*\*\* for  $p < 0.01$ , \*\* for  $p < 0.05$ , \* for  $p < 0.1$ .

Table F9: Beliefs in ancestors, impartible inheritance, and fertility

|                        | Impartible inheritance | Number of children ever born |                  |                     |                     |
|------------------------|------------------------|------------------------------|------------------|---------------------|---------------------|
|                        | (1)                    | (2)                          | (3)              | (4)                 | (5)                 |
| Ancestral beliefs      | 0.0556***<br>(0.0181)  |                              |                  | 1.811***<br>(0.411) | 1.262***<br>(0.387) |
| Impartible inheritance |                        | 0.911<br>(0.594)             | 0.215<br>(0.583) | 0.407<br>(0.590)    | -0.106<br>(0.587)   |
| Influence family       |                        |                              |                  | 1.039***<br>(0.399) | 0.982***<br>(0.371) |
| Controls               | No                     | No                           | Yes              | No                  | Yes                 |
| Mean Y                 | 0.926                  | 8.037                        | 8.072            | 8.037               | 8.072               |
| N                      | 767                    | 767                          | 754              | 767                 | 754                 |

NOTE. Data: First-hand data collected in southern Benin. The table reports OLS estimates. The outcome variable is whether the ethnic group practiced impartible inheritance in column (1), and the total number of children ever born in columns (2), (3), (4) and (5). "Ancestral beliefs" is a dummy variable that equals one if the respondent believes that the spirits of his ancestors influence his life. "Impartible inheritance" equals one if the respondent's ethnic group practiced impartible inheritance. "Influence family" equals one if the respondent has experienced pressure from his extended family to have children. Controls include age, whether the household has electricity, education, number of agricultural fields, whether the household owns a TV, and whether the household has been part of a program to encourage the production of pineapple. Robust standard errors in parenthesis. \*\*\* for  $p < 0.01$ , \*\* for  $p < 0.05$ , \* for  $p < 0.1$ .

#### F.8. Old-age security motive for fertility

Ancestor worship could in part be interpreted as an extension of the well-known "old-age security motive", whereby people's needs for old-age support raise the demand for children (Lambert and Rossi, 2016; Rossi and Godard, 2022; Sage, 2023). In fact, in societies where the continuation of the family line is of central importance due to the influence of ancestors, the question of "security" has two different facets (Goody, 1973). In addition to the standard security in old age, there is the security in the after-life which drives the demand for children in the same way as the former. Ancestors would need their descendants to sustain themselves in the afterlife as much as the living need them in old age. It is therefore important to distinguish between security in old age and security in the after-life. I use detailed field-level information on property rights to show that my results are not affected when I control for the quantity of owned (and rented) land by women (both in terms of area in squared meters and of number). Land ownership should be negatively correlated with fertility when the old-age security motive for fertility is important (in my sample, the median number of owned fields by women is one, and the median area of owned fields is 800 squared meters).

Table F10: Ancestors, land ownership and female fertility

|                               | Number of children ever born |                     |                     |                     |
|-------------------------------|------------------------------|---------------------|---------------------|---------------------|
|                               | (1)                          | (2)                 | (3)                 | (4)                 |
| Men + Women ancestral beliefs | 0.698***<br>(0.179)          | 0.694***<br>(0.178) | 0.714***<br>(0.179) | 0.693***<br>(0.178) |
| Controls                      | Yes                          | Yes                 | Yes                 | Yes                 |
| Area fields                   | No                           | Yes                 | No                  | Yes                 |
| Number fields                 | No                           | No                  | Yes                 | No                  |
| Religion FE                   | Yes                          | Yes                 | Yes                 | Yes                 |
| Mean Y                        | 5.352                        | 5.352               | 5.352               | 5.352               |
| N                             | 840                          | 840                 | 840                 | 840                 |

NOTE. Data: First-hand data collected in southern Benin. The sample is restricted to women. The table reports OLS estimates. The outcome variable is the total number of children ever born. "Men + Women beliefs" is a dummy variable that equals one if both the husband and the wife answer *yes* to the question *Do you believe that the spirits of your ancestors influence the events of your life?*. Column 1 does not include any control related to land. Control 2 controls for the total area of all owned and rented fields. Column 3 controls for the total number of owned and rented fields. Column 4 control for the total area of all fields. Controls include age, age squared, whether the household has electricity, education, whether the household owns a TV, and whether the respondent has received pressure from the extended family to have children. Religion fixed effects includes five categories: Vodoun, Roman Catholic, Evangelic, Celeste and other. Robust standard errors in parenthesis. \*\*\* for  $p < 0.01$ , \*\* for  $p < 0.05$ , \* for  $p < 0.1$ .

#### F.9. Using selection on observables to assess the bias from unobservables

I follow Oster (2019)'s methodology to further reduce concerns related to omitted variable bias. This strategy consists in identifying the degree of omitted variable bias by exploring the sensitivity of treatment effects to the inclusion of observed controls, taking into account movements in the R-squared when the controls are introduced. Two assumptions are required. First, a value for the relative degree of selection on observed and unobserved variables is needed ( $\delta$ ). Second, a value for the R-squared from a hypothetical regression of the outcome on treatment and both observed and unobserved controls ( $R_{\max}$ ).

Regarding  $\delta$ , Oster (2019) suggests equal selection ( $\delta = 1$ ) as an appropriate upper bound on  $\delta$ . A value of  $\delta = 1$  suggests that observables are at least as important as the unobservables. This assumptions seems likely to hold since observed variables are (usually) the factors that are most important.

Regarding  $R_{\max}$ , different bounds that are a function of the R-squared from the regression with observed controls ( $R'$ ) can be chosen. Based on a sample of randomized articles, Oster (2019) suggests  $R_{\max} = 1.3 \times R'$ , which would allow about 90% of ran-



domized results to survive (but only 45% of nonrandomized results).<sup>70</sup>

Table F11 shows that the results are extremely unlikely to be fully explained by unobservables. Column 1 shows that, when looking at the impact on men, we would need that a 310% of the observed sorting on observables applied to unobservables to make the coefficient of ancestral beliefs equal to 0. Since observables include variables that are considered as highly important to explain fertility (i.e., age, education, religion, wealth...), this situation seems unlikely. Even more striking are the results in column 2 where we look at female fertility. Now, a 2352% of the observed sorting on observables applied to unobservables would be needed to make the coefficient of ancestral beliefs (in this case, of both husband and wife) equal to 0.

Table F11: Importance of bias from unobserved variables

|                                                   | $\beta$ Ancestral Beliefs |       |
|---------------------------------------------------|---------------------------|-------|
|                                                   | Men                       | Women |
| Ancestral beliefs only (no controls)              | 1.89                      | 0.72  |
| Ancestral beliefs (Full model)                    | 0.95                      | 0.69  |
| $R_{max} = 1.3 \times R'$ and $\delta=1$          | 0.65                      | 0.66  |
| Adjusted $\beta$ set to zero: $\delta$ solved for | 3.10                      | 23.52 |

NOTE. The table shows the unadjusted  $\beta$  (coefficient associated with ancestral beliefs) from the model without controls (row 1), from the full model (row 2, controls include age, age squared, electricity, education, TV, number of fields, pressure from extended family to have children, religion, and marital status). Row 3 assumes  $R_{max} = 1.3 \times R'$  and  $\delta=1$  and shows the adjusted beta after accounting for unobservables. Row 4 indicates that, if 310% (column 1) of the observed sorting on covariates in the full model applied to unobservables there would be no effect of ancestral beliefs i.e., the ancestral beliefs coefficient would be zero. This figure increases to 2352% in column 2.

<sup>70</sup>There are good reasons to use thresholds below 1. For example, there might be measurement error in the dependent variable, as it is probably the case here.

## G. Democratic Republic of Congo: Robustness

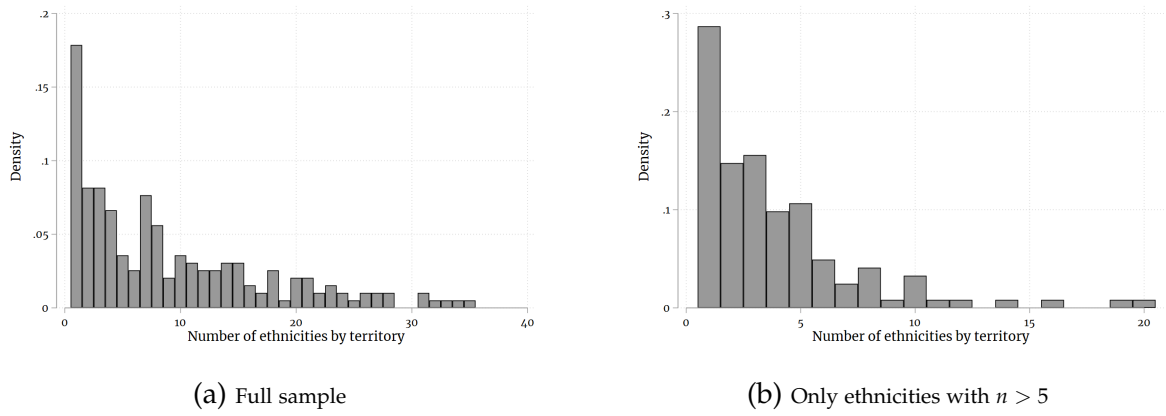
### G.1. *Variation: ethnicities by territory*

A potential concern with the specification that includes fixed effects for territory of birth and city of residence in the urban sample is that there could be almost no variation in the number of people of different ethnicities born in the same territory, and thus the estimates could be driven by very few observations. First, it is important to note that territories are large administrative units.<sup>71</sup> Second, migration on the eve of and immediately after independence was very important in the DRC, and therefore the number of second generation migrants from different territories (and possibly with a different ethnicity) was potentially very high. In fact, about 90% of the sample was born in a rural area and migrated during the last years of the colonial era or after independence. Third, the classification of ethnicities (or, as defined in Vansina (1966), tribes) that I use in this paper is granular: I have about 220 different ethnicities in the urban sample, and the maps of ethnic boundaries drawn by Vansina (1966) generally depict small polygons that can lie within a single territory (see Figure B1a). Figure G1, panel G1a, shows the distribution of the number of ethnicities by territory of birth: it looks at the ethnicity of each person born in a given territory and counts the number of different ethnicities (it does not look at the number of ancestral ethnic boundaries that lie within a territory). However, there may be many ethnicities because only 1 or 2 people born in the territory are of the same ethnicity. Panel G1b shows the distribution of the number of ethnicities by territory of birth, considering only ethnicities with at least five people.

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<sup>71</sup>Territories in the 1970s surveys do not correspond exactly to territories today: while I have about 200 territories (including cities) in my sample, there are 145 territories and 35 large cities today.

Figure G1: Variation in the number of ethnicities by territory of birth



Notes: Panel G1a, shows the distribution of the number of ethnicities by territory of birth: it looks at the ethnicity of each person born in a given territory and counts the number of different ethnicities. Panel G1b shows the distribution of the number of ethnicities by territory of birth, considering only ethnicities with at least five people.

## G.2. Age-specific fertility rates

Using birth calendars, I reconstruct the total number of births women had at ages 35, 40, and 45. The results are reported in Table G1. While fertility is more likely to be completed at older ages, the sample size becomes smaller because there are fewer women in the older age categories. I find a positive effect of ancestor worship on age-specific fertility, ranging from an average effect of 18% of the full sample mean at age 35 to 21% at age 40 and 26% at age 45. Here, I observe an stronger effect on the rural sample (again, note that I include city of residence FE when using the urban sample and region FE when I use the EDOZA sample, which can explain – at least partially – the stronger effect on the rural sample. I find that ancestor worship increases the number of births at age 30 by 27% in the EDOZA sample and by 15% in the urban sample. At age 40, the effect moves up to 31% of the EDOZA sample mean and to 18% in the urban sample. Finally, at age 45, ancestor worship is associated with an increase of 36% of the sample mean in the EDOZA survey and of 23% in the urban sample.

Table G1: Ancestor worship and age-specific fertility in the DRC

|                     | Births at 35        |                     |                     | Births at 40        |                     |                     | Births at 45        |                     |                     |
|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                     | (1)                 | (2)                 | (3)                 | (4)                 | (5)                 | (6)                 | (7)                 | (8)                 | (9)                 |
| Ancestral beliefs   | 0.767***<br>(0.172) | 1.139***<br>(0.349) | 0.692***<br>(0.165) | 0.984***<br>(0.229) | 1.358***<br>(0.403) | 0.902***<br>(0.250) | 1.193***<br>(0.244) | 1.583***<br>(0.402) | 1.113***<br>(0.259) |
| Controls            | Yes                 | Yes                 | Yes                 | Yes                 | Yes                 | Yes                 | Yes                 | Yes                 | Yes                 |
| FE                  | Yes                 | Yes                 | Yes                 | Yes                 | Yes                 | Yes                 | Yes                 | Yes                 | Yes                 |
| City x Territory FE | No                  | No                  | No                  | Yes                 | No                  | No                  | No                  | Yes                 | Yes                 |
| All                 | Yes                 | No                  | No                  | Yes                 | No                  | No                  | Yes                 | No                  | No                  |
| Edoza               | No                  | Yes                 | No                  | No                  | Yes                 | No                  | No                  | Yes                 | No                  |
| Urban               | No                  | No                  | Yes                 | No                  | No                  | Yes                 | No                  | No                  | Yes                 |
| Mean Y              | 4.568               | 4.243               | 4.746               | 4.862               | 4.580               | 5.055               | 4.710               | 4.489               | 4.890               |
| R-squared           | 0.246               | 0.256               | 0.274               | 0.244               | 0.267               | 0.267               | 0.213               | 0.252               | 0.226               |
| N                   | 18997               | 6730                | 12267               | 13447               | 5466                | 7981                | 9678                | 4337                | 5341                |

NOTE. Data: Demographic Survey of 1970s and EDOZA. The table reports OLS estimates. The outcome variables are the total number of births a women had before age 35 (columns 1, 2 and 3), 40 (columns 4, 5 and 6) and 45 (columns 7, 8 and 9). "Ancestor worship" is a dummy variable that equals one if the ethnic group  $e$  practice ancestor worship. Controls include age, whether the place of birth was urban or rural, whether the respondent has primary education, dummy equal to one if migrant, dummy equal to one if the father's respondent is alive, dummy equal to one if the respondent is working, dummy equal to one if the respondent is a farmer (only available in the urban sample), number of household members, and year of installation in the current city. City/region fixed effects include city of residence in the case of the urban sample and region of residence in the case of EDOZA. Standard errors clustered at the Vansina's ethnic group-level in parenthesis. \*\*\* for  $p < 0.01$ , \*\* for  $p < 0.05$ , \* for  $p < 0.1$ .

### G.3. Intensive vs. Extensive margin and estimation of Poisson model

Table G2: Ancestor worship and fertility in the DRC

|                     | Number of children ever born |                      |                     |                      |                      |                      |                      |                       |
|---------------------|------------------------------|----------------------|---------------------|----------------------|----------------------|----------------------|----------------------|-----------------------|
|                     | Full sample                  |                      | EDOZA               |                      | Urban sample         |                      |                      |                       |
|                     | All                          | +30                  | All                 | +30                  | All                  | +30                  | All                  | +30                   |
| Ancestor worship    | 0.136***<br>(0.0338)         | 0.188***<br>(0.0434) | 0.154**<br>(0.0654) | 0.218***<br>(0.0731) | 0.135***<br>(0.0249) | 0.186***<br>(0.0358) | 0.0451**<br>(0.0200) | 0.0829***<br>(0.0263) |
| Controls            | Yes                          | Yes                  | Yes                 | Yes                  | Yes                  | Yes                  | Yes                  | Yes                   |
| FE                  | Yes                          | Yes                  | Yes                 | Yes                  | Yes                  | Yes                  | Yes                  | Yes                   |
| City x Territory FE | No                           | No                   | No                  | No                   | No                   | No                   | Yes                  | Yes                   |
| Mean Y              | 2.883                        | 5.322                | 3.546               | 4.866                | 2.703                | 5.542                | 2.702                | 5.538                 |
| N                   | 60064                        | 23194                | 12825               | 7551                 | 47239                | 15643                | 47014                | 15489                 |

NOTE. Data: Demographic Survey of 1970s and EDOZA. Columns 1 and 2 combine both the Urban Demographic survey and EDOZA. Columns 3 and 4 use only the EDOZA sample. Columns 5-8 use only the urban sample. In columns 2, 4, 6 and 8, the sample is restricted to women older than 30 years old. The table reports Pseudo-Poisson Maximum Likelihood (PPML) estimators. The outcome variable is the total number of children ever born. "Ancestor worship" is a dummy variable that equals one if the ethnic group  $e$  practice ancestor worship. Controls include age, age squared, whether the place of birth was urban or rural, whether the respondent has primary education, dummy equal to one if migrant, dummy equal to one if the father's respondent is alive, dummy equal to one if the respondent is working, dummy equal to one if the respondent is a farmer (only available in the urban sample), number of household members, and year of installation in the current city. City/region fixed effects include city of residence in the case of the urban sample and region of residence in the case of EDOZA. Standard errors clustered at the Vansina's ethnic group-level in parenthesis. \*\*\* for  $p < 0.01$ , \*\* for  $p < 0.05$ , \* for  $p < 0.1$ .

Table G3: Ancestor worship and fertility in the DRC

|            | Full sample           |                           | EDOZA                 |                           | Urban sample          |                           |
|------------|-----------------------|---------------------------|-----------------------|---------------------------|-----------------------|---------------------------|
|            | Has children<br>(1)   | Number of children<br>(2) | Has children<br>(3)   | Number of children<br>(4) | Has children<br>(5)   | Number of children<br>(6) |
| Ancestors  | 0.0701***<br>(0.0127) | 0.585***<br>(0.182)       | 0.0820***<br>(0.0173) | 0.771**<br>(0.345)        | 0.0766***<br>(0.0146) | 0.509***<br>(0.133)       |
| Controls   | Yes                   | Yes                       | Yes                   | Yes                       | Yes                   | Yes                       |
| FE         | Yes                   | Yes                       | Yes                   | Yes                       | Yes                   | Yes                       |
| Ext margin | Yes                   | No                        | Yes                   | No                        | Yes                   | No                        |
| Mean Y     | 0.882                 | 6.037                     | 0.840                 | 5.793                     | 0.902                 | 6.146                     |
| N          | 23194                 | 20448                     | 7551                  | 6342                      | 15643                 | 14106                     |

NOTE. Data: Demographic Survey of 1970s and EDOZA. The sample is restricted to women older than 30 years old. Columns 1 and 2 combine both the Urban Demographic survey and EDOZA. Columns 3 and 4 use only the EDOZA sample. Columns 5-6 use only the urban sample. Columns 1, 3, and 5 look at the extensive margin (probability of having children), while columns 2, 4, and 6 look at the intensive margin (total number of children ever born conditioned on having children). The table reports OLS estimates. "Ancestor worship" is a dummy variable that equals one if the ethnic group  $e$  practice ancestor worship. Controls include age, age squared, whether the place of birth was urban or rural, whether the respondent has primary education, dummy equal to one if migrant, dummy equal to one if the father's respondent is alive, dummy equal to one if the respondent is working, dummy equal to one if the respondent is a farmer (only available in the urban sample), number of household members, and year of installation in the current city. City/region fixed effects include city of residence in the case of the urban sample and region of residence in the case of EDOZA. Standard errors clustered at the Vansina's ethnic group-level in parenthesis. \*\*\* for  $p < 0.01$ , \*\* for  $p < 0.05$ , \* for  $p < 0.1$ .

#### G.4. Unobserved human capital

Table G4: Ancestor worship and unobserved human capital

|                       | Pseudo-Poisson Maximul Likelihood | OLS estimates             |                     |
|-----------------------|-----------------------------------|---------------------------|---------------------|
|                       | Total wage<br>(1)                 | Log(1+ total wage)<br>(2) | Total wage<br>(3)   |
| Ancestor Worship      | -0.107<br>(0.208)                 | 0.236<br>(0.305)          | -949.7<br>(1315.8)  |
| Years of education    | 0.161***<br>(0.0144)              | 0.297***<br>(0.0353)      | 628.8***<br>(92.69) |
| City FE               | Yes                               | Yes                       | Yes                 |
| Territory of birth FE | Yes                               | Yes                       | Yes                 |
| Controls              | Yes                               | Yes                       | Yes                 |
| Mean Y                | 4047.8                            | 5.411                     | 4045.2              |
| R-squared             |                                   | 0.234                     | 0.191               |
| N                     | 15708                             | 15718                     | 15718               |

NOTE. Data: Budgetary Survey of the 1970s and Vansina(1966). Column 1 reports Pseudo-Poisson Maximul Likelihood estimates, and the dependent variable is the total wage. Column 2 reports OLS estimates, and the dependent variable has been transformed to Log(1+wage). Finally, column 3 reports OLS estimates using total wage as dependent variable. Ancestor worship is a dummy variable that equals one if the ethnic group  $e$  of individual  $i$  traditionally practices ancestor worship. Controls include age, age squared, total number of household members, gender, whether respondents were born in urban/rural area, education, dummy equal to one if migrant, year of installation in the current city, dummy equal to one if the father's respondent is alive, and dummy equal to one if the respondent is a farmer. Standard errors () are clustered at the ethnic group level. \*\*\* for  $p < 0.01$ , \*\* for  $p < 0.05$ , \* for  $p < 0.1$ .

#### *G.5. Exposure to missionary presence*

I construct a measure of exposure to all Christian missions opened in the DRC between 1885 and 1948 for all individuals born before 1948 using the West Zaire Demographic surveys and data on missions from Guirkinger and Villar (2022) and Alvarez-Aragon et al. (2023). In the historical individual-level data, information on the territory of birth of each individual is available, so we can construct the exposure to missionary presence of each territory in each year and assign this measure to each respondent according to their year of birth. To do this, I follow Guirkinger and Villar (2022) and Alvarez-Aragon et al. (2023) and construct a continuous measure of proximity at the territory level that controls for mission density. Specifically, I generate 1000 random points within each territory and compute the distance from each random point to its closest mission before averaging over these distances. This process is repeated for each territory, each year between 1885 and 1948, and three types of missions: Catholic, Catholic with nuns, and Protestant. When showing my results, I report negative (log) distances to have an easier interpretation.

As we can observe in Table G5, the coefficients barely move when I control for exposure to missionary presence. The magnitude of the effects remain high. Ancestor worship is associated with an increase of 17% in the number of births at age 35 (using the full sample), of 20% at age 40, and of 25% at age 45. Moreover, I am able to replicate the main findings in Guirkinger and Villar (2022). Exposure at birth to Catholic missions with nuns is associated with higher fertility rates, while exposure to Protestant missionaries generally reduces fertility.

Table G5: Ancestor worship and age-specific fertility in the DRC

|                     | Births at 35        |                     |                      | Births at 40        |                     |                     | Births at 45        |                     |                      |
|---------------------|---------------------|---------------------|----------------------|---------------------|---------------------|---------------------|---------------------|---------------------|----------------------|
|                     | (1)                 | (2)                 | (3)                  | (4)                 | (5)                 | (6)                 | (7)                 | (8)                 | (9)                  |
| Ancestral beliefs   | 0.736***<br>(0.170) | 1.011***<br>(0.359) | 0.641***<br>(0.160)  | 0.943***<br>(0.221) | 1.209***<br>(0.413) | 0.839***<br>(0.233) | 1.133***<br>(0.242) | 1.432***<br>(0.399) | 0.984***<br>(0.233)  |
| Exp to Catholics    | -0.112<br>(0.191)   | -0.153<br>(0.344)   | 0.0199<br>(0.120)    | -0.0308<br>(0.243)  | -0.0971<br>(0.445)  | 0.0520<br>(0.142)   | 0.0574<br>(0.208)   | -0.151<br>(0.393)   | 0.282**<br>(0.130)   |
| Exp to Cath nuns    | 0.234**<br>(0.106)  | 0.484**<br>(0.186)  | 0.281**<br>(0.134)   | 0.254**<br>(0.125)  | 0.548***<br>(0.191) | 0.362**<br>(0.155)  | 0.271*<br>(0.140)   | 0.480**<br>(0.208)  | 0.407***<br>(0.140)  |
| Exp to Protestants  | 0.0154<br>(0.127)   | 0.230<br>(0.288)    | -0.121**<br>(0.0587) | -0.0489<br>(0.164)  | 0.169<br>(0.371)    | -0.202*<br>(0.103)  | -0.0799<br>(0.212)  | 0.247<br>(0.433)    | -0.365***<br>(0.115) |
| Controls            | Yes                 | Yes                 | Yes                  | Yes                 | Yes                 | Yes                 | Yes                 | Yes                 | Yes                  |
| FE                  | Yes                 | Yes                 | Yes                  | Yes                 | Yes                 | Yes                 | Yes                 | Yes                 | Yes                  |
| City x Territory FE | No                  | No                  | No                   | Yes                 | No                  | No                  | No                  | Yes                 | Yes                  |
| All                 | Yes                 | No                  | No                   | Yes                 | No                  | No                  | Yes                 | No                  | No                   |
| Edoza               | No                  | Yes                 | No                   | No                  | Yes                 | No                  | No                  | Yes                 | No                   |
| Urban               | No                  | No                  | Yes                  | No                  | No                  | Yes                 | No                  | No                  | Yes                  |
| Mean Y              | 4.575               | 4.244               | 4.758                | 4.871               | 4.581               | 5.071               | 4.723               | 4.488               | 4.915                |
| R-squared           | 0.249               | 0.262               | 0.273                | 0.247               | 0.274               | 0.267               | 0.218               | 0.259               | 0.229                |
| N                   | 18852               | 6698                | 12154                | 13328               | 5435                | 7893                | 9579                | 4306                | 5273                 |

NOTE. Data: Demographic Survey of 1970s and EDOZA. The table reports OLS estimates. The outcome variables are the total number of births a women had before age 35 (columns 1, 2 and 3), 40 (columns 4, 5 and 6) and 45 (columns 7, 8 and 9). "Ancestor worship" is a dummy variable that equals one if the ethnic group  $e$  practice ancestor worship. Details on how exposure to missions is constructed can be found in ???. Controls include age, whether the place of birth was urban or rural, whether the respondent has primary education, dummy equal to one if migrant, dummy equal to one if the father's respondent is alive, dummy equal to one if the respondent is working, dummy equal to one if the respondent is a farmer (only available in the urban sample), number of household members, and year of installation in the current city. City/region fixed effects include city of residence in the case of the urban sample and region of residence in the case of EDOZA. Standard errors clustered at the Vansina's ethnic group-level in parenthesis. \*\*\* for  $p < 0.01$ , \*\* for  $p < 0.05$ , \* for  $p < 0.1$ .

## H. Ancestral beliefs and contemporary fertility: robustness

### H.1. Alternative definitions of "ancestral beliefs"

Table H1: Ancestor worship and fertility in sub-Saharan Africa

|                      | Number of children   |                      |                      |                      | P( $\geq 5$ children)  |                        |                        |                      |
|----------------------|----------------------|----------------------|----------------------|----------------------|------------------------|------------------------|------------------------|----------------------|
|                      | (1)                  | (2)                  | (3)                  | (4)                  | (5)                    | (6)                    | (7)                    | (8)                  |
| Sacrifices Ancestors | 0.332***<br>(0.0370) | 0.167***<br>(0.0294) | 0.149***<br>(0.0294) | 0.102***<br>(0.0286) | 0.0265***<br>(0.00478) | 0.0117***<br>(0.00442) | 0.00875**<br>(0.00438) | 0.00563<br>(0.00446) |
| Country FE           | Yes                  | Yes                  | Yes                  | Yes                  | Yes                    | Yes                    | Yes                    | Yes                  |
| Controls             | No                   | Yes                  | Yes                  | Yes                  | No                     | Yes                    | Yes                    | Yes                  |
| Ext Controls 1       | No                   | No                   | Yes                  | Yes                  | No                     | No                     | Yes                    | Yes                  |
| Ext Controls 2       | No                   | No                   | No                   | Yes                  | No                     | No                     | No                     | Yes                  |
| Mean Y               | 2.101                | 2.095                | 2.095                | 2.103                | 0.0978                 | 0.0974                 | 0.0974                 | 0.0973               |
| R-squared            | 0.0606               | 0.418                | 0.425                | 0.481                | 0.0426                 | 0.174                  | 0.180                  | 0.188                |
| N                    | 22243                | 22115                | 22115                | 21361                | 22243                  | 22115                  | 22115                  | 21361                |

NOTE. Data: PEW research forum 2008-2009 Survey. The outcome variables are defined as follows: in columns (1)-(4), it is the total number of children, and in columns (5)-(8) it is the probability of having at least 5 children. The explanatory variable is an indicator equal to one if respondent answers "yes" to the following question: "do you believe that sacrifices to spirits or ancestors can protect you from bad things happening?". Basic controls include age, age<sup>2</sup> and sex. Extended controls 1 also include whether the individual lives in a urban or rural area and whether he/she is christian/muslim or from other religion. Finally, extended controls 2 also include a dummy variable that equals one if the respondent has completed primary education, a dummy that equals one if the respondent is married, a dummy variable that equals one if the respondent finds himself/herself in a good economic situation, and a dummy equal to one if the respondent did not have money at some point during the last year to buy food for his/her family. Robust standard errors in parenthesis. \*\*\* for  $p < 0.01$ , \*\* for  $p < 0.05$ , \* for  $p < 0.1$ .

Table H2: Ancestor worship and fertility in sub-Saharan Africa

|                     | Number of children   |                     |                       |                      | P( $\geq 5$ children)  |                      |                       |                      |
|---------------------|----------------------|---------------------|-----------------------|----------------------|------------------------|----------------------|-----------------------|----------------------|
|                     | (1)                  | (2)                 | (3)                   | (4)                  | (5)                    | (6)                  | (7)                   | (8)                  |
| Reincarnation       | 0.118***<br>(0.0334) | 0.0503*<br>(0.0263) | 0.0775***<br>(0.0262) | 0.0632**<br>(0.0253) | 0.00971**<br>(0.00426) | 0.00325<br>(0.00395) | 0.00649*<br>(0.00395) | 0.00577<br>(0.00400) |
| Country FE          | Yes                  | Yes                 | Yes                   | Yes                  | Yes                    | Yes                  | Yes                   | Yes                  |
| Controls            | No                   | Yes                 | Yes                   | Yes                  | No                     | Yes                  | Yes                   | Yes                  |
| Extended Controls 1 | No                   | No                  | Yes                   | Yes                  | No                     | No                   | Yes                   | Yes                  |
| Extended Controls 2 | No                   | No                  | No                    | Yes                  | No                     | No                   | No                    | Yes                  |
| Mean Y              | 2.087                | 2.081               | 2.081                 | 2.087                | 0.0966                 | 0.0961               | 0.0961                | 0.0959               |
| R-squared           | 0.0560               | 0.418               | 0.425                 | 0.482                | 0.0390                 | 0.173                | 0.178                 | 0.187                |
| N                   | 21134                | 21010               | 21010                 | 20328                | 21134                  | 21010                | 21010                 | 20328                |

NOTE. Data: PEW research forum 2008-2009 Survey. The outcome variables are defined as follows: in columns (1)-(4), it is the total number of children, and in columns (5)-(8) it is the probability of having at least 5 children. The explanatory variable is an indicator equal to one if respondent answers "yes" to the following question: "Which, if any, of the following do you believe in? Reincarnation.". Basic controls include age, age<sup>2</sup> and sex. Extended controls 1 also include whether the individual lives in a urban or rural area and whether he/she is christian/muslim or from other religion. Finally, extended controls 2 also include a dummy variable that equals one if the respondent has completed primary education, a dummy that equals one if the respondent is married, a dummy variable that equals one if the respondent finds himself/herself in a good economic situation, and a dummy equal to one if the respondent did not have money at some point during the last year to buy food for his/her family. Robust standard errors in parenthesis. \*\*\* for  $p < 0.01$ , \*\* for  $p < 0.05$ , \* for  $p < 0.1$ .



Table H3: Ancestor worship and fertility in sub-Saharan Africa

|                          | Number of children   |                      |                       |                       | P( $\geq 5$ children)  |                        |                        |                       |
|--------------------------|----------------------|----------------------|-----------------------|-----------------------|------------------------|------------------------|------------------------|-----------------------|
|                          | (1)                  | (2)                  | (3)                   | (4)                   | (5)                    | (6)                    | (7)                    | (8)                   |
| Knowledge trad. religion | 0.314***<br>(0.0352) | 0.102***<br>(0.0281) | 0.0965***<br>(0.0281) | 0.0748***<br>(0.0272) | 0.0293***<br>(0.00452) | 0.0128***<br>(0.00424) | 0.0123***<br>(0.00420) | 0.0107**<br>(0.00426) |
| Country FE               | Yes                  | Yes                  | Yes                   | Yes                   | Yes                    | Yes                    | Yes                    | Yes                   |
| Basic Controls           | No                   | Yes                  | Yes                   | Yes                   | No                     | Yes                    | Yes                    | Yes                   |
| Ext Controls 1           | No                   | No                   | Yes                   | Yes                   | No                     | No                     | Yes                    | Yes                   |
| Ext Controls 2           | No                   | No                   | No                    | Yes                   | No                     | No                     | No                     | Yes                   |
| Mean Y                   | 2.078                | 2.073                | 2.073                 | 2.079                 | 0.0965                 | 0.0960                 | 0.0960                 | 0.0958                |
| R-squared                | 0.0597               | 0.419                | 0.426                 | 0.483                 | 0.0427                 | 0.173                  | 0.178                  | 0.186                 |
| N                        | 21783                | 21664                | 21664                 | 20917                 | 21783                  | 21664                  | 21664                  | 20917                 |

NOTE. Data: PEW research forum 2008-2009 Survey. The outcome variables are defined as follows: in columns (1)-(4), it is the total number of children, and in columns (5)-(8) it is the probability of having at least 5 children. The explanatory variable is an indicator equal to one if the respondent answers "a great deal" or "some" to the question "How much would you say you know about ancestral, tribal, animist, or other traditional African religions?". Basic controls include age, age<sup>2</sup> and sex. Extended controls 1 also include whether the individual lives in a urban or rural area and whether he/she is christian/muslim or from other religion. Finally, extended controls 2 also include a dummy variable that equals one if the respondent has completed primary education, a dummy that equals one if the respondent is married, a dummy variable that equals one if the respondent finds himself/herself in a good economic situation, and a dummy equal to one if the respondent did not have money at some point during the last year to buy food for his/her family. Robust standard errors in parenthesis. \*\*\* for  $p < 0.01$ , \*\* for  $p < 0.05$ , \* for  $p < 0.1$ .

## H.2. Additional controls: region fixed effects and clustered standard errors

Table H4: Ancestor worship and fertility in sub-Saharan Africa

|                     | Number of children   |                      |                      |                      | P( $\geq 5$ children)  |                       |                       |                      |
|---------------------|----------------------|----------------------|----------------------|----------------------|------------------------|-----------------------|-----------------------|----------------------|
|                     | (1)                  | (2)                  | (3)                  | (4)                  | (5)                    | (6)                   | (7)                   | (8)                  |
| Ancestral beliefs   | 0.520***<br>(0.0407) | 0.211***<br>(0.0324) | 0.200***<br>(0.0324) | 0.143***<br>(0.0315) | 0.0369***<br>(0.00534) | 0.0119**<br>(0.00493) | 0.0103**<br>(0.00489) | 0.00713<br>(0.00497) |
| Region FE           | Yes                  | Yes                  | Yes                  | Yes                  | Yes                    | Yes                   | Yes                   | Yes                  |
| Basic Controls      | No                   | Yes                  | Yes                  | Yes                  | No                     | Yes                   | Yes                   | Yes                  |
| Extended Controls 1 | No                   | No                   | Yes                  | Yes                  | No                     | No                    | Yes                   | Yes                  |
| Extended Controls 2 | No                   | No                   | No                   | Yes                  | No                     | No                    | No                    | Yes                  |
| Mean Y              | 2.100                | 2.095                | 2.095                | 2.104                | 0.0997                 | 0.0993                | 0.0993                | 0.0993               |
| R-squared           | 0.0982               | 0.442                | 0.445                | 0.501                | 0.0684                 | 0.201                 | 0.204                 | 0.212                |
| N                   | 21980                | 21845                | 21845                | 21068                | 21980                  | 21845                 | 21845                 | 21068                |

NOTE. Data: PEW research forum 2008-2009 Survey. The outcome variables are defined as follows: in columns (1)-(4), it is the total number of children, and in columns (5)-(8) it is the probability of having at least 5 children. The explanatory variable is an indicator equal to one if the respondent answers "yes" to the question "Do you ever participate in traditional African ceremonies or perform special acts to honor or celebrate your ancestors?". Basic controls include age, age<sup>2</sup> and sex. Extended controls 1 also include whether the individual lives in a urban or rural area and whether he/she is christian/muslim or from other religion. Finally, extended controls 2 also include a dummy variable that equals one if the respondent has completed primary education, a dummy that equals one if the respondent is married, a dummy variable that equals one if the respondent finds himself/herself in a good economic situation, and a dummy equal to one if the respondent did not have money at some point during the last year to buy food for his/her family. Robust standard errors in parenthesis. \*\*\* for  $p < 0.01$ , \*\* for  $p < 0.05$ , \* for  $p < 0.1$ .

Table H5: Ancestor worship and fertility in sub-Saharan Africa

|                     | Number of children   |                      |                      |                      | P( $\geq 5$ children)  |                        |                        |                       |
|---------------------|----------------------|----------------------|----------------------|----------------------|------------------------|------------------------|------------------------|-----------------------|
|                     | (1)                  | (2)                  | (3)                  | (4)                  | (5)                    | (6)                    | (7)                    | (8)                   |
| Ancestral beliefs   | 0.517***<br>(0.0595) | 0.243***<br>(0.0379) | 0.230***<br>(0.0367) | 0.162***<br>(0.0366) | 0.0369***<br>(0.00697) | 0.0146***<br>(0.00531) | 0.0133***<br>(0.00503) | 0.00918*<br>(0.00515) |
| Country FE          | Yes                  | Yes                  | Yes                  | Yes                  | Yes                    | Yes                    | Yes                    | Yes                   |
| Controls            | No                   | Yes                  | Yes                  | Yes                  | No                     | Yes                    | Yes                    | Yes                   |
| Extended Controls 1 | No                   | No                   | Yes                  | Yes                  | No                     | No                     | Yes                    | Yes                   |
| Extended Controls 2 | No                   | No                   | No                   | Yes                  | No                     | No                     | No                     | Yes                   |
| Mean Y              | 2.100                | 2.095                | 2.095                | 2.104                | 0.0997                 | 0.0993                 | 0.0993                 | 0.0993                |
| R-squared           | 0.0648               | 0.419                | 0.425                | 0.484                | 0.0421                 | 0.178                  | 0.183                  | 0.192                 |
| N                   | 21980                | 21845                | 21845                | 21068                | 21980                  | 21845                  | 21845                  | 21068                 |

NOTE. Data: PEW research forum 2008-2009 Survey. The outcome variables are defined as follows: in columns (1)-(4), it is the total number of children, and in columns (5)-(8) it is the probability of having at least 5 children. The explanatory variable is an indicator equal to one if the respondent answers "yes" to the question "Do you ever participate in traditional African ceremonies or perform special acts to honor or celebrate your ancestors?". Basic controls include age, age<sup>2</sup> and sex. Extended controls 1 also include whether the individual lives in a urban or rural area and whether he/she is christian/muslim or from other religion. Finally, extended controls 2 also include a dummy variable that equals one if the respondent has completed primary education, a dummy that equals one if the respondent is married, a dummy variable that equals one if the respondent finds himself/herself in a good economic situation, and a dummy equal to one if the respondent did not have money at some point during the last year to buy food for his/her family. Standard errors clustered at the region level in parenthesis. \*\*\* for  $p < 0.01$ , \*\* for  $p < 0.05$ , \* for  $p < 0.1$ .

### H.3. Additional controls: religiosity and supernatural beliefs

Table H6: Ancestor worship, fertility, and religiosity

|                           | Number of children ever born |                      |                      |
|---------------------------|------------------------------|----------------------|----------------------|
|                           | (1)                          | (2)                  | (3)                  |
| Ancestral beliefs         | 0.184***<br>(0.0292)         | 0.183***<br>(0.0291) | 0.184***<br>(0.0292) |
| High religious attendance |                              | 0.138***<br>(0.0459) | 0.136***<br>(0.0461) |
| Religion is important     |                              |                      | 0.0254<br>(0.0343)   |
| Country FE                | Yes                          | Yes                  | Yes                  |
| Controls                  | Yes                          | Yes                  | Yes                  |
| Mean Y                    | 2.083                        | 2.083                | 2.083                |
| R-squared                 | 0.480                        | 0.481                | 0.481                |
| N                         | 22295                        | 22295                | 22295                |

NOTE. Data: PEW research forum 2008-2009 Survey. The outcome variables is the total number of children ever born. The explanatory variable is an indicator equal to one if the respondent answers "a great deal" or "some" to the question "How much would you say you know about ancestral, tribal, animist, or other traditional African religions?". *Attendance* is a dummy variable that equals one if the respondent attends more than once a week religious events. *Importance* is a dummy variable that equals one if the respondent declares that religion is "very important" in his/her life. Controls include age, age<sup>2</sup>, gender, whether the individual lives in a urban or rural area, whether he/she is christian/muslim or from other religion, a dummy variable that equals one if the respondent has completed primary education, a dummy that equals one if the respondent is married, a dummy variable that equals one if the respondent finds himself/herself in a good economic situation, and a dummy equal to one if the respondent did not have money at some point during the last year to buy food for his/her family. Robust standard errors in parenthesis. \*\*\* for  $p < 0.01$ , \*\* for  $p < 0.05$ , \* for  $p < 0.1$ .

Table H7: Ancestor worship, fertility and other supernatural beliefs

|                      | Number of children ever born |                      |                      |                      |                      |                       |                      |                      |                      |                      |
|----------------------|------------------------------|----------------------|----------------------|----------------------|----------------------|-----------------------|----------------------|----------------------|----------------------|----------------------|
|                      | (1)                          | (2)                  | (3)                  | (4)                  | (5)                  | (6)                   | (7)                  | (8)                  | (9)                  | (10)                 |
| Ancestral beliefs    | 0.395***<br>(0.0411)         | 0.495***<br>(0.0381) | 0.370***<br>(0.0431) | 0.476***<br>(0.0392) | 0.479***<br>(0.0392) | 0.143***<br>(0.0318)  | 0.175***<br>(0.0294) | 0.157***<br>(0.0332) | 0.163***<br>(0.0301) | 0.186***<br>(0.0300) |
| Religious healers    | 0.261***<br>(0.0356)         |                      |                      |                      |                      | 0.0766***<br>(0.0273) |                      |                      |                      |                      |
| Experienced miracle  |                              | 0.131***<br>(0.0315) |                      |                      |                      |                       | 0.0580**<br>(0.0243) |                      |                      |                      |
| Initiation ritual    |                              |                      | 0.282***<br>(0.0422) |                      |                      |                       |                      | 0.0380<br>(0.0322)   |                      |                      |
| Witchcraft           |                              |                      |                      | 0.0880**<br>(0.0347) |                      |                       |                      |                      | 0.0162<br>(0.0265)   |                      |
| Traditional religion |                              |                      |                      |                      | 0.221***<br>(0.0358) |                       |                      |                      |                      | 0.0447<br>(0.0277)   |
| Country FE           | Yes                          | Yes                  | Yes                  | Yes                  | Yes                  | Yes                   | Yes                  | Yes                  | Yes                  | Yes                  |
| Controls             | No                           | No                   | No                   | No                   | No                   | Yes                   | Yes                  | Yes                  | Yes                  | Yes                  |
| Mean Y               | 2.097                        | 2.088                | 2.091                | 2.090                | 2.078                | 2.099                 | 2.092                | 2.094                | 2.093                | 2.078                |
| R-squared            | 0.0673                       | 0.0638               | 0.0660               | 0.0641               | 0.0669               | 0.481                 | 0.482                | 0.481                | 0.483                | 0.486                |
| N                    | 22157                        | 22654                | 22698                | 21974                | 21396                | 21324                 | 21761                | 21808                | 21129                | 20587                |

NOTE. Data: PEW research forum 2008-2009 Survey. "Religious healers" equals one if the respondent's family has ever used traditional religious healers when someone is sick. "Experienced miracle" equals one if the respondent has experienced/witnessed a divine healing of an illness or injury. "Initiation ritual" equals one if the respondent has ever participated in an initiation ritual for friends, relatives or neighbors. "Witchcraft" equals one if the respondent declares that he/she believes in witchcraft. "Traditional religion" is a variable that equals one if the respondent answers "some" or "a great deal" to the question "How much would you say you know about ancestral, tribal, animist, or other traditional African religions?". Controls include age, age<sup>2</sup>, sex, whether the individual lives in a urban or rural area, whether he/she is christian/muslim or from other religion, a dummy variable that equals one if the respondent has completed primary education, a dummy that equals one if the respondent is married, a dummy variable that equals one if the respondent finds himself/herself in a good economic situation, and a dummy equal to one if the respondent did not have money at some point during the last year to buy food for his/her family. Robust standard errors in parenthesis. \*\*\* for  $p < 0.01$ , \*\* for  $p < 0.05$ , \* for  $p < 0.1$ .

## I. Heterogeneity and dynamics

### I.1. *Correlates of beliefs in ancestors in contemporary Benin*

I examine the correlates of believing in the influence of ancestors by plotting the standardized beta coefficients of the following bivariate regression:

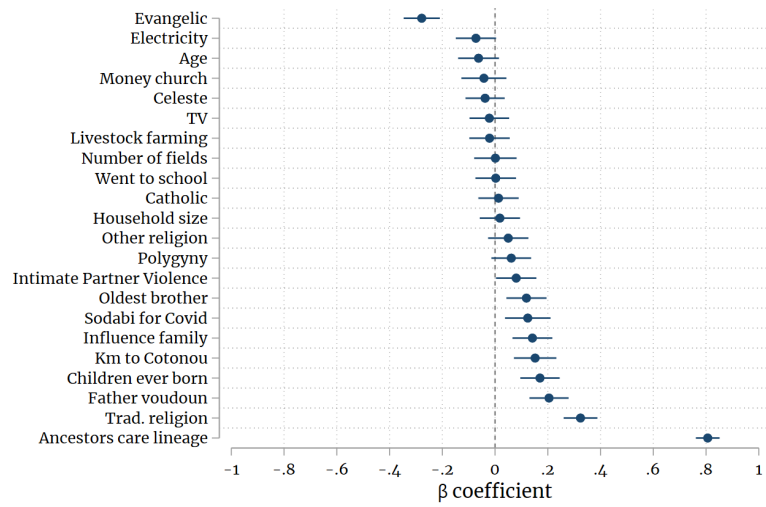
$$Y_i = \alpha + \beta_1 AncestralBeliefs_i + \epsilon_i \quad (8)$$

Where  $Y_i$  is each time a different outcome variable (a row in Figure I1) and  $AncestralBeliefs_i$  is an indicator variable that equals one if the respondent answers *yes* to the question "*do you believe that the spirits of your ancestors have an influence on the events of your life?*"

Believing in the influence of ancestors is positively associated with practicing a traditional religion as your main religion (mostly Voodoo), having a father (or mother) that practice(d) Voodoo, number of children, having received pressure from the extended family to have children, drinking sodabi (traditional local alcoholic drink) to fight against COVID, distance to Cotonou (economic capital of Benin) or being the oldest brother. As expected, there is a very strong correlation between believing in the influence of ancestors on people's lives and believing that ancestors care about the survival of the extended family and the continuation of the family line, as they are the two main characteristics of this belief system. There is no statistically significant association between traditional beliefs in ancestors and variables such as education, age, being Catholic, having a TV, number of fields used for agriculture, amount donated to the church or having livestock. Interestingly, I find a negative and strong correlation with being evangelic. This is in line with previous accounts highlighting that the endogenous nature of African churches (in opposition to the imported missionary churches – as reflected in the null correlation with Catholicism –) has allowed them to challenge certain important aspects of the traditional belief system. As Caldwell and Caldwell (1987) note:

The fact that they (African or endogenous churches, such as most evangelical churches in the context of Benin) are very much a grass-roots phenomenon and do not share the apprehension of African cultural manifestations that has often marked the missionary churches has tended to hide their greater willingness to challenge those aspects of the traditional belief system that they consider unacceptable (Caldwell and Caldwell (1987), p.428)".

Figure I1: Correlates of beliefs in ancestors



*Note:* The figure plots the standardized beta coefficient of regressing beliefs in the influence of ancestors on several explanatory variables. The unit of observation is an individual from the 2024 survey among pineapple producers in southern Benin. Lines around point estimates represent 95% confidence intervals.

## I.2. Heterogeneous effects of ancestral beliefs: PEW

Table I1: Ancestor worship and fertility in sub-Saharan Africa

|                                   | Number of children   |                      |                      |                      |                      |                      |                      |
|-----------------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
|                                   | (1)                  | (2)                  | (3)                  | (4)                  | (5)                  | (6)                  | (7)                  |
| Ancestor Worship                  | 0.184***<br>(0.0292) | 0.228***<br>(0.0369) | 0.154***<br>(0.0494) | 0.204***<br>(0.0503) | 0.220***<br>(0.0333) | 0.191***<br>(0.0421) | 0.198***<br>(0.0309) |
| Ancestor Worship x Urban          |                      | -0.110**<br>(0.0561) |                      |                      |                      |                      |                      |
| Ancestor Worship x Secondary      |                      |                      | 0.0632<br>(0.0613)   |                      |                      |                      |                      |
| Ancestor worship x Post secondary |                      |                      | -0.00534<br>(0.0779) |                      |                      |                      |                      |
| Ancestor Worship x Christian      |                      |                      |                      | -0.0341<br>(0.0601)  |                      |                      |                      |
| Ancestor Worship x Muslim         |                      |                      |                      |                      | -0.108*<br>(0.0637)  |                      |                      |
| Ancestor Worship x Good econ sit  |                      |                      |                      |                      |                      | -0.0155<br>(0.0553)  |                      |
| Ancestor Worship x > 40           |                      |                      |                      |                      |                      |                      | 0.0412<br>(0.0752)   |
| Country FE                        | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  |
| Controls                          | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  |
| Mean Y                            | 2.083                | 2.083                | 2.083                | 2.083                | 2.083                | 2.083                | 2.083                |
| R-squared                         | 0.480                | 0.481                | 0.482                | 0.480                | 0.480                | 0.480                | 0.423                |
| N                                 | 22295                | 22295                | 22295                | 22295                | 22295                | 22295                | 22295                |

NOTE. Data: PEW research forum 2008-2009 Survey. The outcome variable is the total number of children. The explanatory variable is an indicator equal to one if the respondent answers "yes" to the question "Do you ever participate in traditional African ceremonies or perform special acts to honor or celebrate your ancestors?". In column 3, the reference category is having primary or less than primary education. Controls include age, age<sup>2</sup>, sex, whether the individual lives in a urban or rural area, religion, education, marital status, a dummy variable that equals one if the respondent finds himself/herself in a good economic situation, and a dummy equal to one if the respondent did not have money at some point during the last year to buy food for his/her family. Robust standard errors in parenthesis. \*\*\* for  $p < 0.01$ , \*\* for  $p < 0.05$ , \* for  $p < 0.1$ .

## J. Ancestral beliefs and fertility in folklore: robustness

Table J1: Motifs related to *birth* and to *ancestor worship*

|                       | Share of motifs related to <i>birth</i> |                        |                        |                        |                        |                        |                        |                        |
|-----------------------|-----------------------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
|                       | SSA Sample                              |                        |                        |                        | Global Sample          |                        |                        |                        |
|                       | (1)                                     | (2)                    | (3)                    | (4)                    | (5)                    | (6)                    | (7)                    | (8)                    |
| Ancestral beliefs     | 0.0273***<br>(0.00134)                  | 0.0276***<br>(0.00423) | 0.0232***<br>(0.00528) | 0.0229***<br>(0.00503) | 0.0272***<br>(0.00137) | 0.0261***<br>(0.00191) | 0.0262***<br>(0.00264) | 0.0255***<br>(0.00300) |
| SSA Sample            | Yes                                     | Yes                    | Yes                    | Yes                    | No                     | No                     | No                     | No                     |
| Folklore controls     | No                                      | Yes                    | Yes                    | Yes                    | No                     | Yes                    | Yes                    | Yes                    |
| Ethnographic controls | No                                      | No                     | Yes                    | Yes                    | No                     | No                     | Yes                    | Yes                    |
| Geographic controls   | No                                      | No                     | No                     | Yes                    | No                     | No                     | No                     | Yes                    |
| Mean Y                | 4.177                                   | 4.177                  | 3.895                  | 3.908                  | 4.086                  | 4.086                  | 3.877                  | 3.848                  |
| N                     | 407                                     | 407                    | 307                    | 306                    | 1245                   | 1245                   | 951                    | 862                    |

NOTE. Data: Ethnographic Atlas and Folklore. In columns (1)-(4), the sample is restricted to SSA. The table reports Pseudo-Poisson Maximum Likelihood (PPML) estimators. The outcome variable is the share of motifs related to "birth" in the oral tradition of an ethnic group. "Ancestor worship" is the share of motifs related to "ancestor worship" in an ethnic group's oral tradition. Folklore controls include the total number of motifs in an ethnic group's oral tradition, the number of publishers of the sources in the group's oral tradition, and the earliest year of publication in the group's oral tradition. Ethnographic controls include whether the domestic organization is around independent nuclear families, whether people are part of localized clans that live as segmented communities, whether the ethnic group is patrilineal, impartible inheritance, political complexity, whether monogamy is dominant, whether the group practices pastoralism, use of historical plough, historical economic development, practice of intensive agriculture, and the share of motifs in an ethnic group related to "supernatural". Geographic controls include tropical climate, precipitations, ruggedness, land quality (population weighted), and agricultural suitability. Robust standard errors in parenthesis. \*\*\* for  $p < 0.01$ , \*\* for  $p < 0.05$ , \* for  $p < 0.1$ .

Table J2: Motifs related to *fertility* and to *ancestor*

|                       | Share of motifs related to <i>fertility</i> |                    |                      |                      |                       |                       |                        |                       |
|-----------------------|---------------------------------------------|--------------------|----------------------|----------------------|-----------------------|-----------------------|------------------------|-----------------------|
|                       | SSA Sample                                  |                    |                      |                      | Global Sample         |                       |                        |                       |
|                       | (1)                                         | (2)                | (3)                  | (4)                  | (5)                   | (6)                   | (7)                    | (8)                   |
| Ancestral beliefs     | 0.148***<br>(0.00650)                       | 0.153*<br>(0.0841) | 0.203***<br>(0.0658) | 0.215***<br>(0.0674) | 0.126***<br>(0.00466) | 0.112***<br>(0.00621) | 0.0992***<br>(0.00696) | 0.104***<br>(0.00756) |
| SSA Sample            | Yes                                         | Yes                | Yes                  | Yes                  | No                    | No                    | No                     | No                    |
| Folklore controls     | No                                          | Yes                | Yes                  | Yes                  | No                    | Yes                   | Yes                    | Yes                   |
| Ethnographic controls | No                                          | No                 | Yes                  | Yes                  | No                    | No                    | Yes                    | Yes                   |
| Geographic controls   | No                                          | No                 | No                   | Yes                  | No                    | No                    | No                     | Yes                   |
| Mean Y                | 0.674                                       | 0.674              | 0.562                | 0.564                | 0.729                 | 0.729                 | 0.687                  | 0.660                 |
| N                     | 407                                         | 407                | 304                  | 303                  | 1245                  | 1245                  | 951                    | 862                   |

NOTE. Data: Ethnographic Atlas and Folklore. In columns (1)-(4), the sample is restricted to SSA. The table reports Pseudo-Poisson Maximum Likelihood (PPML) estimators. The outcome variable is the share of motifs related to "fertility" in the oral tradition of an ethnic group. "Ancestor worship" is the share of motifs related to "ancestors" in an ethnic group's oral tradition. Folklore controls include the total number of motifs in an ethnic group's oral tradition, the number of publishers of the sources in the group's oral tradition, and the earliest year of publication in the group's oral tradition. Ethnographic controls include whether the domestic organization is around independent nuclear families, whether people are part of localized clans that live as segmented communities, whether the ethnic group is patrilineal, impartible inheritance, political complexity, whether monogamy is dominant, whether the group practices pastoralism, use of historical plough, historical economic development, practice of intensive agriculture, and the share of motifs in an ethnic group related to "supernatural". Geographic controls include tropical climate, precipitations, ruggedness, land quality (population weighted), and agricultural suitability. Robust standard errors in parenthesis. \*\*\* for  $p < 0.01$ , \*\* for  $p < 0.05$ , \* for  $p < 0.1$ .



Table J3: Motifs related to *fertility* and to *ancestor worship*

|                       | Share of motifs related to <i>fertility</i> |                       |                      |                      |                        |                        |                        |                        |
|-----------------------|---------------------------------------------|-----------------------|----------------------|----------------------|------------------------|------------------------|------------------------|------------------------|
|                       | SSA Sample                                  |                       |                      |                      | Global Sample          |                        |                        |                        |
|                       | (1)                                         | (2)                   | (3)                  | (4)                  | (5)                    | (6)                    | (7)                    | (8)                    |
| Ancestral beliefs     | 0.0745***<br>(0.00305)                      | 0.0746***<br>(0.0239) | 0.139***<br>(0.0441) | 0.143***<br>(0.0479) | 0.0701***<br>(0.00229) | 0.0635***<br>(0.00291) | 0.0570***<br>(0.00369) | 0.0591***<br>(0.00405) |
| SSA Sample            | Yes                                         | Yes                   | Yes                  | Yes                  | No                     | No                     | No                     | No                     |
| Folklore controls     | No                                          | Yes                   | Yes                  | Yes                  | No                     | Yes                    | Yes                    | Yes                    |
| Ethnographic controls | No                                          | No                    | Yes                  | Yes                  | No                     | No                     | Yes                    | Yes                    |
| Geographic controls   | No                                          | No                    | No                   | Yes                  | No                     | No                     | No                     | Yes                    |
| Mean Y                | 0.674                                       | 0.674                 | 0.562                | 0.564                | 0.729                  | 0.729                  | 0.687                  | 0.660                  |
| N                     | 407                                         | 407                   | 304                  | 303                  | 1245                   | 1245                   | 951                    | 862                    |

NOTE. Data: Ethnographic Atlas and Folklore. In columns (1)-(4), the sample is restricted to SSA. The table reports Pseudo-Poisson Maximum Likelihood (PPML) estimators. The outcome variable is the share of motifs related to "fertility" in the oral tradition of an ethnic group. "Ancestor worship" is the share of motifs related to "ancestor worship" in an ethnic group's oral tradition. Folklore controls include the total number of motifs in an ethnic group's oral tradition, the number of publishers of the sources in the group's oral tradition, and the earliest year of publication in the group's oral tradition. Ethnographic controls include whether the domestic organization is around independent nuclear families, whether people are part of localized clans that live as segmented communities, whether the ethnic group is patrilineal, impartible inheritance, political complexity, whether monogamy is dominant, whether the group practices pastoralism, use of historical plough, historical economic development, practice of intensive agriculture, and the share of motifs in an ethnic group related to "supernatural". Geographic controls include tropical climate, precipitations, ruggedness, land quality (population weighted), and agricultural suitability. Robust standard errors in parenthesis. \*\*\* for  $p < 0.01$ , \*\* for  $p < 0.05$ , \* for  $p < 0.1$ .

Table J4: Ancestor worship and fertility in Folklore

|                       | Share of motifs related to <i>birth</i> |                     |                     |                     |                     |                     |                     |                     |
|-----------------------|-----------------------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                       | SSA Sample                              |                     |                     |                     | Global Sample       |                     |                     |                     |
|                       | (1)                                     | (2)                 | (3)                 | (4)                 | (5)                 | (6)                 | (7)                 | (8)                 |
| Ancestral beliefs     | 0.619***<br>(0.128)                     | 0.609***<br>(0.133) | 0.541***<br>(0.156) | 0.536***<br>(0.156) | 0.472***<br>(0.116) | 0.461***<br>(0.117) | 0.420***<br>(0.141) | 0.432***<br>(0.147) |
| SSA Sample            | Yes                                     | Yes                 | Yes                 | Yes                 | No                  | No                  | No                  | No                  |
| Folklore controls     | No                                      | Yes                 | Yes                 | Yes                 | No                  | Yes                 | Yes                 | Yes                 |
| Ethnographic controls | No                                      | No                  | Yes                 | Yes                 | No                  | No                  | Yes                 | Yes                 |
| Geographic controls   | No                                      | No                  | No                  | Yes                 | No                  | No                  | No                  | Yes                 |
| Mean Y                | 4.177                                   | 4.177               | 3.895               | 3.908               | 4.086               | 4.086               | 3.877               | 3.848               |
| N                     | 407                                     | 407                 | 307                 | 306                 | 1245                | 1245                | 951                 | 862                 |

NOTE. Data: Ethnographic Atlas and Folklore. In columns (1)-(4), the sample is restricted to SSA. The table reports OLS estimates. The outcome variable is the share of motifs related to "birth" in the oral tradition of an ethnic group. "Ancestor worship" is the share of motifs related to "ancestors" in an ethnic group's oral tradition. Folklore controls include the total number of motifs in an ethnic group's oral tradition, the number of publishers of the sources in the group's oral tradition, and the earliest year of publication in the group's oral tradition. Ethnographic controls include whether the domestic organization is around independent nuclear families, whether people are part of localized clans that live as segmented communities, whether the ethnic group is patrilineal, impartible inheritance, political complexity, whether monogamy is dominant, whether the group practices pastoralism, use of historical plough, historical economic development, practice of intensive agriculture, and the share of motifs in an ethnic group related to "supernatural". Geographic controls include tropical climate, precipitations, ruggedness, land quality (population weighted), and agricultural suitability. Robust standard errors in parenthesis. \*\*\* for  $p < 0.01$ , \*\* for  $p < 0.05$ , \* for  $p < 0.1$ .

## K. Conceptual Framework: Woman as decision-maker

From the woman's point of view, the optimization problem is slightly different. Specifically, the total number of children belonging to the woman's lineage is defined differently than the total number of children belonging to the man's lineage. This is because women in patrilineal societies become part of the husband's lineage upon marriage. On the contrary, men in matrilineal societies do not join their wives' lineage upon marriage. Now, the total number of children in a woman's lineage is:

$$N = \begin{cases} n_i + \sum_{j \neq i} n_j & \text{if patrilineal, } j \text{ indexes husband's brothers} \\ n_i + \sum_{k \neq i} n_k & \text{if matrilineal, } k \text{ indexes own sisters} \end{cases}$$

As we can see, nothing changes in the patrilineal case, and the motive to continue the family line is the same for both the husband and the wife. In the matrilineal case, however, a woman's lineage is now extended through her own children as well as through her sisters' children. We will now look at the matrilineal case to see the differences with respect to the husband's situation.

### K.1. Matrilineal case

Then, the maximization problem in the matrilineal case becomes:

$$\begin{aligned} \max_{c_i, n_i} \quad & U_i = u(c_i) + b(n_i) + \beta \ell(n_i + \sum_{k \neq i} n_k) \\ \text{s.t.} \quad & c_i + n_i \leq y_i \end{aligned}$$

From the first order conditions follow that:

$$u'(c_i) = b'(n_i) + \beta \ell'(n_i + \sum_{j \neq i} n_j)$$

And, as before, the study of the interaction between  $n_i$  and  $n_k$  gives us:

$$\frac{dr_i(n_k)}{dn_k} = -\frac{\frac{\partial F}{\partial n_k}}{\frac{\partial F}{\partial r_i(n_k)}} = -\frac{\frac{\partial^2 U_i(c_i, n_i, n_i + \sum n_k)}{\partial n_i \partial n_k}}{\frac{\partial^2 U_i(c_i, n_i, n_i + \sum n_k)}{\partial n_i^2}} < 0$$

If the woman takes part in the decision-making process regarding the number of children, which is possibly more likely to happen in matrilineal societies (Lowes, 2022), there is a negative relationship between own fertility and the number of children of the woman's sisters.

## L. Backgroup on kinship systems

Two persons are kin when one is descended from the other. Kinship results from the recognition of a *social* relationship between parents and children.<sup>72</sup> A system of kinship and marriage can be looked at as an arrangement which enables persons to live together and co-operate with one another in an orderly social life. Kinship systems determine the set of people to whom an individual is considered related and their social obligations to this group (Radcliffe-Brown and Forde, 1950).

A kinship system that is reckoned by each individual with reference to ascendance and descendance, without distinction between male and female lines, is called a *cognatic system*. Persons are cognatic kin or cognates when they are descended from a common ancestor or ancestress counting descent through males and females (Lorimer, 1954). In a cognatic system, the emphasis in kinship relations rests on the nuclear family. In this type of kinship, kin, as defined by descendance through both the paternal and maternal sides, is not a cohesive kinship group. It includes individuals on maternal and paternal sides, at each ascending and descending juncture, who may not be related to one another

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On the other hand, descent groups in which one belongs exclusively to one's father's or mother's line are called *unilineal systems*. In unilineal descent groups, one's descent is traced either exclusively through male ancestors (patrilineal descent groups), or exclusively through female ancestors (matrilineal descent groups). These groups form a cohesive and continuing kinship structure. In fact, the unity of the sibling group is preserved (especially as regards those who by virtue of their sex are part of a continuing lineage). A group of brothers, or a group of sisters, acquire hereditary rights and transmit them to their descendants of the same sex. There is, however, an important difference between patrilineal and matrilineal descent systems that matters for the analysis in this paper, known as asymmetric marital allegiances (Berggreen and Gokmen, 2023). In pa-

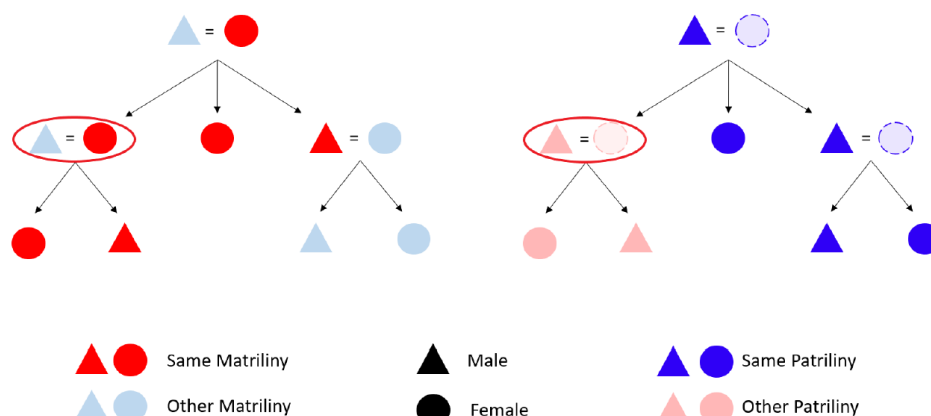
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<sup>72</sup>The stress of the word social is key to distinguish kinship from consanguinity (physical relationship). In fact, an illegitimate child has a physical father but may not have a social father (he does not belong to the same kin group as his biological father). Similarly, an adopted child may be part of the same kin group as his adoptive parents while they are not consanguineous.

<sup>73</sup>As Lorimer (1954) notes: "I may feel myself to be related to the children and grandchildren of my paternal grandfathers brothers and sisters, and to those of my paternal grandmother, my maternal grandfather, and my maternal grandmother; but these four sets of relatives have no necessary relationship to one another".

trilineal societies, the children belong to the father's lineage, and a patrilineal daughter who marries becomes part of her husband's lineage. Then, both spouses and children are part of the same kin group in patrilineal societies. In matrilineal societies, children belong exclusively to the mother's kin group, and a patrilineal son who marries maintains his birth lineage (in fact, inheritance often passes from the maternal uncle to his sisters' children). Figure L1 summarizes these differences:

Figure L1: Matrilineal and patrilineal kinship systems, from Berggreen and Gokmen (2023)



Source: Berggreen and Gokmen (2023). The figure shows the kinship structure of both patrilineal and matrilineal societies. The sign = symbolizes marriage, while the bracket over the triangle/circle indicates siblingship.

## M. Kinship structure and fertility behavior in the DRC

Although this paper suggests that patrilineality could positively affect fertility through the stronger motive to continue one's lineage in patrilineal societies as compared to matrilineal societies (for a given degree of ancestral beliefs in the society) little is known about the effect of the nature of group membership – either determined by men (patrilineal societies) or by women (matrilineal societies) – on fertility outcomes. Interestingly, those few studies that have examined the influence of lineage or clan on fertility emphasize that this influence is particularly important in patrilineal societies (Okafor et al., 2021; Church et al., 2023), although the mechanism driving these results remained unclear. Although a systematic exploration of the relationship between kinship systems and fertility is beyond the scope of this paper, I slightly advance in this direction in this direction and provide hints for future research.

To the extent of my knowledge, there are only two recent papers in economics that mention the potential relationship between kinship structure and fertility. First, BenYishay et al. (2017) show that matrilineal inheritance leads to smaller average household size and total population village, but they do so in the very specific context of the Solomon Islands, and only when exploring the implications of their main research question (whether reef density, their proxy for the quality of the marine environment, systematically predicts the prevalence of female land inheritance). Their explanation is related to inclusive fitness (Holden et al., 2003): "under a patrilineal inheritance system, the additional number of offspring that can result from transmitting an asset to sons needs to outweigh the loss in terms of paternal certainty". Second, Fontenay et al. (2024) include patrilineality as a control in their main regression (they are interested in the effect of impartible inheritance on fertility in SSA), showing a positive and significant coefficient. However, they do not provide any explanation regarding the mechanisms underlying this relationship.

Table M1 shows the relationship between ideal family size and patrilineality. I find that, on average, individuals in patrilineal societies want 0.5 (10%) more children (column 1), although there are no important differences between men and women (column 2). However, this relationship is completely driven by societies with strong beliefs in ancestors (columns 3 and 4).<sup>74</sup>

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<sup>74</sup>Note that the difference between columns (3)-(4) and columns (5)-(6) is not only statistical significance. Also point estimates are much closer to zero when the ethnic group does not practice ancestor worship.

Table M1: Ancestor worship and fertility preferences in the DRC

|                      | Ideal Nb of Children |                       |                      |                       |                      |                      |
|----------------------|----------------------|-----------------------|----------------------|-----------------------|----------------------|----------------------|
|                      | Full sample          |                       | Ancestor Worship = 1 |                       | Ancestor Worship = 0 |                      |
|                      | (1)                  | (2)                   | (3)                  | (4)                   | (5)                  | (6)                  |
| Patrilineal          | 0.491***<br>(0.117)  | 0.558***<br>(0.130)   | 0.637***<br>(0.163)  | 0.655***<br>(0.174)   | 0.0748<br>(0.266)    | 0.138<br>(0.336)     |
| Female               |                      | -0.443***<br>(0.0485) |                      | -0.421***<br>(0.0499) |                      | -0.725***<br>(0.206) |
| Patrilineal x Female |                      | -0.0981<br>(0.0717)   |                      | -0.0272<br>(0.0792)   |                      | -0.0903<br>(0.239)   |
| Province FE          | Yes                  | Yes                   | Yes                  | Yes                   | Yes                  | Yes                  |
| Controls             | Yes                  | Yes                   | Yes                  | Yes                   | Yes                  | Yes                  |
| Mean Y               | 6.413                | 6.413                 | 6.428                | 6.428                 | 6.561                | 6.561                |
| R-squared            | 0.238                | 0.238                 | 0.256                | 0.256                 | 0.179                | 0.179                |
| N                    | 35626                | 35626                 | 28904                | 28904                 | 4284                 | 4284                 |

NOTE. Data: Demographic and Health Surveys of DRC (2007 and 2013-2014). The table reports OLS estimates. The outcome variable is the ideal number of children. The sample is restricted to ethnic groups with ancestor worship in columns (3) and (4) and to ethnic groups without ancestor worship in columns (5) and (6). "Patrilineal" is a dummy variable that equals one if the ethnic group  $e$  has patrilineal descent. Controls include age, age squared, gender, a dummy variable that equals one if the respondent is catholic, single years of education, whether the DHS cluster is urban, whether the respondent works, whether the respondent works in agriculture and a dummy variable that equals one if the respondent is in the top 40% of the wealth distribution. Standard errors clustered at the DHS cluster-level in parenthesis. \*\*\* for  $p < 0.01$ , \*\* for  $p < 0.05$ , \* for  $p < 0.1$ .

The absence of men-women differences may be surprising since, as mentioned in 5, the motive to continue the family line for men only applies in patrilineal societies, while it applies independently of the kinship structure for women. The lack of differences between the ideal number of children for women and for men might be due to the important correlation that has been documented between desired and actual number of children. This might be because people may adjust their ideal number of children upwards as their actual number of children increases. In Tables M2 and M3 I explore these results separately for sons and daughters, and the differences between men and women emerge. Moreover, if we look at whether respondents want an additional child (see Table M4), we observe that the influence of patrilineality is stronger for men, specially when beliefs in ancestors are strong.

Table M2: Ancestor worship and fertility preferences in the DRC

|                      | Ideal Number of Sons |                       |                      |                       |                      |                      |
|----------------------|----------------------|-----------------------|----------------------|-----------------------|----------------------|----------------------|
|                      | Full sample          |                       | Ancestor Worship = 1 |                       | Ancestor Worship = 0 |                      |
|                      | (1)                  | (2)                   | (3)                  | (4)                   | (5)                  | (6)                  |
| Patrilineal          | 0.260***<br>(0.0932) | 0.528***<br>(0.114)   | 0.311**<br>(0.127)   | 0.555***<br>(0.147)   | -0.00435<br>(0.183)  | 0.227<br>(0.278)     |
| Female               |                      | -0.744***<br>(0.0527) |                      | -0.752***<br>(0.0549) |                      | -0.977***<br>(0.246) |
| Patrilineal x Female |                      | -0.397***<br>(0.0742) |                      | -0.362***<br>(0.0833) |                      | -0.333<br>(0.271)    |
| Province FE          | Yes                  | Yes                   | Yes                  | Yes                   | Yes                  | Yes                  |
| Controls             | Yes                  | Yes                   | Yes                  | Yes                   | Yes                  | Yes                  |
| Mean Y               | 2.931                | 2.931                 | 2.917                | 2.917                 | 3.114                | 3.114                |
| R-squared            | 0.131                | 0.133                 | 0.135                | 0.136                 | 0.131                | 0.132                |
| N                    | 36146                | 36146                 | 29322                | 29322                 | 4367                 | 4367                 |

NOTE. Data: Demographic and Health Surveys of DRC (2007 and 2013-2014). The table reports OLS estimates. The outcome variable is the ideal number of sons. The sample is restricted to ethnic groups with ancestor worship in columns (3) and (4) and to ethnic groups without ancestor worship in columns (5) and (6). "Patrilineal" is a dummy variable that equals one if the ethnic group  $e$ , defined as in Murdock (1967), has patrilineal descent. Controls include age, age squared, gender, a dummy variable that equals one if the respondent is catholic, single years of education, whether the DHS cluster is urban, whether the respondent works, whether the respondent works in agriculture and a dummy variable that equals one if the respondent is in the top 40% of the wealth distribution. Standard errors clustered at the DHS cluster-level in parenthesis. \*\*\* for  $p < 0.01$ , \*\* for  $p < 0.05$ , \* for  $p < 0.1$ .

Table M3: Ancestor worship and fertility preferences in the DRC

|                      | Ideal Number of Daughters |                       |                      |                       |                      |                   |
|----------------------|---------------------------|-----------------------|----------------------|-----------------------|----------------------|-------------------|
|                      | Full sample               |                       | Ancestor Worship = 1 |                       | Ancestor Worship = 0 |                   |
|                      | (1)                       | (2)                   | (3)                  | (4)                   | (5)                  | (6)               |
| Patrilineal          | 0.0983<br>(0.0760)        | 0.300***<br>(0.0892)  | 0.149<br>(0.101)     | 0.342***<br>(0.114)   | -0.0365<br>(0.211)   | 0.0816<br>(0.244) |
| Female               |                           | 0.0492<br>(0.0428)    |                      | 0.0416<br>(0.0445)    |                      | -0.128<br>(0.161) |
| Patrilineal x Female |                           | -0.298***<br>(0.0599) |                      | -0.288***<br>(0.0666) |                      | -0.170<br>(0.186) |
| Province FE          | Yes                       | Yes                   | Yes                  | Yes                   | Yes                  | Yes               |
| Controls             | Yes                       | Yes                   | Yes                  | Yes                   | Yes                  | Yes               |
| Mean Y               | 2.624                     | 2.624                 | 2.609                | 2.609                 | 2.799                | 2.799             |
| R-squared            | 0.0831                    | 0.0843                | 0.0870               | 0.0881                | 0.0589               | 0.0591            |
| N                    | 36177                     | 36177                 | 29346                | 29346                 | 4374                 | 4374              |

NOTE. Data: Demographic and Health Surveys of DRC (2007 and 2013-2014). The table reports OLS estimates. The outcome variable is the ideal number of daughters. The sample is restricted to ethnic groups with ancestor worship in columns (3) and (4) and to ethnic groups without ancestor worship in columns (5) and (6). "Patrilineal" is a dummy variable that equals one if the ethnic group  $e$  has patrilineal descent. Controls include age, age squared, gender, a dummy variable that equals one if the respondent is catholic, single years of education, whether the DHS cluster is urban, whether the respondent works, whether the respondent works in agriculture and a dummy variable that equals one if the respondent is in the top 40% of the wealth distribution. Standard errors clustered at the DHS cluster-level in parenthesis. \*\*\* for  $p < 0.01$ , \*\* for  $p < 0.05$ , \* for  $p < 0.1$ .



Table M4: Ancestor worship and fertility preferences in the DRC

|                      | Wants another child    |                       |                       |                      |
|----------------------|------------------------|-----------------------|-----------------------|----------------------|
|                      | Full sample            |                       | Ancestor Worship = 1  | Ancestor Worship = 0 |
|                      | (1)                    | (2)                   | (3)                   | (4)                  |
| Patrilineal          | 0.0652***<br>(0.0170)  | 0.0401***<br>(0.0153) | 0.0705***<br>(0.0161) | -0.0229<br>(0.0587)  |
| Patrilineal x Female | -0.0585***<br>(0.0134) | -0.0262**<br>(0.0122) | -0.0233*<br>(0.0134)  | -0.0271<br>(0.0426)  |
| Province FE          | Yes                    | Yes                   | Yes                   | Yes                  |
| Controls             | Yes                    | Yes                   | Yes                   | Yes                  |
| Mean Y               | 0.755                  | 0.756                 | 0.761                 | 0.750                |
| R-squared            | 0.0124                 | 0.223                 | 0.229                 | 0.186                |
| N                    | 32120                  | 31665                 | 25576                 | 3992                 |

NOTE. Data: Demographic and Health Surveys of DRC (2007 and 2013-2014). The table reports OLS estimates. The outcome variable is a dummy variable that equals one if the respondent wants an additional child. The sample is restricted to ethnic groups with ancestor worship in column (3) and to ethnic groups without ancestor worship in column (4). "Patrilineal" is a dummy variable that equals one if the ethnic group  $e$  has patrilineal descent. Controls include age, age squared, gender, a dummy variable that equals one if the respondent is catholic, single years of education, whether the DHS cluster is urban, whether the respondent works, whether the respondent works in agriculture and a dummy variable that equals one if the respondent is in the top 40% of the wealth distribution. Standard errors clustered at the DHS cluster-level in parenthesis. \*\*\* for  $p < 0.01$ , \*\* for  $p < 0.05$ , \* for  $p < 0.1$ .

Overall, these results suggest that there is a positive relationship between patrilineality and fertility, that may be driven by a stronger motive to continue one's lineage in patrilineal societies. In addition, women are compelled to have at least one son to gain respectability in marriage, as well as to inherit property and maintain the family line in patrilineal societies (Okafor et al., 2021).<sup>75</sup> The decision to reproduce is rarely an individual one, and a woman's reproduction is a matter for her husband and his lineage, who, on the other hand, bear almost no costs of additional children and whose reproductive obligations are solely to their lineage and ancestry.<sup>76</sup>

<sup>75</sup>As Okafor et al. (2021) notes in the context of southeastern Nigeria, which is under patrilineal descent: "Maintaining a limited family size without a male child may not be widely supported among economically independent women here compared to other places in the world where women's economic independence is gradually eroding the idea of large family size. As has been pointed out in relation to women's economic independence and son preference, in a traditional system where patrilineal descent dominates inheritance and lineage continuity, no woman will risk her marriage and family pride. Therefore, in such kinship systems, where all major decisions are usually in the hands of a woman's relatives, women can only increase their status and security through their sons.

<sup>76</sup>In this sense, marriage in an institution that goes beyond the couple. In patrilineal societies, marriage is considered stronger than in matrilineal societies, and divorce is low or forbidden. The fact that marriage belong to kinship and lineage structure is made clear by the fact that the death of the husband is usually not enough to break it. In such cases, the wife remains in her husband's family and is married to the deceased husband's brother (levirate marriage).

## N. Ancestral Beliefs and contraceptive use

### N.1. Ancestral Beliefs and contraceptive use in the DRC

Table N1: Ancestral beliefs and contraceptive use in the DRC DHS

|                   | Contraceptive use    |                       |                       | Children condom<br>(4) | Intention to use<br>(5) | Fear side-effects<br>(6) | Knowledge<br>(7)    |
|-------------------|----------------------|-----------------------|-----------------------|------------------------|-------------------------|--------------------------|---------------------|
|                   | Currently<br>(1)     | Traditional<br>(2)    | Modern<br>(3)         |                        |                         |                          |                     |
| Ancestral Beliefs | -0.0242*<br>(0.0130) | -0.0249**<br>(0.0109) | 0.000698<br>(0.00588) | -0.0704***<br>(0.0272) | -0.00576<br>(0.0207)    | 0.0219**<br>(0.00887)    | 0.00571<br>(0.0234) |
| Province FE       | Yes                  | Yes                   | Yes                   | Yes                    | Yes                     | Yes                      | Yes                 |
| Controls          | Yes                  | Yes                   | Yes                   | Yes                    | Yes                     | Yes                      | Yes                 |
| Mean Y            | 0.185                | 0.105                 | 0.0800                | 0.523                  | 0.333                   | 0.0574                   | 0.174               |
| R-squared         | 0.0965               | 0.0470                | 0.0514                | 0.0617                 | 0.0833                  | 0.0410                   | 0.136               |
| N                 | 24674                | 24674                 | 24674                 | 20640                  | 19961                   | 5031                     | 24674               |

NOTE. Data: Demographic and Health Surveys of DRC (2007 and 2013-2014). The table reports OLS estimates. The sample is restricted to women. The outcome variable in column (1) is a dummy that equals one if the respondent was using contraceptives at the time of the survey. In column (2) a dummy that equals one if the respondent was using traditional methods. In column (3) a dummy that equals one if the respondent was using modern methods. In column (4) a dummy that equals one if the respondent thinks that children should be taught about condoms to avoid aids. In column (5) a dummy that equals one if the respondent intends to use contraceptives in the future. In column (6) a dummy that equals one if the respondent does not use contraceptive because of fear to side effects. In column (7) a dummy that equals one if the respondent does not know about any contraceptive method. "Ancestral beliefs" is a dummy variable that equals one if the DHS cluster  $c$  of individual  $i$  belongs to the ethnic homeland of an ethnic group that traditionally practice ancestor worship as reported in Vansina (1966). Controls also include age, age squared, a dummy variable that equals one if the respondent is catholic, single years of education, whether the DHS cluster is urban, whether the respondent works, whether the respondent works in agriculture and a dummy variable that equals one if the respondent is in the top 40% of the wealth distribution. Standard errors clustered at the DHS cluster-level in parenthesis. \*\*\* for  $p < 0.01$ , \*\* for  $p < 0.05$ , \* for  $p < 0.1$ .

### N.2. Ancestral Beliefs and ideal family size in Benin DHS

As a first stage, I first verify that the main results of the paper – that is, the positive influence of traditional beliefs in ancestors on the demand for children – hold in this context when I use Voodoo religion as a proxy for beliefs in ancestors. Tables N2 and N3 show that followers of the Voodoo religion want more children, either estimating a lineal regression model by ordinary least squares or by 2SLS.

Table N2: Traditional beliefs and ideal family size

|                     | Ideal Number of children |                     |                      |                      |                     |                      |
|---------------------|--------------------------|---------------------|----------------------|----------------------|---------------------|----------------------|
|                     | All<br>(1)               | Men<br>(2)          | Women<br>(3)         | All<br>(4)           | Men<br>(5)          | Women<br>(6)         |
| Traditional beliefs | 0.966***<br>(0.0556)     | 1.550***<br>(0.138) | 0.440***<br>(0.0413) | 0.295***<br>(0.0598) | 0.675***<br>(0.146) | 0.188***<br>(0.0491) |
| Controls            | No                       | No                  | No                   | Yes                  | Yes                 | Yes                  |
| Round FE            | Yes                      | Yes                 | Yes                  | Yes                  | Yes                 | Yes                  |
| Cluster FE          | Yes                      | Yes                 | Yes                  | Yes                  | Yes                 | Yes                  |
| Mean Y              | 5.444                    | 6.698               | 4.997                | 5.375                | 6.593               | 4.937                |
| N                   | 79641                    | 20930               | 58681                | 64038                | 16920               | 47095                |

NOTE. Data: Demographic and Health Surveys of Benin (1996, 2001, 2006, 2011 and 2017). The table reports OLS estimates. The sample is restricted to women. The outcome variable is the ideal number of children. "Traditional beliefs" is a dummy variable that equals one if the respondent declares African traditional religion as his/her main religion. Controls also include age, gender, education, whether the DHS cluster is urban, whether the respondent works, marital status, polygamy, and wealth quintiles. Standard errors clustered at the DHS cluster-level in parenthesis. \*\*\* for  $p < 0.01$ , \*\* for  $p < 0.05$ , \* for  $p < 0.1$ .

Table N3: Traditional beliefs and ideal number of children

|                     | Ideal Number of children |                     |                     | First stage           |                      |                       |
|---------------------|--------------------------|---------------------|---------------------|-----------------------|----------------------|-----------------------|
|                     | (1)                      | (2)                 | (3)                 | (4)                   | (5)                  | (6)                   |
| Traditional beliefs | 2.707***<br>(0.229)      | 3.191***<br>(0.431) | 2.895***<br>(0.227) |                       |                      |                       |
| Adja Dahomey        |                          |                     |                     | 0.222***<br>(0.00609) | 0.387***<br>(0.0138) | 0.168***<br>(0.00640) |
| Controls            | Yes                      | Yes                 | Yes                 |                       |                      |                       |
| Round FE            | Yes                      | Yes                 | Yes                 |                       |                      |                       |
| Mean Y              | 5.377                    | 6.588               | 4.938               |                       |                      |                       |
| N                   | 63321                    | 16831               | 46490               | 63321                 | 16831                | 46490                 |
| F-stat              |                          |                     |                     | 1324.7                | 785.2                | 693.9                 |

NOTE. Data: Demographic and Health Surveys of Benin (1996, 2001, 2006, 2011 and 2017). The table reports OLS estimates. The sample is restricted to women. The outcome variable is the ideal number of children. "Traditional beliefs" is a dummy variable that equals one if the respondent declares African traditional religion as his/her main religion. Controls also include age, gender, education, whether the DHS cluster is urban, whether the respondent works, marital status, polygamy, and wealth quintiles. Standard errors clustered at the DHS cluster-level in parenthesis. \*\*\* for  $p < 0.01$ , \*\* for  $p < 0.05$ , \* for  $p < 0.1$ .

### N.3. Benin DHS: Instrumental Variable Strategy

As a first stage to verify whether using the Adja ethnicity as an instrument is valid, I use the first-hand data that I collected in Benin to show that the Adja people effectively hold

much stronger beliefs in the influence of ancestors than other ethnic groups, while other characteristics do not seem to be different. Table N4 reports the results of regressing different outcomes on ethnicity fixed effects (Adja is the reference group). I find that Adja people have significantly more children, have less land (the economic benefit of having children is therefore not likely to explain the higher fertility levels of the Adja), are not less educated, are not less Catholic, are not more polygamous, do not have more electricity at home, and do not own more physical assets than the other ethnic groups. However, they believe much more (up to 38 percentage points, if we compare with the Yoruba) in the influence of their ancestors in their lives (column 9), and have experienced more pressure from their extended family to have children. These simple correlations reinforce the choice of the instrument, and point in the same direction as the main findings of Sections 4 and 6, namely that cultural factors and family structures might be important determinants of differences in fertility levels. The results are shown in Table N4.

Table N4: Ethnicities in Benin and their social, economic and cultural characteristics

|           | Children<br>(1)      | Schooling<br>(2)     | Land<br>(3)         | Ag.Products<br>(4)  | Catholic<br>(5)      | Polygyny<br>(6)      | Electricity<br>(7)   | Assets<br>(8)       | Ancestors<br>(9)      | Clan<br>(10)          |
|-----------|----------------------|----------------------|---------------------|---------------------|----------------------|----------------------|----------------------|---------------------|-----------------------|-----------------------|
| Mean Adja | 10.07***<br>(0.701)  | 0.776***<br>(0.0388) | 2.879***<br>(0.182) | 2.879***<br>(0.196) | 0.209***<br>(0.0380) | 0.457***<br>(0.0464) | 0.681***<br>(0.0434) | 1.310***<br>(0.116) | 0.626***<br>(0.0469)  | 0.720***<br>(0.0435)  |
| Fon       | -2.293***<br>(0.731) | -0.0800*<br>(0.0428) | 0.749***<br>(0.212) | -0.0977<br>(0.214)  | 0.0405<br>(0.0416)   | -0.0806<br>(0.0501)  | -0.0543<br>(0.0473)  | -0.143<br>(0.128)   | -0.218***<br>(0.0511) | -0.313***<br>(0.0480) |
| Yoruba    | -2.833***<br>(0.901) | 0.0312<br>(0.0652)   | -0.0372<br>(0.314)  | 0.121<br>(0.318)    | 0.0895<br>(0.0716)   | 0.0343<br>(0.0809)   | 0.00318<br>(0.0754)  | 0.198<br>(0.245)    | -0.376***<br>(0.0746) | -0.398***<br>(0.0762) |
| Other     | -2.507***<br>(0.834) | -0.0228<br>(0.0617)  | 0.837***<br>(0.313) | -0.515*<br>(0.283)  | 0.112*<br>(0.0644)   | -0.0495<br>(0.0717)  | 0.0350<br>(0.0663)   | 0.245<br>(0.211)    | -0.146**<br>(0.0739)  | -0.343***<br>(0.0704) |
| Mean Y    | 8.011                | 0.718                | 3.491               | 2.770               | 0.254                | 0.397                | 0.645                | 1.242               | 0.432                 | 0.438                 |
| R-squared | 0.0228               | 0.00687              | 0.0138              | 0.00492             | 0.00425              | 0.00558              | 0.00417              | 0.00890             | 0.0314                | 0.0494                |
| N         | 833                  | 905                  | 905                 | 833                 | 903                  | 905                  | 905                  | 905                 | 833                   | 833                   |

NOTE. Data: First-hand data collected in southern Benin. The table reports the OLS estimates of regressing each outcome on ethnicity fixed effects. The sample is restricted to men. "Mean Adja" is the omitted category in the regression. The outcome variable is the total number of children ever born in column 1, whether the respondent attended school in column 2, number of agricultural fields in column 3, number of agricultural products cultivated by the respondent in column 4, whether the respondent is Catholic in column 5, polygamous union in column 6, electricity at home in column 7, a dummy variable that equals one if the respondent owns a speaker or a TV or a DVD or a fridge or a bicycle, a dummy variable that equals one if the respondent believes that his ancestors have an influence on his life, and a dummy variable that equals one if the respondent has experienced pressure from his extended family or clan to have children. Robust standard errors in parenthesis. \*\*\* for  $p < 0.01$ , \*\* for  $p < 0.05$ , \* for  $p < 0.1$ .

The results are shown in Table N5. As expected, the instrument is strong (F-statistic always over 100). In the second stage, the results are confirmed, although of higher magnitude. The increase in the coefficients' magnitude is expected as attenuation bias was likely important in this context (high degree of syncretism), and now the treatment effect is identified for those beliefs were shifted by the instrument, who are likely those with the strongest beliefs in ancestors.

Table N5: Traditional beliefs and contraceptive use

|              | Second stage outcome variables: |                    |                       |                     |                     |                     |                   |                       |                       |                       |                       |                       |                      |                       |
|--------------|---------------------------------|--------------------|-----------------------|---------------------|---------------------|---------------------|-------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|----------------------|-----------------------|
|              | Contraceptive use               |                    |                       |                     | Intention           | Side-effects        | witchcraft        | First stage           |                       |                       |                       |                       |                      |                       |
|              | Currently                       | Traditional        | Modern                | Ever                |                     |                     |                   | (8)                   | (9)                   | (10)                  | (11)                  | (12)                  | (13)                 | (14)                  |
|              | (1)                             | (2)                | (3)                   | (4)                 | (5)                 | (6)                 | (7)               |                       |                       |                       |                       |                       |                      |                       |
| Trad beliefs | -0.121***<br>(0.0385)           | 0.0171<br>(0.0262) | -0.123***<br>(0.0300) | -0.0376<br>(0.0383) | -0.0592<br>(0.0563) | 0.574***<br>(0.102) | 0.102<br>(0.0669) |                       |                       |                       |                       |                       |                      |                       |
| Adja x D     |                                 |                    |                       |                     |                     |                     |                   | 0.157***<br>(0.00646) | 0.157***<br>(0.00646) | 0.157***<br>(0.00646) | 0.157***<br>(0.00646) | 0.157***<br>(0.00700) | 0.134***<br>(0.0106) | 0.149***<br>(0.00741) |
| Controls     | Yes                             | Yes                | Yes                   | Yes                 | Yes                 | Yes                 | Yes               |                       |                       |                       |                       |                       |                      |                       |
| Round FE     | Yes                             | Yes                | Yes                   | Yes                 | Yes                 | Yes                 | Yes               |                       |                       |                       |                       |                       |                      |                       |
| Cluster FE   | Yes                             | Yes                | Yes                   | Yes                 | Yes                 | Yes                 | Yes               |                       |                       |                       |                       |                       |                      |                       |
| Mean Y       | 0.166                           | 0.0589             | 0.104                 | 0.556               | 0.381               | 0.159               | 0.471             |                       |                       |                       |                       |                       |                      |                       |
| N            | 43064                           | 43064              | 43064                 | 43064               | 35915               | 14347               | 26856             | 43064<br>588.3        | 43064<br>588.3        | 43064<br>588.3        | 43064<br>588.3        | 35915<br>501.4        | 14347<br>159.2       | 26856<br>405.0        |
| F-stat       |                                 |                    |                       |                     |                     |                     |                   |                       |                       |                       |                       |                       |                      |                       |

NOTE. Data: Demographic and Health Surveys of Benin (1996, 2001, 2006, 2011 and 2017). The table reports OLS estimates. The sample is restricted to women. The outcome variable in column (1) is a dummy that equals one if the respondent was using contraceptives at the time of the survey. In column (2) a dummy that equals one if the respondent was using traditional methods. In column (3) a dummy that equals one if the respondent was using modern methods. In column (4) a dummy that equals one if the respondent has ever used contraceptives. In column (5) a dummy that equals one if the respondent intends to use contraceptives in the future. In column (6) a dummy that equals one if the respondent does not use contraceptive because of fear to side effects. In column (7) a dummy that equals one if the respondent thinks that aids may be caused by witchcraft. "Trad beliefs" is a dummy variable that equals one if the respondent declares African traditional religion as his/her main religion. "Adja x D" if a dummy variable that equals one if the respondent is Adja and lives within the boundaries of the old Dahomey Kingdom. Controls also include age, education, whether the respondent works, marital status, polygamy, and wealth quintiles. Standard errors clustered at the DHS cluster-level in parenthesis. \*\*\* for  $p < 0.01$ , \*\* for  $p < 0.05$ , \* for  $p < 0.1$ .